JAMAICA

PRACTICAL CASE MANAGEMENT OF COMMON STI SYNDROMES

Specially adapted for use in Primary Health Care Centres

2020

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Ministry of Health & Wellness

Jamaica

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Sexually Transmitted Infections (STI) and STI Syndromes

Specially adapted for use in Primary Health Care Centres

National HIV/STI/TB Programme Ministry of Health and Wellness, Jamaica

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Commonly Used Acronyms and Abbreviations

AIDS	-	Acquired Immune Deficiency Syndrome
AIDSCAP	-	AIDS Control and Prevention
ANC	-	Antenatal Clinic
ARV	-	Antiretroviral
BCC	-	Behaviour Change Communication
BV	-	Bacterial Vaginosis
CA	-	Candida Albicans
CAREC	-	Caribbean Epidemiology Centre
CBO	-	Community Based Organisation
CDC	-	Centers for Disease Control and Prevention
CI	-	Contact Investigator
CIFGM	-	Contact Investigation Field Guide Manual
CISSCM	-	Contact Investigation Self-Study Course Module

CPS	-	Contraceptive Prevalence Survey
CSTI	-	Conventional Sexually Transmitted Infection
CT/Ct	-	Chlamydia Trachomatis
ELISA	-	Enzyme-Linked Immuno-Sorbent Assay
EPI UNIT	-	Epidemiology Unit
FHI	-	Family Health International
FP	-	Family Planning
FU	-	Follow-up
GC	-	Gonorrhoea
Goj/GOJ	-	Government of Jamaica
GPA	-	Global Programme on AIDS
GUD	-	Genital Ulcer Disease
HBV	-	Hepatitis B Virus
HIV	-	Human Immunodeficiency Virus
HPPB	-	Health Promotion and Protection Branch

HPV	-	Human Papilloma Virus
HTLV-I/II	-	Human T-Cell Lymphotrophic Virus Type I/II
ICU	-	Intensive Care Unit
IM	-	Intra-Muscular
IUCD/IUD	-	Intra-Uterine Contraceptive Device
KOH	-	Potassium Hydroxide
LED	-	Leucocyte Esterase Dipstick (test)
MCH	-	Maternal and Child Health
MoHW	-	Ministry of Health and Wellness
MO(H)MOH	-	Medical Officer of Health
MPC	-	Muco-Purulent Cervicitis
MTCT	-	Mother to Child Transmission (of HIV)
MWM	-	Men Who Have Sex With Men
NAC	-	National AIDS Committee
NGU	-	Non-Gonococcal Urethritis
NHCP	-	National HIV/STD Control Programme

NGO	-	Non-Governmental Organisation
NHPL	-	National Public Health Laboratory
NSP	-	National Strategic Plan (formerly Medium Term Plan)
OI	-	Opportunistic Infections
ON	-	Ophthalmia Neonatorum
PAC	-	Parish AIDS Committee
PAHO	-	Pan American Health Organisation
PID	-	Pelvic Inflammatory Disease
PLHIV	-	Persons Living with HIV/AIDS
PIOJ	-	Planning Institute of Jamaica
PPNG	-	Penicillinase Producing Neisseria Gonorrhoeae
RHS	-	Reproductive Health Survey
RTI	-	Reproductive Tract Infection
SMO(H)	-	Senior Medical Officer of Health
STAT	-	At Once

STD/STI	-	Sexually Transmitted Disease/Infection
STICA	-	Sexually Transmitted Infection Clinic Attender
STS	-	Screening (Standard) Test for Syphilis
SW	-	Sex Worker
TRUST	-	Toluidine Red Unheated Serum Test (for syphilis)
T/SMZ	-	Trimethroprim/sulphamethoxazole (or Co-trimoxazole)
TV	-	Trichomonas Vaginalis
USAID	-	United States Agency for International Development
UNAIDS	-	Joint United Nations Programme on HIV/AIDS
VDRL	-	Venereal Disease Research Laboratory (test for syphilis)
WHO	-	World Health Organisation

STI CODING

Helps to ensure **confidentiality** of patient's records and contact investigation.

Cases	Code
1. New Cases of Syphilis	
Primary	A1
Secondary	A2
Early latent (in first year of infection)	A3
Cardiovascular	A4
Neurological	A5
All other late and late latent stages	A6
Congenital aged under 1 year	A7
Congenital aged 1 year and over	A8
2. New cases of gonorrhoea	
Post-pubertal infection	B1
Pre-pubertal infection	B2
Ophthalmia Neonatorum	B3
3. New Cases of other STIs	
Chancroid	C1
Lymphogranuloma venereum	C2
Granuloma inguinale	C3
Non-gonococcal urethritis/cervicitis	C4
Non-specific genital infection with arthritis	C5
Trichomoniasis	C6
Candidiasis	C7

Cases	**Code**
Scabies	C8
Pubic lice (pediculosis pubis)	C9
Herpes simplex virus	C10
Warts (condylomata acuminata/human papilloma virus)	C11
Molluscum contagiosum	C12
HIV infection (latent)	C13.1
HIV infection (minimal symptoms. Score < 6 points)	C13.2
AIDS (major symptoms. Score >/= 6 points)	C13.3
Paediatric HIV/AIDS (</= 9 years)	C13.4
Hepatitis B virus	C14
HTLV-1	C15
4. New cases of other conditions	
Other treponemal disease e.g., yaws	D1
Other conditions requiring treatment in the centre	D2
Other conditions not requiring treatment in the centre	D3

PREFACE TO THE THIRD EDITION

The Ministry of Health and Wellness (MoHW) has made important strides towards improving the surveillance and treatment of sexually transmitted infections (STIs). Prevention messaging, promoting condom use and the treatment of sexual partners have been important strategies in seeking to prevent transmission of STIs. Patients with STIs, including those that can be cured, have a higher risk of HIV infection due to similarity in modes of transmission, enhanced ease of transmission of HIV in the presence of an STI, and similarities in risk factors, such as inconsistent condom use, unprotected sexual intercourse with multiple sex partners, and stigma, discrimination and other barriers to accessing prompt treatment. Untreated STIs have significant long-term sequelae, including infertility, cancer, and may negatively impact quality of life.

Significant medical advances have resulted in new approaches for STI and HIV prevention. Mortality due to HIV/AIDS has decreased significantly because of increased access to testing and antiretroviral therapy. Pre-exposure prophylaxis (PrEP) is another important prevention method that is used to prevent HIV infections in serodiscordant couples, persons who have high risk sexual behaviours, and among patients with repeated STI infections who are at high risk of contracting HIV.

However, significant challenges to controlling STIs still exist. Correct and consistent condom use is an effective measure in limiting STI spread, but nearly 1 in 5 persons who reported having multiple sexual partners in the preceding 12 months reported not using a condom the last 10 times they had sex based on data in the 2017 Knowledge, Attitudes and Behaviour (KAPB) Survey (Ministry of Health and Wellness, Jamaica). The emergence of antimicrobial resistant Neiserria gonorrhoeae in certain parts of the world warrants close STI monitoring locally.

The MOHW has invested in a robust Contact Investigators' programme further emphasising the importance of the treatment of sexual partners of index cases, and reinforcing prevention messaging.

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INTRODUCTION

This guide for the case management of common STI syndromes and conditions is intended for use as a handbook to be kept for consultation mainly at primary health care centres (public and private) providing care to clients with STIs (including HIV).

The approach to clinical management that is emphasised is one based on history taking and clinical examination based on the presenting complaint. Patients who do not respond adequately to the protocols described herein should be referred to an STI Treatment Site or Infectious Diseases Clinic. There is at least one such in each parish. A list of these with telephone numbers is given in the Appendices.

The management of each syndrome is organised in a logical and easy sequence to follow:

- A case definition of the syndrome
- The main aetiologic agents comprising the syndrome
- Treatment protocols for each agent
- STI management decisions using simple algorithms
- In some instances, other important facts about the syndrome

Adequate Case Management of STIs is an important aspect of STI control since it:

- Reduces the period of infection and therefore spread of these diseases
- Prevents the development of expensive and permanent complications and sequelae
- Reduces the risk of transmission of HIV, HBV, HTLV-I/II, and other incurable viral STIs

Algorithms:

For the most effective use of algorithms they should be based, as best as possible, on surveillance data obtained from each geographic area. Where data is quoted in this handbook it is derived from country data.

Where two or more syndromes coexist, the health provider should assess carefully whether they may share a common aetiology (e.g., a GUD and a urethral discharge caused by herpes simplex), or may be caused by different agents. If the latter, a knowledge of the recommended treatments for specific agents should then be used to devise the most cost-effective syndromic approach to management (e.g., ceftriaxone for gonorrhoea and chancroid).

The standard for STI case management is based on adequate laboratory investigations and specific treatment. The aim of the STI programme is to make available at STI Treatment Sites a minimum of microscopy for urethral discharge in males (and other selected categories), darkfield examination for GUD, a rapid test for syphilis on antenatal clients and STI patients – all these while the patient waits.

STI/HIV updates will be shared periodically (quarterly, semi-annually) by the MOHW.

PART 1

STI Case Management

SECTION 1

STI CASE MANAGEMENT

INTRODUCTION

Why is STI Case Management So Important?

STIs are a major public health concern as they are common and serious conditions, and may often be asymptomatic. Effective case management is therefore a cornerstone of STI control since it can prevent physical and psychological complications and sequelae.

Additionally, if managed at the point of first encounter with the health service it can decrease the spread and repeat infections when preventive counselling and services are also offered at that time to index cases and their sexual partners.

Points to remember:

Components of Effective Case Management

- Making an accurate diagnosis
- Giving appropriate therapy to achieve cure where possible, relieve symptoms or distress and reduce infectivity
- Patient education to reduce/prevent future high risk behaviours
- Promotion (and provision) of condoms and other prevention services
- Clinical follow-up as necessary
- Ensuring sexual partners are appropriately notified and treated

STI Case Management may be:

I. ***AETIOLOGIC*****:** based on the laboratory isolation of the causative organism of each STI. There are over twenty distinct aetiologic agents responsible for STIs. Laboratory testing may however be costly and inaccessible.

II. ***CLINICAL*****:** studies have shown that a clinical diagnosis of STI (e.g., based on the clinical appearance of a urethral discharge) may be inaccurate and cannot diagnose mixed infections (e.g., coexisting gonorrhoea and chlamydia), which are very common. Attempts at a “clinical assessment” of aetiology can result in failure to treat appropriately with serious consequences for the patient.

III. ***SYNDROMIC*****:** based essentially on syndromes, i.e., clinical symptoms and signs are used to make a probable diagnosis of the STI for treatment and management. It may include simple laboratory tests, especially those that give immediate results for same-day management of patients, e.g., a rapid plasma reagin (RPR) test for syphilis. Management of STIs based on syndromic presentation has been shown to be feasible, affordable, practical, and rapid for most STI syndromes and may reduce the incidence of HIV.

According to the WHO, chlamydia accounts for the most common STI among men 15-49 years of age in Latin America and the Caribbean, estimated at 3.7% in 2016. Among women in this age group, trichomoniasis is the most common STI at 7.7%, closely followed by chlamydia at 7% (2016). The Ministry of Health and Wellness, Jamaica estimates that the prevalence of HIV among STI clinic attendees in 2017 was 4.61% (95% C.I., 3.76 – 5.60%).

AETIOLOGIC CASE MANAGEMENT

This is the traditional approach to STI case management. However, this approach presents a number of problems for countries with limited resources.

Table 1: ADVANTAGES OF STI AETIOLOGIC MANAGEMENT

- Screening for asymptomatic disease
- Definitive diagnosis to guide partner management and counselling
- Potential for antimicrobial susceptibility testing
- Epidemiologic research studies to guide syndromic management

DISADVANTAGES OF STI AETIOLOGIC MANAGEMENT

- Based on demonstration of a single potential STI pathogen. Currently, there are over 20 STI organisms
- Multiple pathogens often cause similar STI syndromes
- Not all patients with an aetiologically defined STI will have symptoms
- Range of existing tests are limited, majority time consuming to perform, and require repeat visits leading to default
- Delays between testing, reporting, diagnosis and treatment lead to patient dissatisfaction and lack of compliance

- Laboratory diagnosis will vary depending on performance (sensitivity and specificity) of the test* being used
- Tests may be negative in incubating period and inadequately treated disease
- Sophisticated equipment and skilled manpower required is costly, not available in primary health care settings where most patients seek care and therefore inaccessible to the vast majority.

Table 2: ADVANTAGES OF STI SYNDROMIC CASE MANAGEMENT

- Rapid and effective
- Uses Flow Charts (algorithms) which facilitate training and monitoring
- Simple and can be implemented on a large scale
- Can be used by clinicians at any level
- Treatment at point of first encounter
- Patient satisfaction and compliance
- High sensitivity in most cases
- Laboratory tests not necessary
- Cost effective
- Comprehensive care: history, risks assessment, treatment, counselling, partner management
- Treatment of multiple pathogens e.g., gonorrhoea and chlamydia in male urethral discharge
- Simplifies data collection and surveillance

*For example: the ligase chain reaction (LCR) test is twice as sensitive as Thayer Martin culture for gonococcal organisms in local antenatal and family planning studies

DISADVANTAGES OF STI SYNDROMIC CASE MANAGEMENT

- Tends to over-treat, justifiable in high prevalence settings – decreased specificity
- Overuse of expensive drugs – international cooperation for price reduction
- May not fully address asymptomatic cases even with risk factor assessment
- Management of cervical infection may be problematic
- No potential for antimicrobial susceptibility testing
- Lack of definitive diagnosis to guide partner management and counselling
- Vaginal discharge algorithm does not perform well in low prevalence settings such as antenatal clinic and family planning.

SYNDROMIC CASE MANAGEMENT

A syndromic approach to the management of STIs has been adopted for Jamaica due to limited resources for laboratory testing, and to maximise the opportunity to offer treatment and ensure compliance at the time of patient presentation.

How common are STIs?

I. Globally:

- **WHO (2019)** reported 376 million cases of four curable STIs (trichomonas 156 M, chlamydia 127 M, gonorrhoea 87 M, syphilis 6M).

- **UNAIDS/WHO (2019)** has estimated that 74.9 million persons have become infected with HIV since the start of the epidemic and 32 million persons have died from AIDS-related illnesses since. In the Caribbean, based on 2016 data, over 310,000 persons are estimated to be living with HIV. In the Caribbean, 64% of persons living with HIV know their status, and 81% of persons diagnosed with HIV are receiving antiretroviral medications. Among persons on ART, 61% have attained viral suppression.

- **Ministry of Health and Wellness, Jamaica (2019)** estimates that 32,617 persons are living with HIV in Jamaica. Persons between the ages of 20 and 39 years account for the highest number of persons living with HIV/AIDS. The proportion of PLHIV diagnosed is estimated to be 84%, and those receiving antiretroviral therapy (ART) is estimated at 53%, with 65% of those on ART being virally suppressed.

II. Locally:

I. As at 2015, 89% of antenatal clinic attendees were tested for syphilis, among whom 1.5% were found to be positive (Pan American Health Organisation, 2017).

II. Cumulative data between 1982 and 2017 by the Ministry of Health and Wellness, Jamaica show that among PLHIV, 34.7% of men and 43% of women had a history of a sexually transmitted infection.

III. The age group 20 to 29 years accounted for the highest number of newly diagnosed HIV infections in 2017, with 160 and 149 new cases among men and women, respectively.

IV. The most common risk factors for HIV identified among men were having multiple sex partners (26.8%), STI (21.5%), and sex with a sex worker (15.8%), according to the 2017 HIV Epidemiology Report for Jamaica.

V. The most common risk factors for HIV identified among women were STI (41.2%), multiple sex partners (22.9%), and sexual assault (7.2%).

VI. Based on surveillance data among STI clinic attendees for 2017, the prevalence of HIV was 4.61% (95% CI, 3.76 – 5.60%).

VII. 2015 data from STI clinics and antenatal clinics showed that the most prevalent STIs were chlamydia and gonorrhoea, accounting for 27% and 19%, respectively, of those sampled overall. Trichomoniasis was the most common STI among antenatal clients with a prevalence of 25%, and chlamydia was the most common STI among STI clinic clients with a prevalence of 29%. Although men comprised 35% of clients studied, men accounted for 66% of gonorrhoea infections overall.

VIII. The behavioural findings of early sexual initiation, multiple sex partners, relatively low use of condoms with non-regular partners, and poor self-assessment for HIV risk in the general population contribute to a sustained high prevalence of STIs. Effective, efficient and early STI case management, given at the point of first contact with the health system, is a cardinal intervention for the control of STI/HIV in the Jamaican population. This is achievable in part with syndromic case management of the curable STIs.

Table 3: Common STI Signs and Symptoms

MEN	WOMEN
Urethral discharge	Vaginal discharge
Pain on urination	Pain on urination
Scrotal pain/swelling	Lower abdominal pain
Rashes (generalised/localised)	Rashes (generalised/localised)
Swollen glands (generalised/localised)	Swollen glands (generalised/localised)
Sores	Sores
	Menstrual abnormalities

TYPICAL STI RISK ASSESSMENT FACTORS

Development of a local Risk Assessment protocol, based on the local situation is essential.

DEMOGRAPHIC FACTORS

I. AGE: (< 20 years)

- Biological – immature reproductive tract
- Behavioural – multiple partners, sense of invulnerability, low condom use, low awareness of STI/HIV

II. PARTNERSHIP STATUS: (monogamous versus non-monogamous relationship)

- Consider gender norms and power imbalances based on such gender norm expectations**.** If married or in a common-law relationship, may be more likely to be in monogamous relationship.
- If single, may be more likely to have increased risk due to multiple partners and partners with multiple partners.

BEHAVIOURAL

I. NEW SEX PARTNER OR MORE THAN 1 SEXUAL PARTNER IN THE LAST 3 MONTHS

- Clients with a recent history of new or multiple partners are at increased risk for STIs, especially if they do not use condoms.

II. PARTNER HAS OTHER OR HAS MULTIPLE SEX PARTNERS

- Clients with multiple sexual partners are at increased risk for STIs, but it may be extremely difficult, especially for women, to assess their partners' behaviour.

CLINICAL

I. HISTORY OF STI OR PID OR PREVIOUS SYNDROMIC TREATMENT

- Clients with prior STIs are at increased risk, especially if partners were not treated and underlying risk behaviours persist.

II. PARTNER HAS SYMPTOMS OF STI: (urethral discharge, genital sore, dysuria).

- Clients whose partner(s) has (have) symptoms of an STI are at increased risk of infection. It may be difficult for persons to assess their partners' symptoms as frank discussions about sex are not commonplace.

III. CURRENT SIGNS OR SYMPTOMS

- May indicate but are not specific for an STI; e.g., vaginal discharge, vaginal sore, dyspareunia, bleeding after sex, dysuria, lower abdominal pain.

Clients with symptoms or signs of a reproductive tract infection (RTI) should be evaluated and their condition addressed guided by the algorithms for:

- STI screening
- Vaginal discharge
- Lower abdominal pain
- Genital ulcer

- IUD insertion
- Urethral discharge

USES OF STI RISK ASSESSMENT IN REPRODUCTIVE HEALTH

For Family Planning Clients:

As a tool to aid in counselling about contraceptive options. Clients at high risk of current or future STIs will be poor candidates for IUD insertion, but good candidates for barrier methods, e.g., condoms.

For STI Screening of Asymptomatic Clients:

As a tool to help identify clients who are at greater risk of being infected. Such clients may be good clients for further laboratory evaluation or treatment of presumptive infections.

For Syndromic Management of Clients with Symptoms and/or Signs of an RTI:

Syndromic management of urethral discharge in men, and genital ulcers in men and women, has been shown to have a high positive predictive value. However, this is not so for vaginal discharge and STI risk assessment may help to predict more specific treatment.

COMMON STI SYNDROMES IN JAMAICA

STIs produce only a limited number of syndromes. The prevalence of the most common syndromes/conditions seen among public sector STI clinic attenders in Jamaica are shown in Table 4.

Table 4: Proportion of Common STI Syndromes in Public STI Clinic Attendees (STI Prevalence Study, MOHW, 2015).

Chlamydia	29%
Gonorrhoea	25%
Mycoplasma	8%
Trichomonas	7%

ORGANISMS CAUSING SEXUALLY TRANSMITTED INFECTIONS

Bacteria

- Neiserria gonorrhoeae (GC)
- Treponema pallidum (Syphilis)
- Haemophillus ducreyi (Chancroid)
- Granuloma inguinale (GI)
- Lymphogranuloma venereum (LGV)
- Bacterial vaginosis (BV)
- Group B haemolytic streptococcus (GBS)
- Enteric (oro-anal practices) e.g, shigella, salmonella

Mycoplasmas

- Ureaplasma urealyticum
- Mycoplasma hominis

Viruses

- Condylomata acuminata (genital warts, HPV) [>60 types*]
- Hepatitis** B Infection (HBV) - [also types A, C, D, E]
- Herpes virus*** Genitals (HSV) [types I and II]
- Cytomegalovirus Infection (CMV)
- Human immunodeficiency virus infection (HIV) [types I and II]

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* Types 6 and 11 most common; also 16, 18, 31, 33 and 35
** Includes at least 5 types, A to E
*** Includes some 8 types, 1 to 8

- Molluscum contagiosum
- Human T-cell lymphotropic virus type I/II (HTLV-I/II)

Protozoa

- Trichomoniasis (Trich)
- Enteric (oro-anal practices) e.g., Entamoebahistolytica and Giardia lamblia

Fungi

- Candidiasis (yeast, thrush)

Metazoa

- Pediculosis pubis (pubic itch)

SECTION 2

History-Taking and Examination

STIs account for a significant disease burden and source of distress for patients. If unrecognised and untreated, STIs can have serious negative sequelae, including infertility and cancer. This section of the handbook attempts to simplify and systematise the various presentations of STIs.

1. The use of an **STI Medical Record** such as the one at Appendix I will help to ensure that major aspects of an STI history and examination are not overlooked.

2. Identification of the type of patient at risk for, and on whom an STI history is necessary, will include the following categories:

   (a) Sexually active adolescents and young adults (<30 years)
   (b) Patients with ano-genital or other important STI syndromes
   (c) Patients with 2 or more sex partners or a partner change in the past 3 months
   (d) Patients with STI symptoms following a recent partner change (<3 months)
   (e) Patients with a sex partner who has an STI
   (f) Patients engaged in high-risk (unprotected) sexual practices, i.e., not using condoms or other barrier methods
   (g) Patients with a past history of STI

Table 5: Important STI Syndromes*

(I)	Urethral discharge	(VII)	Congenital syphilis
(II)	Vaginal discharge	(VIII)	Anal discharge
(III)	Lower abdominal pain – PID	(IX)	Scrotal pain/swelling
(IV)	Genital/anal ulcers – GUD (with or without adenopathy)	(X)	Balanoposthitis
		(XI)	Enteritis (recurrent/chronic)
(V)	HIV/AIDS or related conditions	(XII)	Acute Arthritis
(VI)	Ophthalmia neonatorum	(XIII)	Genital/skin rashes

*Syndromes I-IX are described in the management sections of this text.

3. Clinicians should use a systematic approach consisting of:

History: To obtain (subjective) information on patient's sexual behaviour and sexual partners as well as history of present and past complaints.

Examination: To confirm physical (objective) evidence of disease and provide specimens for laboratory investigation.

Diagnosis: To determine (or assess) the aetiology or cause of the patient's complaints.

Counsel*: To attempt/plan to prevent/reduce risk-behaviour and notify sex partners by client or provider.

* Order may be interchangeable depending on clinic routine or special problems

Treatment*: To effectively break the chain of transmission.

Follow-up: To plan for test of cure (TOC) by laboratory tests or clinical reassessment if tests are not available.

CARING FOR CLIENT WITH STI - RULE OF "C's"

- **Comfort** – Be at ease, non-judgmental, open-minded
- **Confidence** – Establish trust
- **Confidentiality & Privacy** - Essential
- **Communicate effectively** – Keep it simple, clear
- **Counsel** – Listen, understand, clarify, explain, learn
- **Consider options** – Diagnosis, what treatment, how to locate contacts
- **Compliance = adherence** – Best to give the treatment
- **Contacts** - Main (expedited treatment), others (trace)
- **Condoms** – Build skills "pinch, leave an inch, roll"
- **Continue** – Follow up

Adapted from Professor J. Peter Figueroa

History-Taking

ADDRESS PATIENT'S CONCERNS

Patient may be anxious about:

- Personal implications
- Complications – infertility, HSV, HIV
- Health of a current or future pregnancy
- Compliance with medication
- Self-treatment, e.g., with tetracycline, ampicillin, etc.
- Attitude of some health workers
- Embarrassment: complains of symptoms away from the genital organs
- Privacy
- CONFIDENTIALITY

"Listen with your ears and your eyes. The patient's STI problem may have nothing to do with an infection".

GENERAL POINTS

- Maintain friendly interactions and good rapport with patient at all times
- Adopt a non-judgmental attitude
- Use mainly open-ended questioning
- Maintain eye contact during history taking
- Take history with both clinician and patient comfortably seated
- Initial history: focus on concerns, complaint, and/or reason for visit.
- Complete history: full STI history, record **all** items in the objective medical record (see Appendix I)

A summary of steps in history-taking and examination is as follows:

A. HISTORY TAKING

1. Medical Case History

- Demographic information:
  - name, age, sex, address, marital status, occupation
- Present medical history – reason for visit/current complaint:
  - When did they first notice problem?
- Exposure to infection:
  - date, place
  - site of exposure
- Does partner(s) have a problem?
  - what and how long?
- Any antibiotic use in past 2 weeks?
  - what and how taken?
- Past medical history:
  - previous STI, what and roughly when; systemic diseases e.g., diabetes?
- Family history:
  - number and health of children, siblings, parents
- Specific gender history:
  - pregnancy, menstrual period, contraception, douching
- Drug allergy?

2. Sexual History

(a) Indications for taking a sexual history e.g., classical syndromes, past medical history of STI, age

(b) Objectives of a sexual history e.g., to develop provider/client rapport, determine method and sites for examination.

(c) Give reassurance and comfort

(d) Establish client's responsibility for partner notification e.g., notification by client is preferred to notification by provider, especially in marital situations. Brief client on how to do this and have him/her repeat.

Establish:

- Last sexual contact:
    - Partner type: regular, casual, paid or transactional sex?
- Number of sexual partners:
    - In last month
    - In last 3 months
    - In last 12 months
- Sex with men, women, both?
- Site(s) of exposure? (vaginal, anal, oral, etc.)
- Condom use?
    - If yes: how often, how used? (demonstrate proper use)

Table 6: Major Complications of STI (including HIV)

Men	Women	Neonates
Epididymitis	PID	Prematurity
Prostatitis	Tubal infertility	Low birth weight
Stricture	Ectopic pregnancy	Congenital syphilis
HIV/AIDS	Abortion/miscarriage	Ophthalmia neonatorum
Death	Stillbirth	Chlamydial pneumonia
	Cervical cancer	Septicaemia
	HIV/AIDS	HIV/AIDS
	Death	Death

A sexual history is important for all patients:

- to identify those at risk for sexually transmitted diseases, including HIV
- to guide risk-reduction counselling
- to identify what anatomic sites are suitable for STI screening

The sexual history should be normalised in the context of overall health.

Example of how the clinician could introduce him/herself:

"I am going to ask you some questions about your sexual health. I ask these questions of all my patients regardless of age or marital status, and they are just as important as other questions about your physical and mental health. Like the rest of this visit, this information is strictly confidential."

The five "Ps"of Sexual Health:

- ✓ **Partners**
- ✓ **Practices (sexual)**
- ✓ **Past STIs**
- ✓ **Pregnancy history and plans**
- ✓ **Protection from STIs**

1. **Partners**

   Start by determining the number and gender of a patient's sexual partners. **Remember: Never make assumptions about the patient's sexual orientation.** If multiple partners, explore for more specific risk factors, such as patterns of condom use and partner's risk factors (i.e., other partners, drug use, history of STIs). If one partner, ask about length of the relationship and partner's risk, such as other partners and injection drug use.

- ***"Are you currently sexually active? (Are you having sex?)"***
- ***"If no, have you ever been sexually active?"***
- ***"Do you have sex with men, women, or both?"***

If a patient answers "both" repeat first two questions for each specific gender.

"In the past 12 months, how many sexual partners have you had?"

2. **Practices** (sexual)

   In addition to determining the gender and number of partners, it is also important to ask about sexual practices and condom use. Asking about sex practices will guide risk-reduction strategies and identify anatomical sites from which to collect specimens for STI testing.

   Example:

   - ***"I am going to be more specific about the kind of sex you may have been having over the last year so I understand your risks for STIs."***
   - ***"Do you have vaginal sex, meaning penis in vagina?"***
   - ***"Do you have anal sex, meaning penis in rectum/anus?"***
   - ***"Do you have oral sex, meaning mouth on penis/vagina?"***
   - ***"How often do you use condoms? (never, sometimes, most of the time, always)?"***
   - ***"In what situations, or with whom, do you not use condoms?"***

3. **Past History of STIs**

   A history of prior gonorrhoea or chlamydia infections increases a person's risk of repeat infection. Recent past STIs indicate higher risk behaviours.

   Example:

   "What STIs have you had in the past, if any?"

   Enquire about treatment of STIs and treatment of sexual partners

Example:

"Have you ever had an STI, such as chlamydia, gonorrhea, herpes, or warts?"

4. **Pregnancy plans**

Based on partner information already obtained, you may determine that the patient is at risk of becoming pregnant or of causing a pregnancy. If so, determine first if pregnancy is desired.

Example:

- ***"What are your current plans or desires regarding pregnancy?"***
- ***"Are you concerned about getting pregnant or getting your partner pregnant?"***
- ***"Are you (or you and a partner) trying to get pregnant?"***
- ***"What are you doing to prevent a pregnancy?"***

5. **Protection from STIs**

Example:

"What do you do to protect yourself from sexually transmitted infections and HIV?"

With this open-ended question, you allow different avenues of discussion: condom use, monogamy, patient self-perception of risk, and perception of partner's risk.

- ***"Have you had the hepatitis B vaccine (all three doses)?"***
- ***"Have you had the vaccine against HPV (the virus that causes cancer of the cervix or of the anal or genital areas)?"***
- ***"Have you ever been tested for HIV, the virus that causes AIDS?"***

By the end of this section of the interview, the patient may have information or questions that she/he might not have been ready to discuss earlier.

- ***"Is there anything else about your sexual practices that I need to know about to ensure your good health care?"***
- ***"Do you have any questions for me?"***

At this point, review and reinforce positive, protective behaviours. Address specific concerns regarding higher risk practices. Undertake risk-reduction counseling or a counselling referral.

Table 7. Common Signs and Symptoms of STIs by Sex

In Women	In Men
• Foul smelling/unusual vaginal discharge or itching • Dysuria • Frequency • Dyspareunia • Unusual bleeding • Painful sores in/on genitals • Dull or acute pain in the lower abdomen • Changes in the normal pattern and the characteristics of the monthly period • Sore throat, pharyngeal exudates, cervical lymphadenopathy would suggest that the oral cavity is the anatomical site of exposure	• Urethral discharge • Dysuria • Frequency • Irritation of the skin of the penis, glans, or urethra • Painful sores in/on the penis • Rash (which may indicate secondary syphilis)

3. Five Points in a STI History to Remember Always

i. Previous STI
ii. Potential site of infection
iii. Detailed symptom history
iv. Detailed partner history
v. Allergies

B. CLINICAL EXAMINATION

I. Examination of the Male

- Give brief explanation of examination procedure
- Ask patient to undress – chest down to knees
- Examine standing or lying down

Inspection – in good light

- *Skin* – over abdomen, thigh, buttocks
- *Genitals* – penis, scrotum
- *Lymph nodes* – in groin
- Look for – rashes, pubic lice, sores, urethral discharge

Palpation – with gloves:

- *Foreskin* – retract – if no discharge seen, gently milk urethra forwards
- *Groin* – for nodes – ask if painful
- *Scrotum* – feel for testes, epididymis, spermatic cord
- *Ulcers* – painful or deep-seated

II. Examination of the Female

- Give brief explanation of examination procedure
- Ask patient to undress
- Examine lying on couch in good light
- Cover with draw sheet
- Expose only parts to be examined

Inspection and Palpation – with gloves:

- *Abdomen* – inspect for rashes and obvious swellings
- *Abdomen* – palpate gently and note any tenderness, masses or swellings
- *Groin* – for enlarged lymph nodes

- *Pubic area* – inspect for rashes, pubic lice, sores, glands
- *Thighs* – inspect for rashes
- *Buttocks and Anus* – rashes, warts

Vulva Inspection and Palpation – with gloves

- Inspect vulva, perineum and perianal skin for rashes, sores and swellings
- Part and inspect labia, vestibule, meatus; note lesions, vaginal discharge
- Note colour, odour, and nature of any vaginal discharge

Speculum Examination – with gloves

- Surgically **clean** procedure
- Needs good source of light
- Pass closed bivalve speculum at a 45-degree angle to avoid injury to urethra
- Open blades and inspect cervix – note inflammation, discharge, ulcer, polyps and other lesions
- Slowly withdraw speculum while inspecting vaginal epithelium for sores, warts, inflammation and discharge

Note: It is often necessary to discuss history again after examination, but avoid attempting to take a history while the patient is undressed.

Abdomino-pelvic Pain

May be experienced during examination; observe:

- Abdominal examination:
    - Rebound tenderness
    - Guarding
- Speculum examination:
    - Vaginal discharge

- Cervical mucopurulent discharge
- Bimanual examination
  - Cervical motion tenderness
  - Adnexal tenderness
  - Adnexal mass

Table 8: **Summary of Common STI-related Findings**

• Skin rashes	• Lower abdominal tenderness
• Local or generalised adenopathy	• Pain on cervical movement
• Alopecia (moth-eaten)	• Adnexal tenderness or pain
• Oral lesions	• Pelvic masses
• Genital ulcers, rashes or swelling	• Acute arthritis
• Genital discharge	• HIV-related conditions

C. LABORATORY INVESTIGATIONS – MINIMAL

- Serological screening for syphilis
- HIV screening
- Leucocyte esterase dipstick (LED)
- Urethral/Cervical Gram stain
- Wet mount, KOH (10%)
- Vaginal pH
- Urine examination (glass test, sugar, albumin)

D. EVALUATION

(a) Diagnosis/treatment/counselling (inclusive of patient education)

(b) Follow-up clinical examination

(c) Contact referral

Note:1. If specimens are to be collected, this should be done before administering antimicrobial treatment.

2. A brief history of presenting complaints, presumptive diagnosis, test results with dates, and types of treatments should accompany clients who are being referred for consultation.

The type of approach taken in the case of each client will depend on the resources available to the individual clinician. The clinician is advised to use the clinical algorithm protocol for the relevant syndrome in the management section of this handbook. Where laboratory facilities are available, the approach will be tailored to suit what exists.

Note: An evaluation of these guidelines will be done periodically with a view to improving both the procedures and the quality of STI care in Jamaica.

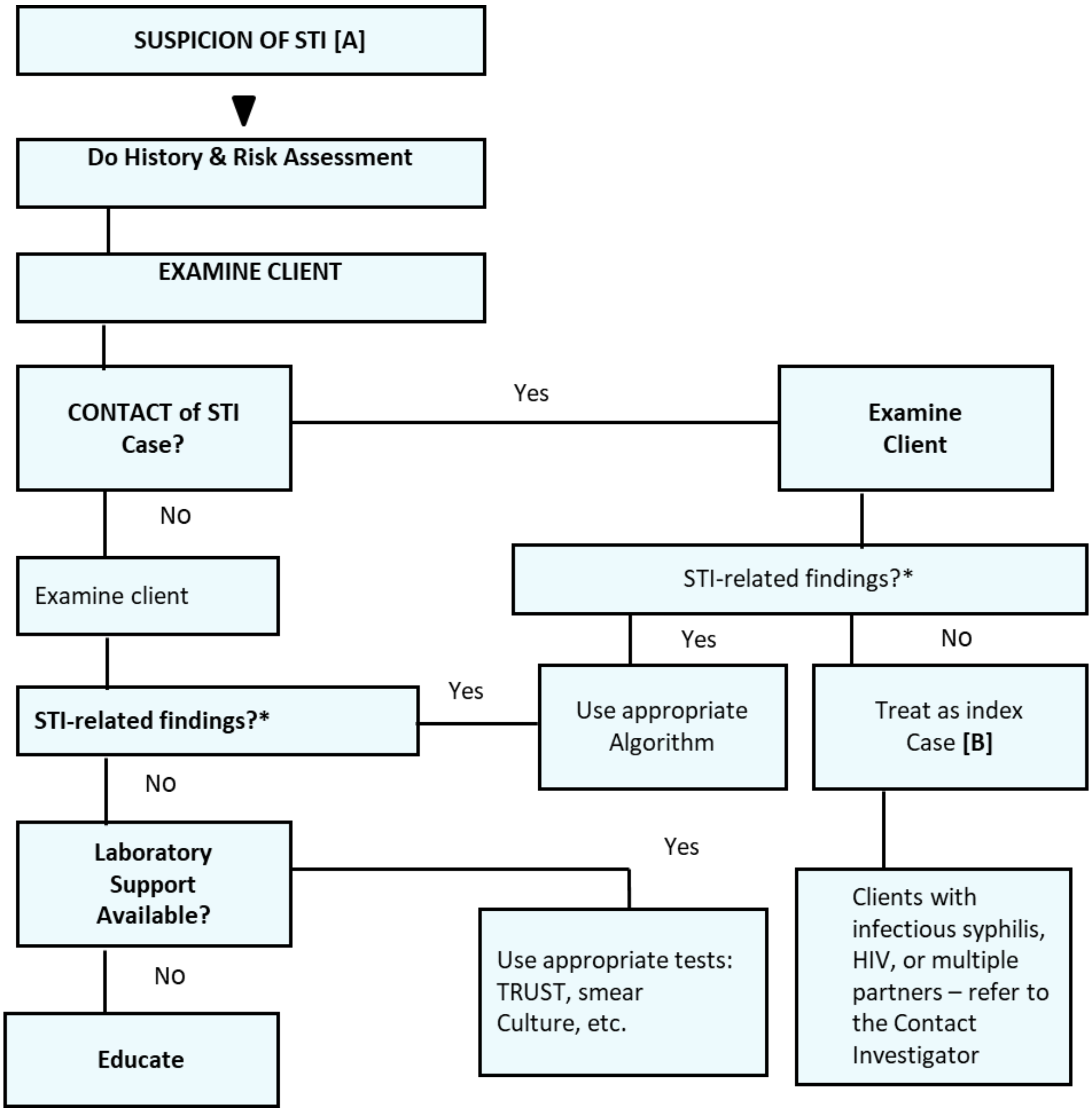
SELECTED REFERENCES

1. Chlamydia, gonorhoea, trichomoniasis and syphilis: global prevalence and incidence estimates, 2016. Rowley, J., et al. Bull World Health Organ 2019; 97:548-562P.
2. Global AIDS Update. UNAIDS; 2019.
3. HIV Epidemiology Report for the Ministry of Health, Jamaica; 2017.
4. National Family Planning Board, Jamaica. (2017). National Integrated Strategic Plan for Sexual and Reproductive Health & HIV 2014-2019.

SECTION 3

Approach to the Patient Who Suspects an STI

FIGURE 3.1: ALGORITHM FOR STI SCREENING



Notes to Figure 3

*See **Table 8** (Summary of Common STI-related Findings)

[A] If the patient has signs, use the appropriate algorithm

[B] Appropriate explanation should be given as to why (s)he needs treatment

SECTION 4

STI Case Management by Syndromes

4.1 URETHRAL DISCHARGE

A. Case Definition

This is the most frequent STI syndrome seen in men. They usually have dysuria (a burning sensation in the urethra), tingling sensation and mucoid to purulent urethral discharge and urinary frequency. Some men have asymptomatic infection (so check history in partner).

B. Recommended Treatments (Urethral Discharge by Aetiology and Syndrome)

I. Gonococcal Urethritis

Definition: Acute uncomplicated infection of the urethra. These regimens are all effective against penicillinase-producing N. gonorrhoeae (PPNG) and high level tetracycline-resistant N. gonorrhoeae (TRNG) strains.

Table 9. Treatment of Gonococcal Urethritis	
Ceftriaxone 250 mg IM stat **PLUS** Azithromycin 1g PO stat (preferred choice)	Effective against infection at all sites. Low toxicity associated with beta-lactam antibiotics. Treponemicidal.
OR	
Doxycyline 100 mg PO bd for 1 week **PLUS** • Metronidazole 2g STAT 12-24 hours later **OR** • Cefixime 400 mg PO STAT **OR** • Ciprofloxacin 500 mg PO stat **OR** • Ofloxacin 400 mg PO stat	Avoid doxycycline in pregnancy, lactation and persons 16 years and under

Notes:

1. If oro-pharyngeal infection is suspected (in persons with a history of oral sex) Azithromycin AND Doxcycline to treat.

2. Epidemiologic (preventive) treatment of partner – same as for uncomplicated gonorrhoea.

3. Do not use less than the recommended dosages as resistance to some regimens is already emerging. This is of special importance when providing oral medication.

II. Non-gonococcal urethritis

Acute uncomplicated chlamydial infection of the urethra.

III. **Syndromic management**

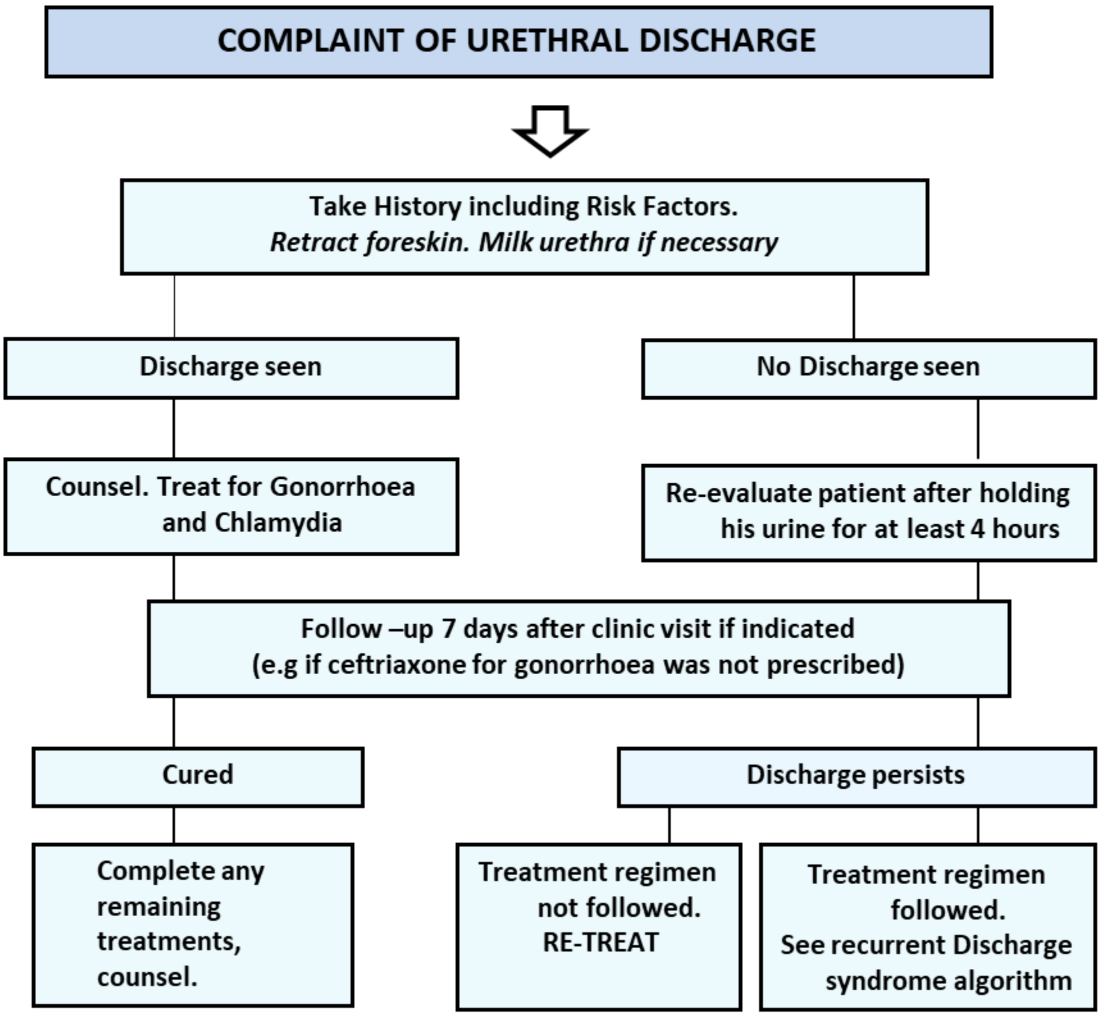
Treat for both gonorrhoea and chlamydia. See urethral discharge algorithm.

Note:

1. Avoid sex, alcohol, urethral milking for 2-3 weeks
2. Examine and treat sexual partner(s)
3. Discharge may not be seen if patient passes urine

ALGORITHM: MANAGEMENT OF URETHRAL DISCHARGE

FIGURE 4.1a

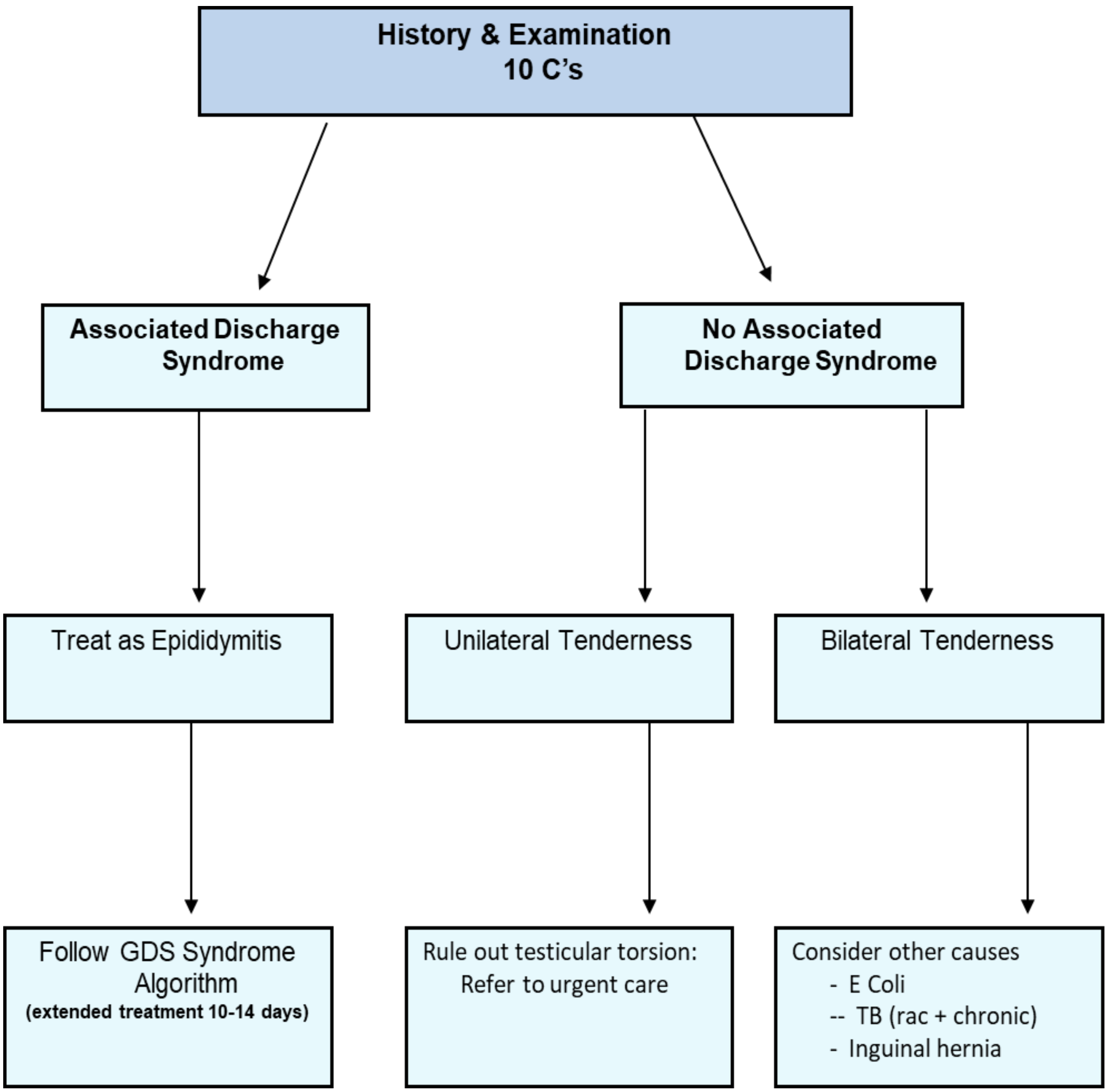


Notes to Figure 4.1a:

1. Early morning bead of pus; urinary "threads" in first urine passed; WBCs>4/HPF on Gram stained smear - suggests chlamydial/non-gonococcal urethritis.
2. Refer to urologist or hospital, especially patients >35 years: non STI causes of urethritis may be presen

ALGORITHM: TESTICULAR PAIN

FIGURE 4.1b



THE 4 Cs OF STI COUNSELLING

1. ***Comply with instructions for taking medication.***
2. ***Counsel for risk reduction i.e., reduce number and type of sex partners, etc.***
3. ***Condom use: properly and with each sex act***
4. ***Contact(s): ensure referral of sexual contacts***

References:

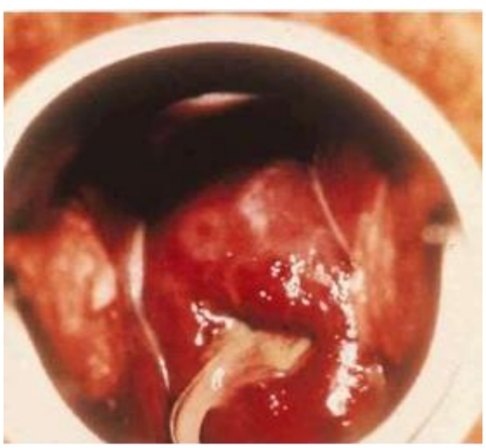
WHO Guidelines for the Treatment of Neisseria gonorrhoea, 2016.

WHO Guidelines for the Treatment of Chlamydia trachomatis, 2016.

IMAGES OF COMMON STIS

No. 1 Gonococcal Urethritis

No. 2 Non-gonococcal Urethritis



No. 3 Gonococcal Cervicitis

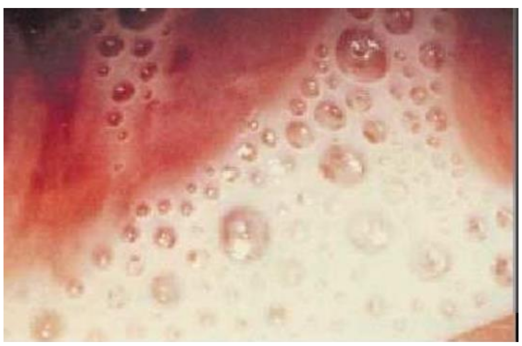
No. 4 Non-gonococcal cervicitis

No. 1: Gonococcal Urethritis in the male. There is an associated meatitis around the bead of pus.

No. 2: Non-gonococcal Urethritis – Chlamydial urethritis typically gives a scant mucoid to mucopurulent discharge with mild or no associated symptoms. A bead of pus, seen on first arising in the morning, may be the only sign.

No. 3: Gonococcal Cervicitis – Mucopurulent endocervical discharge with bleeding and oedema is typical of gonococcal infection.

No. 4: Non-gonococcal/Chlamydial cervicitis – Classically the discharge is slight, purulent and tenacious, but is mucoid or mucopurent in many cases. In rare cases, may be absent.



No. 5 Trichomonal vaginitis

No. 6 Candidal vaginitis

No. 7 Balanoposthitis

No. 8 Bacterial Vaginosis

No. 5: Trichomonal vaginitis –The vaginal discharge is profuse, purulent and malodorous in typical cases. Occasionally it may be frothy with multiple, small, red erosions seen on the vaginal portion of the cervix (“strawberry cervix”)

No. 6: Candidal vaginitis – The discharge is clumped or flocculent (“cheesy”)

No. 7: Balanoposthitis – The glans penis and surrounding foreskin may be inflamed.

No. 8: Bacterial (Gardnerella) Vaginosis – The vaginal discharge is greyish white and homogenous. When moderate to profuse, it may be seen flowing out over the posterior fourchette.

ALGORITHM - PERSISTENT DISCHARGE

FIGURE 4.1c

Re-Evaluate patient: 10 C's History Taking Examination

Appropriate regime not followed OR unprotected sex with untreated partner

Re-treat

Ceftriaxone 250mg stat PLUS

Azithromycin 1g stat

Or

Doxycycline 100 mg bd x 1/52

Appropriate regime followed & no unprotected sex

Gonorrhea swab & culture and

Then treatment

Ceftriaxone 500 mg IM AND

Azithromycin 2 g PO stat

OR

Cefixime 800mg po stat AND

Azithromycin 2g po stat

Refer to STI Clinic

Notes to Figure 4.1c:

1. Abstain from all sexual activity OR use a condom with ALL types of sexual activity for 7 days OR until completion of course of medication to prevent spread to partner(s).

4.2 VAGINAL DISCHARGE SYNDROME

A. Case Definition

The most common STI syndrome seen in women. The discharge may come from the cervix (cervicitis), vagina (vaginitis) or both. Clients may complain of increased (abnormal) vaginal discharge, with an associated odour or colour. There may be vulval itching, dysuria and dyspareunia. Antibiotic or steroid use, diabetes mellitus, douching, immunosuppression, oral contraceptives, pregnancy and menstruation are important predisposing factors that must be elicited in the patient's history.

N.B A subjective complaint of vaginal discharge must be supported by objective evidence on examination. Some women translate a wetness of their underclothing as being due to a vaginal discharge when the source may be non-vaginal. Before taking swabs, ascertain that the patient has not douched recently or used vaginal medication.

B. Recommended Treatments [Abnormal Vaginal Discharge by Aetiology and Syndrome]

Table 10: Treatment of Abnormal Vaginal Discharge Syndrome

CONDITION	**TREATMENT**
CERVICITIS	
1. CHLAMYDIA TRACHOMATIS (CT)	• Azithromycin 1 g PO stat **OR** • Doxycycline 100 mg PO bid x 7 days. **OR** • Erythromycin 500 mg PO qid x 7 days. **OR** • Ofloxacin 300 mg PO bid x 7 days.
2. GONORRHOEA (GC)	• Ceftriaxone 250 mg IM stat **PLUS** Azithromycin 1g PO stat. **OR** • Cefixime 400 mg PO single dose **PLUS** Azithromycin 1 g PO single dose **OR** • Ciprofloxacin 500 mg PO stat. **OR** • Ofloxacin 400 mg PO stat. **OR** • Norfloxacin 800 mg PO stat. (PLEASE NOTE THAT THIS IS COMBINATION TREATMENT FOR CHLAMYDIA TRACHOMATASIS AND GONORRHOEA)

3. TRICHOMONIASIS (TV)	• Metronidazole or Tinidazole 2g PO stat for acute cases and contacts **OR** • Metronidazole or Tinidazole 500 mg PO bid x 7 days (depending on chronicity) In pregnancy: Clotrimazole vaginal tabs in the first trimester and metronidazole regime second & third trimester. Avoid alcohol 48 hours pre- and post-treatment.
4. HERPES SIMPLEX VIRUS (HSV)	• Acyclovir 400mg PO: Acute primary episode: 3 x daily for 7-10 days. Recurrent episode: 3 x daily for 5 days. More than 6 episodes per year requires suppressive therapy at 400 mg twice daily for 6 months • Saline (sitz) baths prn • Pap smear 6-12 monthly • Refer to a gynaecologist if pregnant.
5. HUMAN PAPILLOMA VIRUS (HPV)	• Electrocautery or cryotherapy • Pap smear 6-12 monthly. Refer OBGYN for treatment if pregnant.

VAGINITIS	
1. BACTERIAL VAGINOSIS (BV)	• Metronidazole 500 mg PO bid x 7 days. **OR** • Metronidazole gel (.75%) 1 full applicator/5 g intra-vaginally for 5 days **OR** • Clindamycin cream (2%) 1 full applicator/5 g intra-vaginally for 3 nights **OR** • Clindamycin 300 mg PO bid x 7 days
2. CANDIDIASIS (CA)	• Clotrimazole 200 mg vaginal preparations for 6 nights **OR** • Miconazole 200 mg vaginal preparations for 6 nights **OR** • Ketoconazole 200 mg vaginal preparations for 3 nights **OR** • Fluconazole 150 mg PO stat **OR** • Itraconazole 200 mg PO q12h x 1 day

III. Sex partners share these organisms during sexual activity – infection and clinical disease can be transmitted from a non-diseased infected carrier to a susceptible host.

1. **Clinical disease –** can occur by the acquisition of a new microorganism, redistribution of the existing genital microbiological flora, changes in the pathogenicity of endogenous organisms, or changes in host susceptibility.

2. **Vulvovaginal infections –** four major groups:
   - **i.** Protozoal: Trichomonas vaginalis
   - **ii.** Bacterial: Bacterial vaginosis
   - **iii.** Fungal: Candida albicans
   - **iv.** Nonspecific: Aetiology varied and complex – drug-resistant organisms, sensitivity to soaps, douches, contraceptives, lubricants, semen, etc.

3. **Vaginal Discharge Syndrome**

Vaginal discharge may be:

1. **Normal (physiological):** occurring pre and post-menstrually, or following excessive douching or coitus.
2. **Abnormal (pathological):** resulting from infection or irritation of the vulva (vulvitis), vagina (vaginitis), and cervix (cervicitis).

FEATURES

1. **Terms vulvo-vaginitis and cervico-vaginitis –** sometimes used to describe the condition of abnormal vaginal discharge.

2. **Normal genital microflora in women –** comprise a variety of microorganisms that seem to be commensal, but under certain conditions can produce genital disease.

3. **Cervical infections; three major groups:**
   - **i.** Bacterial: Chlamydia trachomatis, Neisseria gonorrhoeae, Trichomoniasis
   - **ii.** Viral: Herpes simplex, Human papilloma virus, HIV
   - **iii.** Other: Aetiology varied and complex – neoplasia, hormonal, trauma, etc.

4. **Occurrence:** vaginal discharge probably accounts for more physician visits by women of reproductive age than any other sexually transmitted infection syndrome (see local prevalence data).

5. **Importance:** serious severe sequelae can follow the agents causing cervicitis, including bacterial or viral. There is also a growing body of evidence which suggests that serious upper genital tract complications (including postpartum endometritis, post abortion PID, preterm delivery and HIV transmission) may follow infection with the organisms causing bacterial vaginosis.

Table 11: Other Causes of Symptomatic Vaginal Discharge

Viral Causes

- Herpes simplex virus
- Human papilloma virus
- Human immonodeficiency virus

Less Common Infective Causes

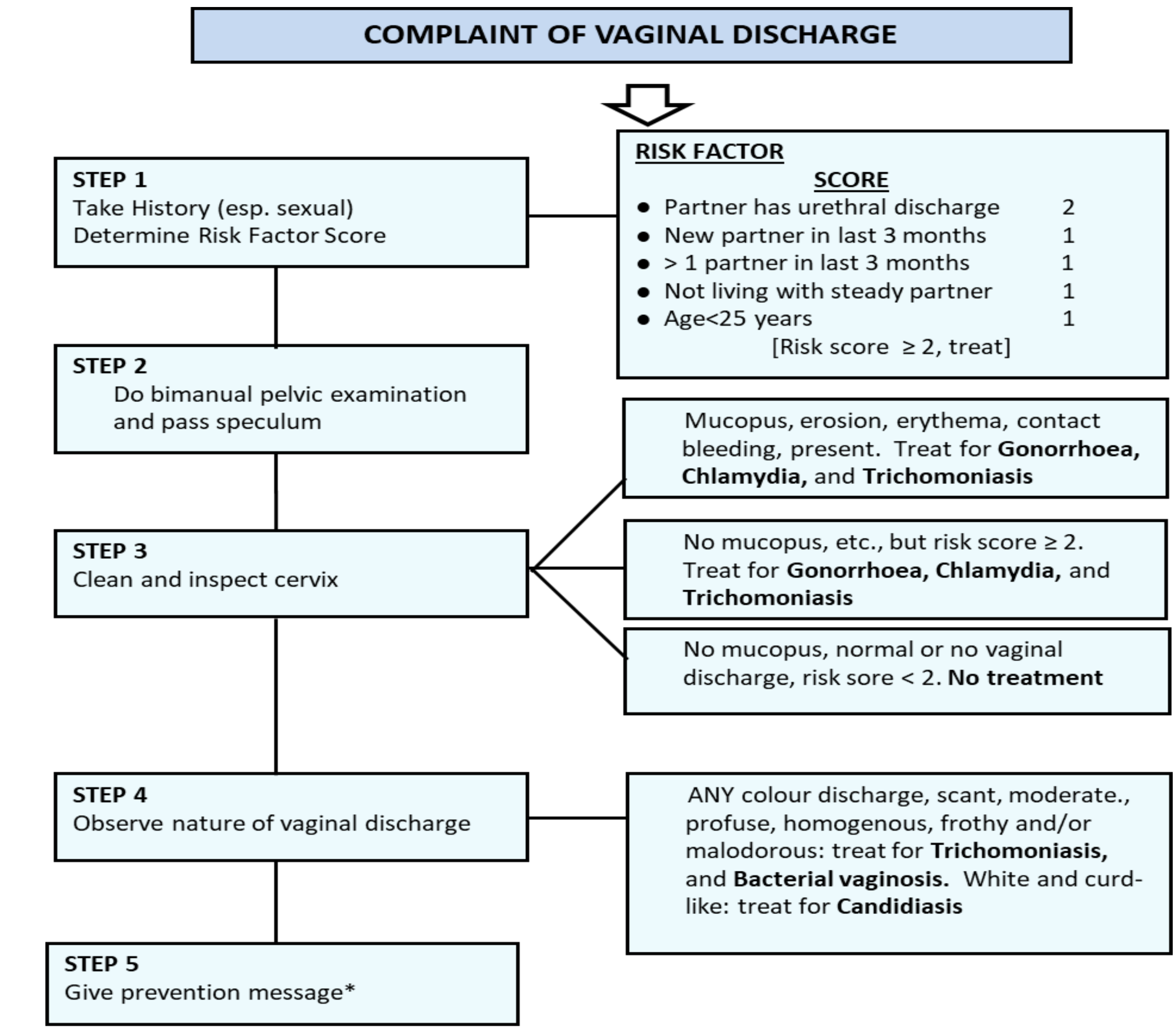
- Staphylococcus aureus (toxic shock)
- Streptococcal (β-haemolytic) vaginitis

Non-infective/non-specific Causes

- Broad spectrum antibiotic, steroid, antiviral or antifungal therapy
- Pregnancy
- Diabetes
- Ovulation
- Excessive coitus
- Atrophic vaginitis (premenarche) postpartum and postmenopausal)
- Foreign bodies
- Neoplasm
- Potential irritants (deodorants, soaps, vaginal douches and sprays, detergents, tampons, pads, spermicides, condoms, perfumed or coloured toilette tissue.
- Chemotherapy
- Allergic and chemical reaction
- Erosive lichen planus
- Immunosuppression
- Radiation therapy

ALGORITHM: MANAGEMENT OF VAGINAL DISCHARGE

FIGURE 4.2a



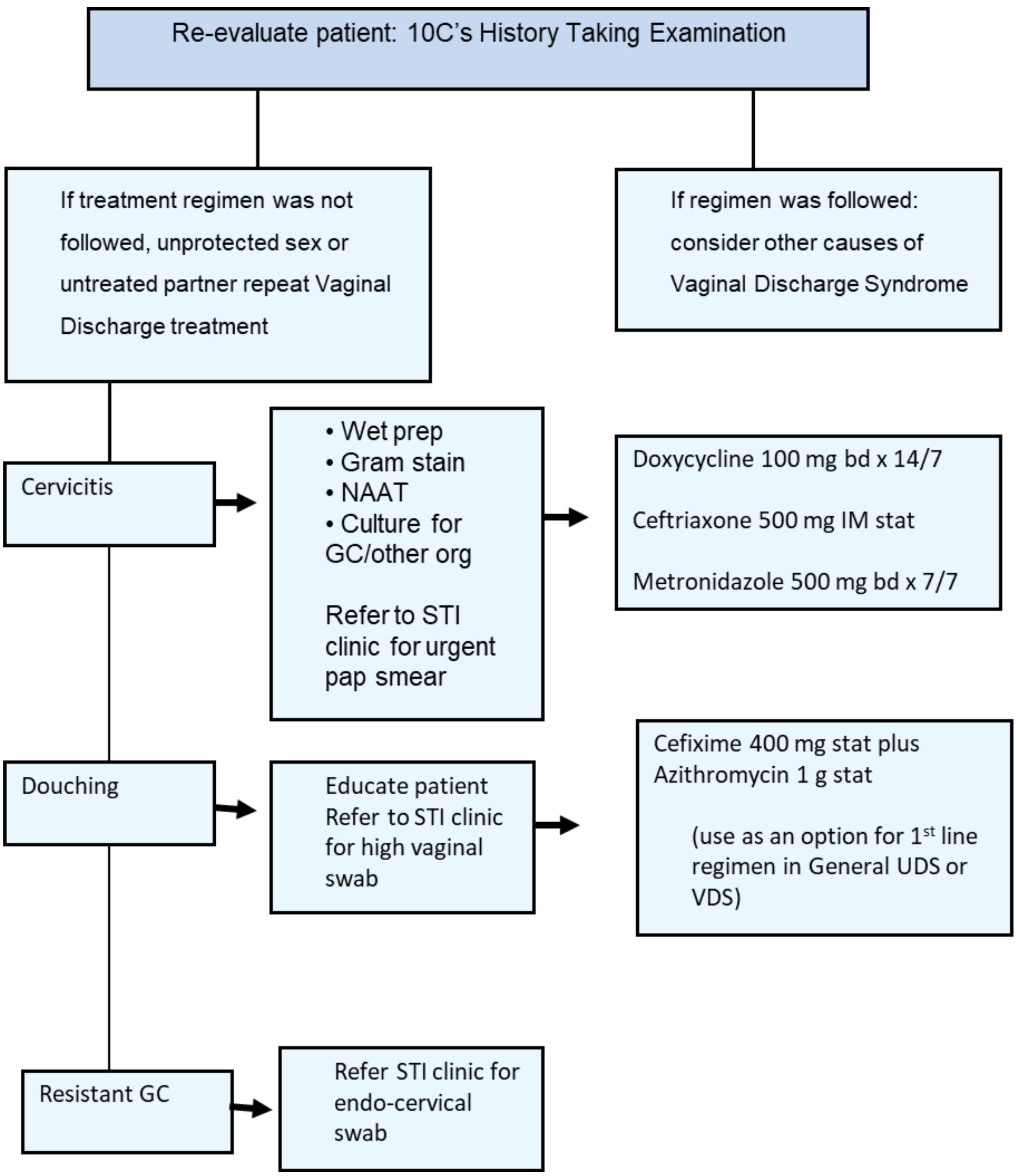
Notes to Figure 4.2a:

***Comply with medication; condoms; contacts, counsel re risk reduction.

[Screen all patients for HIV/Syphilis. Do pap smear and refer persistent cases of discharge to gynaecologist]

ALGORITHM: PERSISTENT VAGINAL DISCHARGE

FIGURE 4.2b



Notes to Figure 4.2b: NAAT: nuclear acid amplification test

Table 12: Diagnostic Features of Vaginal Infection in Premenopausal Women*

CLINICAL CONDITION				
	NORMAL VAGINA	**CANDIDIASIS**	**TRICHOMONIASIS**	**BACTERIAL VAGINOSIS (BV)**
Aetiology	Uninfected. Lactobacilli Predominant	Candida albicans and other yeasts	Trichomonas vaginalis	Uncertain; associated with Gardnerella vaginalis, various anaerobic bacteria, and Mycoplasma hominis
Typical Symptoms	None	Vulvar pruritus and/or irritation; external dysuria, increased discharge	Profuse discharge, often malodorous; external dysuria and genital irritation often present. Dyspareunia may be present	Malodorous especially after ejaculation of male, increased discharge (commonly present at the introitus). Often a chronic history of discharge

*Adapted from Stamm, W. E. et al. (1981). Treatment of the acute urethral syndrome. New England Journal of Medicine, 304, 956-8.

Table 12: Diagnostic Features of Vaginal Infection in Premenopausal Women*

CLINICAL CONDITION				
	NORMAL VAGINA	**CANDIDIASIS**	**TRICHOMONIASIS**	**BACTERIAL VAGINOSIS (BV)**
Discharge Amount	Variable; Usually scant	Scant to moderate	Profuse (in posterior fornix)	Moderate to profuse
Colour	Clear or white	White	White, yellow or green	White or grey
Consistency	Non-homogenous, flocculent	Clumped, “cheesy”, adherent plaques	Homogenous, watery, often frothy or foamy	Homogenous, uniformly coating vaginal walls; occasionally frothy
Associated Inflammatory Signs	None	Erythema, oedema and/or erosions of vagina or external genitalia; vulvar	Erythema of vaginal mucosa, introitus; occasional cervical petechiae; occasional vulvar dermatitis	None

Table 12: Diagnostic Features of Vaginal Infection in Premenopausal Women*				
CLINICAL CONDITION				
	NORMAL VAGINA	**CANDIDIASIS**	**TRICHOMONIASIS**	**BACTERIAL VAGINOSIS (BV)**
		dermatitis common		

Table 12: Diagnostic Features of Vaginal Infection in Premenopausal Women* (cont'd)

CLINICAL CONDITION

	NORMAL VAGINA	**CANDIDIASIS**	**TRICHOMONIASIS**	**BACTERIAL VAGINOSIS (BV)**
pH of Secretions	Usually <4.5	<4.5	Usually ≥ 5.0	Usually ≤ 4.5
Amine test ("fishy") odour with 10% KOH	None	None	Usually present	Present
Microscopy (if available)	Normal epithelial cells; Lactobacilli predominate	Leukocytes; epithelial cells; yeasts, mycelia or psuedomycelia In up to 80% of symptomatic patients seen	Leukocytes; motile trichomonads seen in 80-90% of symptomatic patients.	Clue cells; rare leukocytes; lactobacilli outnumbered by profuse mixed flora, nearly always including Gardnerella vaginalis plus anaerobes.

Table 12: Diagnostic Features of Vaginal Infection in Premenopausal Women* (cont'd)				
CLINICAL CONDITION				
	NORMAL VAGINA	**CANDIDIASIS**	**TRICHOMONIASIS**	**BACTERIAL VAGINOSIS (BV)**
Basic Management of Sex Partners	None	Examination and treatment usually are not necessary, but may be indicated in some cases of recurrent infection	**1. Regular partners or source contacts:** Routine STI examination; treat with Metronidazole 2 g in a single dose. If history of STIs treat with metronidazole 500 mg bd x 7/7 **2. Casual contacts:** Routine examination; individualise treatment depending on clinical history, examination and epidemiologic information	**1. Regular partners of source contacts;** Routine STI examination; treatment usually not indicated; in the event of recurrent BV without evidence of other STI, treat partner(s), and patient simultaneously. **2. Casual contacts:** Routine STI examination; do not treat.

Notes to Table 12:

1. Colour of discharge is determined against a white background or on a swab.
2. pH determination is not useful if blood is present.

3. Gram stain is excellent for detecting yeasts and pseudomycelia and for distinguishing normal flora from the mixed flora seen in bacterial vaginosis, but is less sensitive than saline preparation for detection of T. vaginalis.

NB: Treatment for TV should not be dispensed or prescribed for unexamined partners, nor given to patients for distribution to partners.

Vaginal pH: Determination of vaginal pH is a quick and easy method of assessing vaginal health during the speculum examination. Vaginal secretion may be transferred to the pH paper by swab or paper may be applied directly to the vaginal wall. Ensure that it is not contaminated by alkaline cervical mucous.

Algorithms for the syndromic management of cervico-vaginal discharge are difficult since many gonococcal and chlamydial infections of the cervix are asymptomatic (even on speculum examination of the cervix). The sensitivity and specificity of the algorithm is enhanced, and decision to treat supported, if the patient has a historical risk score of 2 or more (see Figure 4.2b).

Vaginal Discharge Syndrome in Non-STI Clients

- *The foregoing management of abnormal vaginal discharge applies essentially to patients attending public STI clinics in which the prevalence of chlamydia and gonorrhoea is high.*
- *Use of simple leucocyte esterase dipstick (LED) test, when positive, gives useful additional support to a decision to treat for cervicitis.*

4.3 ANAL DISCHARGE SYNDROME

A. Case Definition

Anal discharge is any discharge from the anus that is not stool or blood. Rectal discharge can occur for many reasons, including anal fissure, anal fistula (an abnormal connection between two organs) or abscess, other infections including STIs, or chronic inflammatory diseases.

A purulent anal discharge with a history of receptive anal intercourse or prior sexually transmitted infections (STIs) is suggestive of proctitis. Purulent discharge can also be a manifestation of a draining perianal abscess, which may be associated with fever or chills, or an anal fistula. Rectal bleeding and pain that occur with defecation are characteristic of anal fissure or hemorrhoids. Patients with anal fissure may have a history of receptive anal intercourse, chronic constipation, or prior anal fissures.

Common clinical presentations of anorectal STIs are characterised by anatomical involvement and causative organism into a distal proctitis and proctocolitis. Acute proctitis, can be infectious in origin or, less commonly, a manifestation of inflammatory bowel disease.

In MSM, proctitis is most often reported in persons who practice receptive anal intercourse and may be caused by gonorrhoea, chlamydial infection, HSV infection, and/or syphilis.

Gonorrheal rectal infection frequently results in proctitis, whereas chlamydial infection is sometimes asymptomatic. However, outbreaks of proctitis related to lymphogranuloma venereum (LGV), caused by *Chlamydia trachomatis* serovars L1, L2, and L3, have been reported in MSM. LGV proctitis causes ulcers, which present with hematochezia and rectal pain. Infectious proctitis occurs more commonly than non-infectious proctitis.

Table 13: Causes of Proctitis

Bacterial	**Virus**	**Other**
Neisseria gonorrhoeae (GC)	Herpes simplex virus (HSV)	Tissue damage resulting from insertion of foreign bodies
Chlamydia trachomatis (CT) – Lymphogranuloma Venereum (LGV)		Ulcerative colitis, Crohn's disease
Treponema pallidum (syphilis)		

B. Recommended Treatment

Table 14. Treatment of Anal Discharge Syndrome

Ceftriaxone 250 mg IM stat **PLUS** Azithromycin 1 g PO stat **OR** Cefixime 400 mg PO stat **PLUS** Azithromycin 1g PO stat.

C. CLINICAL EXAMINATION

ANOGENITAL EXAMINATION

Anogenital Examination Procedures

A complete anogenital examination includes three components: external visual examination, digital rectal examination, and anoscopy.

Preparing the Patient:

Greet the patient warmly. Ensure patient privacy (visual and auditory). Clearly explain the procedure before and as you proceed so that the patient is aware of each action as it occurs. Position the patient appropriately and ensure they are comfortable. Reassure the patient throughout the exam, as needed.

External Visual Inspection of Anus

To conduct a thorough visual examination, the provider should retract the buttocks with both gloved hands to expose the perianal skin. Check the perianal region (anus and perineum) for external lesions, skin irritation (acute or chronic), dermatitis, fistula, erythema, abscess, or thrombosed external hemorrhoid.

If anal fissure is present, ***do not proceed*** with a digital rectal examination or anoscope examination. These procedures are very painful for a patient with an anal fissure, and can be postponed until the anal fissure is treated and resolved.

Digital Rectal Examination (DRE)

The buttocks should be gently spread apart, and the anus, posterior perineum, and gluteal folds visually inspected to identify pathologic conditions such as condylomata (warts), external hemorrhoids, abrasions, ulcers, abscesses or cellulitis, and, occasionally, malignancies (e.g., melanoma and anal or rectal carcinoma).

A hand is then placed on the patient's anterior pelvic bone to provide counter-traction while the dominant hand, with the help of generous lubrication, slowly advances through the sphincter and into the rectum. After a few seconds, the sphincter should relax slightly, at which point the digit is advanced further (see the image below). Note should be made of sphincter tone. Palpation of the internal structures then proceed in a systematic fashion.

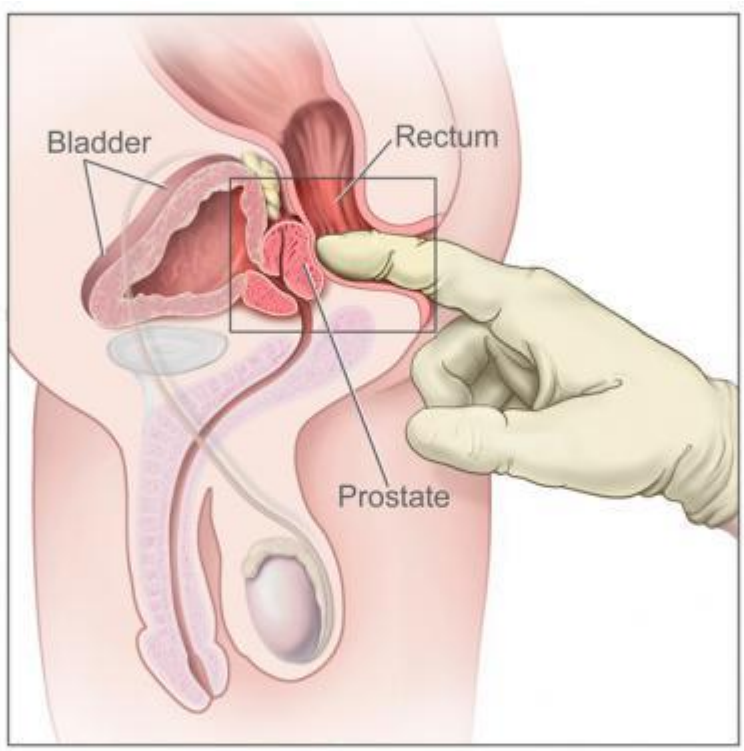


Figure 4.2c. ***Digital rectal examination. Drawing shows gloved and lubricated finger inserted into rectum to feel prostate. Image courtesy of National Cancer Institute. Medscape Gallery image.***

Palpation begins at the apex of the prostate and progresses toward the base to determine the size of the gland and assess its consistency, which, in a normal gland, resembles that of the thenar eminence when the thumb and little finger are opposed.

Palpate the circumference of the rectal vault to identify any internal hemorrhoids, to determine whether the consistency is smooth, and to detect any stool present and assess its consistency. Upon removal of the finger, the stool on the finger is evaluated for blood and can be sent for studies, including occult blood.

After the examination, ideally a generous supply of paper towel or wipes should be made available to the patient, along with a sink with soap and water, privacy for cleaning up, and space for dressing.

Anoscopy

Anoscopy is recommended for patients with rectal pain, bleeding, and/or bloody or purulent discharge and in patients with a palpable abnormality on digital rectal exam. It is performed using a small, rigid, tubular instrument called an anoscope. Lubricant is applied, and the instrument is inserted into the anus, after which the obturator (central conical structure) is removed. Each quadrant of the anal canal is examined with the aid of a bright light as the instrument is slowly removed. Lesions of the anal canal, including internal hemorrhoids, fissures, mucosal abnormalities (e.g., warts, proctitis), abscesses, and tumors, can be visualised and biopsied as indicated.

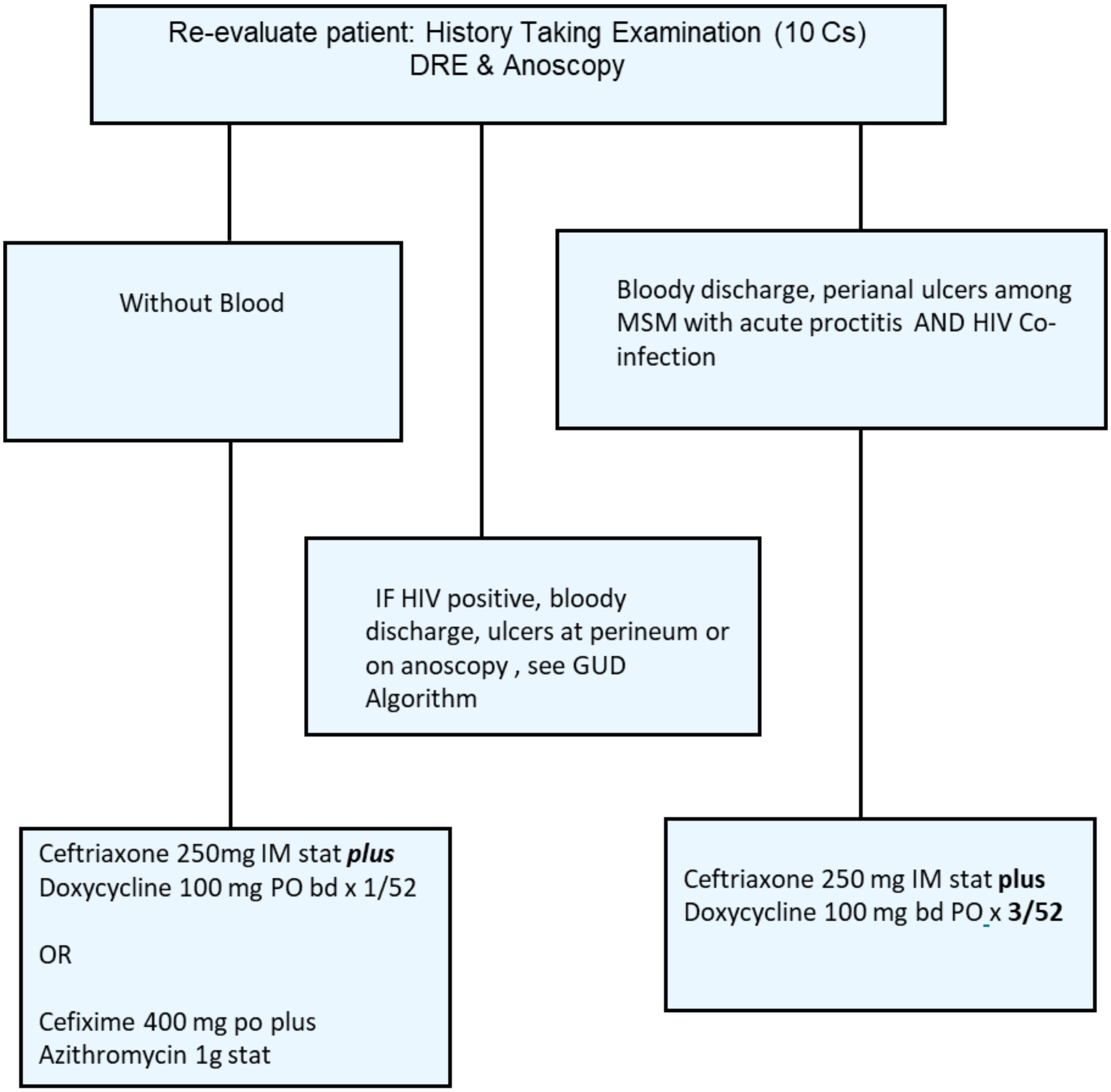
When using an anoscope with an obturator, it is important to ensure that the obturator of the anoscope is completely inserted before beginning the anoscopy evaluation. Generously lubricate the anoscope with standard lubricating jelly or lidocaine jelly. Introduce the anoscope gently, and advance it slowly with a slight side-to-side twisting motion while the patient bears down. If resistance due to contraction of the external anal sphincter is significant, constant pressure on the anoscope eventually fatigues the muscles and permits insertion.

Maintain pressure over the obturator with the thumb during insertion to keep the obturator from slipping out. To avoid pinching the anal mucosa, completely remove the anoscope, and reinsert the device if the obturator slips or falls out during insertion. Some anoscope models have small tabs at the operator end of the device. These tabs should be aligned along the rostral-to-caudal axis of the patient to allow complete insertion of the device. Once the anoscope is completely inserted, remove the obturator.

As the anoscope is slowly withdrawn, the anal mucosa can be visualised over the entire circumference of the canal. Any debris or blood can be swabbed for analysis, if desired. As the instrument is withdrawn at the anal verge, spasm of the external sphincter may lead to rapid expulsion. Firm counter-pressure prevents expulsion. Reinsertion may be required for adequate visualisation of the anal verge.

ALGORITHM: ANAL DISCHARGE

FIGURE 4.2d



4.3 LOWER ABDOMINAL PAIN (LAP)/PELVIC INFLAMMATORY DISEASE (PID) SYNDROME

A. Case Definition

LAP is the syndromic marker for PID and is the second most common STI syndrome seen in women. PID is a term used to describe ascending spread of lower genital tract infection. It may involve the endometrium (endometritis), fallopian tubes (salpingitis), and pelvic peritoneum (peritonitis). The patient may present with:

- Lower abdominal pain and tenderness (usually bilateral)
- Vaginal discharge and/or bleeding (endometritis)
- Dyspareunia, dysuria and or menometrorrhagia
- Onset of pain usually 5-10 days postmenstrual
- May be associated fever ($\geq$ 100.4 F), chills, nausea or vomiting

Abdominal and bimanual examination makes the probability of PID highly likely when a woman with the above complaint also shows the following:[**]

- Direct or rebound abdominal tenderness
- Cervical motion tenderness
- Enlargement or induration of one or both fallopian tubes or a tender adnexa or pelvis mass

The diagnosis is also more likely if the woman:

- Is in the high risk age group, i.e., age < 25 years (previous STI, multiple sex partners)
- Has had a previous episode of PID

[**] Ideally all three should be present

- Gives history of disease in partner
- Uses a IUCD
- Has multiple sex partners
- Has very frequent coitus with same partner

B. Aetiologic Agents

- **N. gonorrhoea**
- **C. trachomatis**
- **Anaerobes**

The prevalence of various agents in the Jamaican population is unknown and warrants a local study.

C. Recommended Treatments (Ambulatory)

Based on aetiologic agents

Table 15: Pelvic Inflammatory Disease (Ambulatory) Treatment – Out Patient
Ceftriaxone 250 IM stat (or equivalent) ***PLUS*** Doxycycline 100 mg PO bid for 14 days ***PLUS*** Metronidazole* 500mg bid x 14 days. Give anti-inflammatory for 3 days

*Not required for epi treatment of male partners

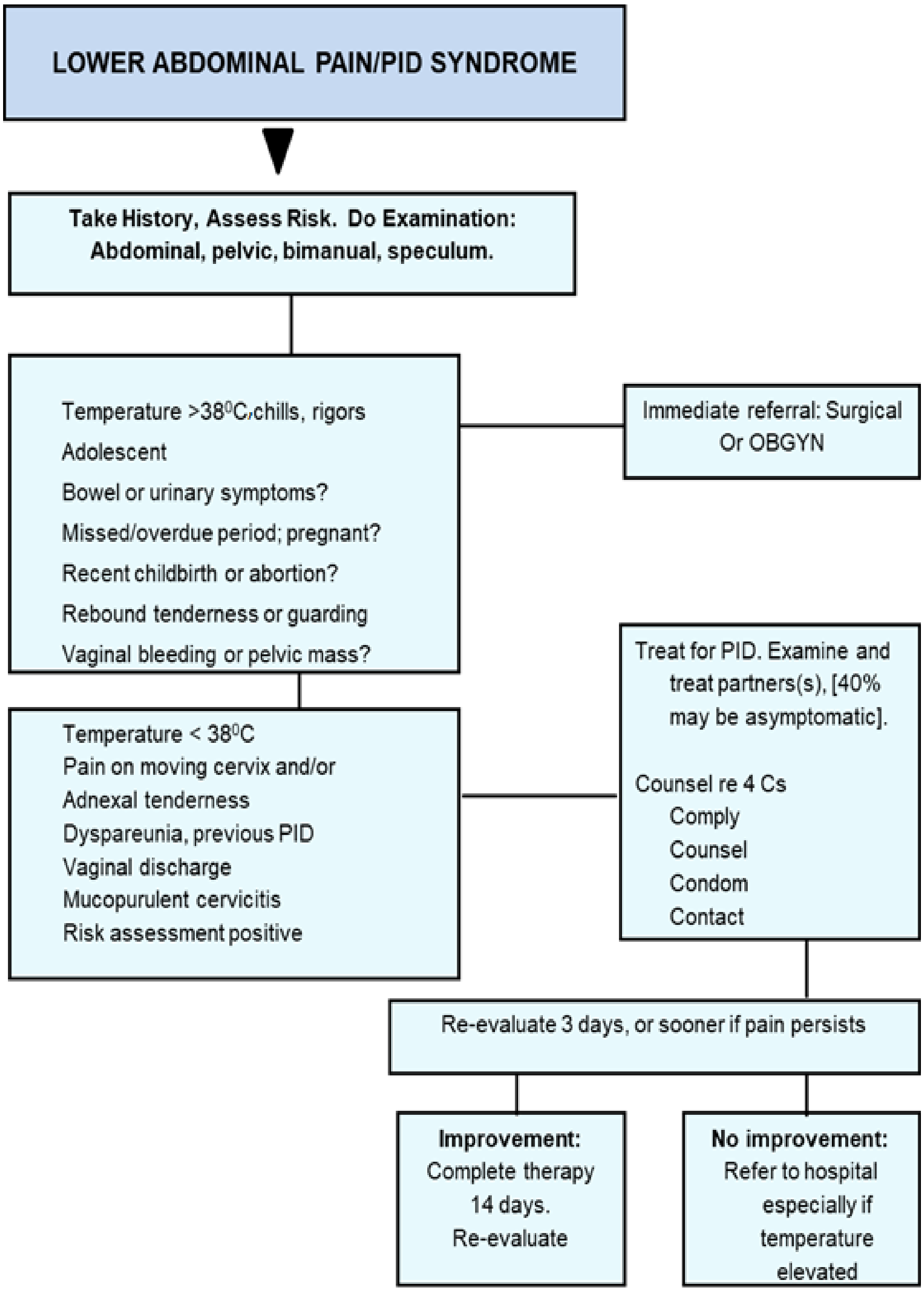
Note:

1. A proportion of early PID cases are asymptomatic. Chlamydia and Neisseria gonorrhoea are the most common causes of PID (CDC, 2015). However, only about a third of women with Chlamydia infection may complain of lower abdominal pain in the Family Planning setting and about 50% may be symptomatic in STI clinics or gynaecology clinics (Dowe et al., 1999). **Routine bimanual examinations on women with suspected STI are necessary to detect these.**

2. IUCDs should be removed in cases of PID and endocervicitis and another family planning method recommended.

FIGURE 4.3

ALGORITHM: MANAGEMENT OF LOWER ABDOMINAL PAIN/PID



Test all patients for syphilis and HIV; offer pap smear if none within past year.

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4.4 GENITAL ULCER DISEASE (GUD)

The **genital ulcer disease (GUD)** syndrome refers to conditions of the anogenital region (with or without lymphadenopathy) which cause a break or dissolution of the epithelial lining of the skin or mucous membrane in this area. In Jamaica, common specific causes of this syndrome include:

- Herpes simplex virus
- Syphilis
- Other less common
    - Chancroid
    - Granuloma inguinale
    - Lymphogranuloma venereum

Trauma may also be a presenting complaint; however, patients should be investigated for specific causes of genital ulcer.

Prevalence of GUD syndromes seen in primary health care clinics

Studies (both local and abroad) have shown that **GUD is an important risk factor in the transmission of HIV infection** and is also important in the **prognosis** of the disease. **Coincident HIV infection also affects the clinical manifestation and the response to treatment of GUD.** More recently it has been shown that GUD similarly affects the transmission of Hepatitis B virus (HBV) and HTLV-I/II infection and is, therefore, a condition to be given **high priority** in an STI/HIV control programme. Available data suggests that GUD is seen in about 7% of patients attending Primary Health Care based STI Centres.

CLINICAL CASE DEFINITIONS

- Lymphadenopathy is often associated with GUD and is of some help in distinguishing various types of ulcers. However, any GUD may be complicated by secondary infection and produce tender to painful local lymph glands.

Do STS for syphilis on all patients with GUD

- If available, do Darkfield (DF) examination on all genital ulcers to rule out syphilis. Care should be taken in the interpretation of DF results as false negatives may occur with the recent use of systemic or local anti-treponemal drugs, or with super-infection. A careful history and examination are required to exclude this possibility.

- **All patients with genital ulcers should be counselled concerning the risk of HIV** (three times higher than in other STI patients), and encouraged to be tested voluntarily for HIV. This is imperative if the response to treatment is unfavourable.

- Aetiologies, based on clinical appearances, are important in the syndromic management of GUD (Figures 4.4a-b).

- Simple and rapid tests are available for syphilis

- Clinical case definitions and differential diagnostic features (Table 16) are given for GUD conditions

- Other important clinical aspects of GUD are summarised in Table 18

TABLE 16: Case Definition and Aetiologic Agents*

DISEASE	CASE DEFINITION	AETIOLOGIC AGENT	INCUBATION PERIOD	DURATION	SITE	ADENOPATHY
Syphilis (Primary)	Non-tender, eroded papule with clean base and raised firm, indurated borders; multiple lesions occasionally seen.	Treponema pallidum	3 – 4 weeks (Range 9 - 90 days)	3 – 6 weeks	Genital, extragenital, perianal or rectal	Regional, unilateral or bilateral, firm, movable, non-suppurative, painless inguinal adenopathy
Syphilis (Secondary)	Painless eroded papules, mucous patches, condylomata. May be highly variable, symmetrical, skin rashes (mostly non-irritant). Generalised lymphadenopathy. May be other systemic signs.	T. pallidum	6 – 9 weeks. (Range 6 wks – 6 mths)	6 – 9 weeks	Genital, perianal, rectal, extragenital	Generalised, mobile, painless, discrete and rubbery

*There is a long list of both STI and non-STI agents. Only the most commonly reported are presented here.

DISEASE	CASE DEFINITION	AETIOLOGIC AGENT	INCUBATION PERIOD	DURATION	SITE	ADENOPATHY
Chancroid	Single or multiple, round to oval, deep ulcers with irregular outlines, ragged and undermined borders and purulent base lesions are tender; erythematous border, non-indurated.	Haemophilus ducreyi	12 hrs – 3 days	Undetermined (months)	Genital or perianal	Regional, unilateral or bilateral, tender, matted, fixed adenopathy that may become soft and fluctuant; rupture through fistulous openings with reactive edges, (25-60% of cases).

TABLE 16: Case Definition and Aetiologic Agents (cont'd)

DISEASE	CASE DEFINITION	AETIOLOGIC AGENT	INCUBATION PERIOD	DURATION	SITE	ADENOPATHY
Herpes Simplex Virus	Primary (initial) lesions are multiple, edematous, vesicles or painful erosions with yellow-white membranous coating; recurrent episodes may have grouped vesicles on an erythematous base and be itchy.	Herpes simplex virus (Types I and II)	3 -10 days	Average: Primary (initial) 12 days: recurrent 4 days	Genital, perianal or oral	Bilateral, tender inguinal adenopathy usually present with primary vulvo-vaginitis and may or may not be present with recurrent genital lesions.
Lympho-granuloma venereum	Evanescent ulcer. Non-tender, single, minute.	Chlamydia trachomatis (Types L_1, L_2, and L_3)	3 days – several weeks (about 6)	2-6 days	Genital, perianal or rectal	Usually regional, unilateral or bilateral, multilocular, firm, painful inguinal adenopathy with overlying "dusky skin"; may become fluctuant and develop "grooves in the groin". Fistulous opening clean.

DISEASE	CASE DEFINITION	AETIOLOGIC AGENT	INCUBATION PERIOD	DURATION	SITE	ADENOPATHY
Granuloma inguinale	Soft to slightly indurated, non-tender papules that form irregular ulcers with beefy-red, friable base and raised, “rolled” borders. Surface clean, shiny and bleeds easily to touch.	Klebsiella granulomatis (Donovania). “Donovan Body”	3 -6 weeks to 6 – 8 plus months	Undetermined (years)	Genital, perianal, or inguinal	Regional. subcutaneous perilymphatic granulomatous lesions produce inguinal swelling but are not lymphadenitis (pseudobuboes)

A. RECOMMENDED TREATMENTS

Table 17. Genital Ulcer Disease (GUD) – Treatment

CONDITION	**TREATMENT**
Syphilis (Primary – ulcer or chancre at site of infection/inoculation, and early latent [duration less than 1 year without evidence of treponemal infection])	Benzathine Penicillin G (BPG) 2.4 MU I.M weekly for 2 doses **OR** Doxycycline 100 mg P.O bid x 14 days **OR** Erythromycin 500 mg P.O qid x 14 days
Syphilis (Secondary – late latent [duration of more than 1 year without evidence of treponemal infection])	Benzathine Penicillin G (BPG) 2.4 MU I.M weekly for 3 doses **OR** Doxycycline 100 mg P.O bid x 30 days **OR** Erythromycin 500 mg P.O qid x 30 days
Chancroid	Erythromycin 500 mg P.O qid x 7 days **OR** Ceftriaxone* 250 mg I.M stat **OR**

* Not fully evaluated.

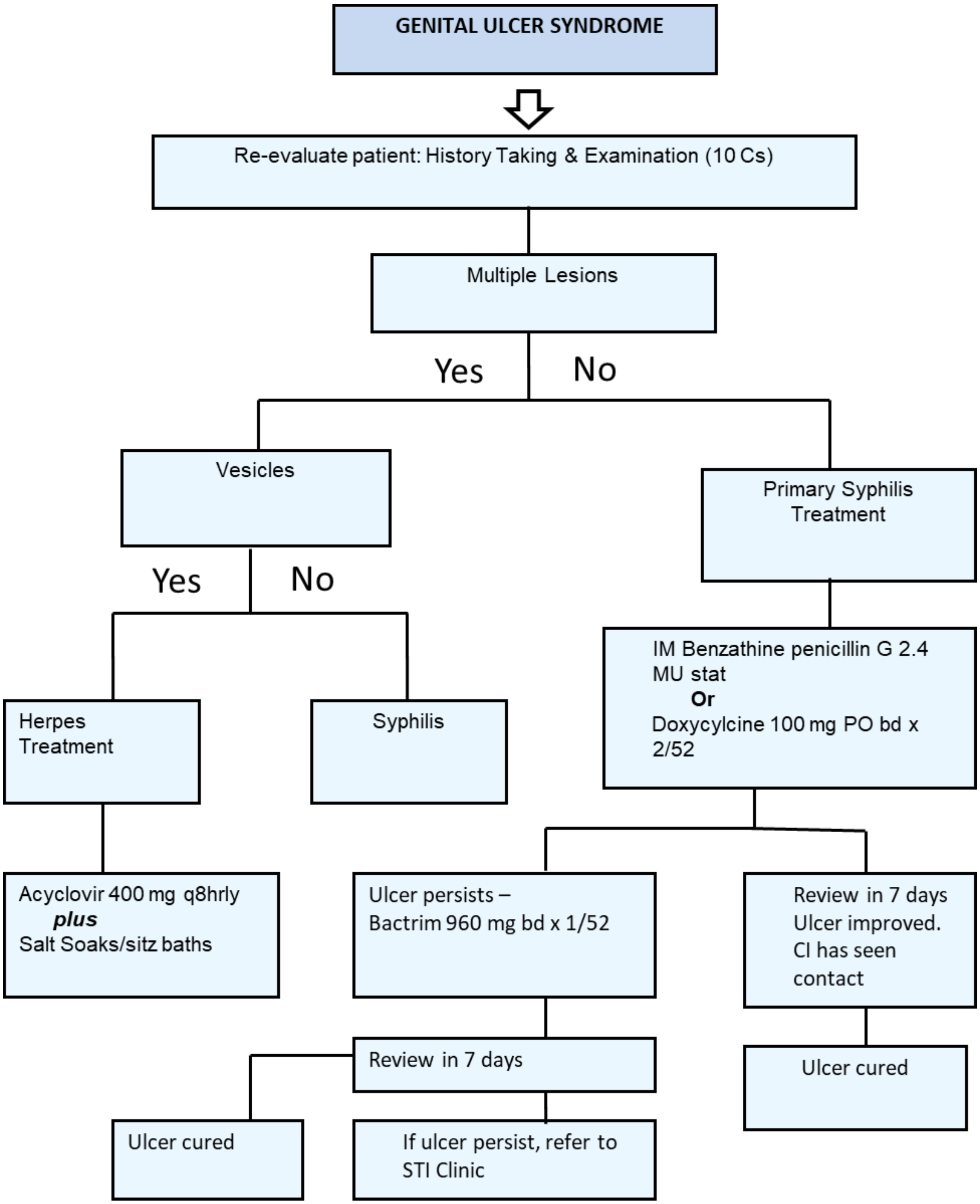
Table 17. Genital Ulcer Disease (GUD) – Treatment

CONDITION	**TREATMENT**
	T/SMZ** (160/800 mg) P.O. bid x 7 days Use saline sitz baths (prn)
Herpes simplex	Symptomatic primary episode - Acyclovir 400mg PO tds x 7-10/7 Recurrent episodes: Acyclovir 400 mg PO tds x 3 days **OR** Acyclovir 800 mg PO bd x 5/7. Saline sitz baths (prn). Chronic suppressive therapy for frequent recurrence or causing significant distress to patient, treat with Acyclovir 400 mg PO bd for 3 - 6 Months or valacyclovir 500 mg PO od for 3 - 6 months. (MANAGEMENT MAY EXTEND UP TO A YEAR)
Lymphogranuloma venereum	Doxycycline 100 mg bid x 14 days **OR** T/SMZ** (160/800 mg) P.O. bid x 14 days
Granuloma inguinale***	Doxycycline 100 mg PO bid x 2-3 weeks **OR** T/SMZ** two tabs bid x 2 -3 weeks*

** T/SMZ=Trimethroprim/sulphamethoxazole (or Co-trimoxazole).
*** Treat until completely cured. Follow up monthly for at least 6 months.

FIGURE 4.4a

ALGORITHM: MANAGEMENT OF GENITAL ULCER DISEASE (GUD)



NB: Footnotes for Care of Genitalia: Do not use bleach, disinfectants or other chemicals to area.

PERSISTENT ULCER REVIEW SYNDROME

Re-evaluate patient: 10C's History Taking & Examination

Consider less common causes:

1. Chancroid
2. Lymphogranuloma venereum
3. Granuloma inguinale
4. Chronic Herpes (consider co-morbidities)
5. Non-benign lesion/malignancy

IV. IMPORTANT ASPECTS OF GUD

Table 18: Genital Ulcer Disease (GUD) Summary		
1. GENITAL ULCER DISEASE (usual causes) • Syphilis • Chancroid • Herpes simplex • Lymphogranuloma venerum (LGV) • Granuloma inguinale	**2. GENITAL ULCER DISEASE (Unusual causes)** • Scabies • Candida • Human Papilloma Virus • Non-STI Ulcers, e.g., 1. Trauma 2. Carcinoma 3. Fixed Drug Eruption 4. Stevens-Johnson 5. Behcet's 6. Other Dermatoses 7. Chemical, e.g., dettol	**3. FACTORS INFLUENCING PROPORTION OF PATIENTS WITH GENITAL ULCERS SEEN AT STI CLINICS** (b) Prevalence of other STIs (c) Health seeking habits (d) Diagnostic and treatment standards (e) Lack of Circumcision (f) Self-treatment (g) Sexual Practices

Table 18: Genital Ulcer Disease (GUD) Summary (Continued)		
4. GENITAL ULCERS – HISTORY 1. Number of Ulcers 2. Duration 3. Degree of discomfort 4. Relation to: • Sexual intercourse • Trauma • Lesions elsewhere 5. Induration 6. Lymphadenopathy	**5. GENITAL ULCERS – GUIDELINES** 1. Darkfield all lesions 2. Serological testing for Syphilis on all patients 3. Beware of dual pathology Follow-up all patients weekly until lesions are completely healed	

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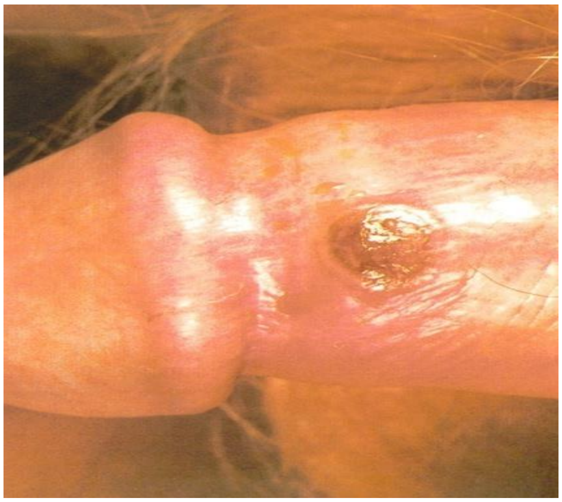
IMAGES OF COMMON STIS CONTINUED



No. 9 Penile Chancre

No. 10 Penile Chancre

No. 11 Anal Chancre

No. 12 Vulval Chancre

No. 9: Penile Chancre – The primary lesion of syphilis is a hard, painless, well-defined papule or ulcer. A careful history is necessary to determine sites which should be examined for chancres.

No. 10: Penile Chancre – Chancres vary greatly in size and may be single or multiple. Suspect all genital lesions until syphilis is ruled out by darkfield and/or serology.

No. 11: Anal Chancre – May occur in either sex practising anal coitus

No. 12: Vulval Chancre– In women, chancres occur more commonly on the cervix or vagina where they are easily overlooked. Contact interviewing of male patients with infectious syphilis is essential.

No. 13 Annular and Crescentic Rash

No. 14 Palmar Rash

No. 15 Maculopapular Rash

No. 16 Condylomata Lata

No. 13: Annular and Crescentic Rash – This is a characteristic lesion of secondary syphilis when it occurs on the lower face and upper neck in dark-skinned individuals.

No. 14: Palmar Rash – Palmar and plantar rashes are characteristic of the secondary stage of syphilis. Lesions are usually symmetrical.

No. 15: Maculopapular Rash – Rashes in secondary syphilis vary greatly in size, shape and appearance. A blood test for syphilis is indicated in all sexually active individuals with skin lesions.

No. 16: Condylomata Lata – This characteristic rash of secondary syphilis occurs in the warm and moist regions of the body. The lesions consist of confluent papules.

No. 17 Chancroid of Penis

No. 18 Chancroid of Penis

No. 19 Herpes of Penis

No. 20 Herpes of Vulva

No. 21 Penile Warts

No. 22 Vulval Warts

No. 23 Lymphogranuloma Venereum

No. 24 Lymphogranuloma Venereum

No. 25 Granuloma Inguinale of Perpuce

No. 26 Granuloma Inguinale

No. 17: Chancroid of Penis – The lesion is usually painful, soft and irregular with undermined or shelving edges. Secondary infection produces a necrotic base with adherent slough.

No. 18: Chancroid of Penis - The lesion of Chancroid is intensely destructive and so should be treated promptly and effectively. Phimosis and paraphimosis are common.

No. 19: Herpes of Penis – Showing typical vesicles, oedema and ulceration. Primary lesions are painful.

No. 20: Herpes of Vulva – Showing extensive ulcerations and vesicles. Such lesions should be kept clean (sitz baths) and adhesions of the vulva prevented by coating lightly with an appropriate surface antiseptic.

No. 21: Penile Warts – Condylomata accuminata are caused by the human papilloma virus (HPV). Lesions are usually filiform and pedunculated and flattened lesions must be differentiated from condylomata lata.

No. 22: Vulval Warts – Female contacts to HPV should be followed up at 6-12 monthly intervals with pap smears. Warts will commonly be on the cervix as well.

No. 23: Lymphogranuloma Venereum – The primary lesion is evanescent and patients usually first present with painful buboes. Systemic manifestations may be present.

No. 24: Lymphogranuloma Venereum – The inguinal glands are painful and matted. Tethering of the skin between the upper and lower rows of glands produces the “sign of the groove” in some 30% of patients.

No. 25: Granuloma Inguinale of Prepuce – Lesions are painless and beefy red, with a clean, shiny surface which bleeds easily on touch. The edges are raised and everted. Lesions may be single or multiple. Neoplastic change is a serious complication.

No. 26: Granuloma Inguinale – The anal orifice and buttocks are also common sites of GI and suggest anal coitus.

4.6 HUMAN IMMUNODEFICIENCY VIRUS (HIV) INFECTION

A. Case Definition

HIV infection may present in a variety of ways, including asymptomatic infection, acute retroviral syndrome, advanced HIV, or with AIDS-defining illnesses.

Acute Infection: Some patients experience infectious mononucleosis-like symptoms at about the time of seroconversion, i.e., about 6-12 weeks after infection.[*] The symptoms (flu-like) include fever, malaise, headaches, flitting arthralgia, a maculopapular rash and transient lymphadenopathy. This stage lasts for about 2 weeks.

Asymptomatic HIV Infection: The patient is infected with HIV but has no symptoms. This stage may last for a few months to more than 10 years. Without diagnosis, lack of awareness of HIV status contributes to onward transmission of HIV.

Symptomatic HIV Related Conditions: The patient may have a combination of the following signs and symptoms:

(i) Severe malaise, lethargy and fatigue > one month
(ii) Chronic cough > one month
(iii) Night sweats > one month
(iv) Diarrhoea > one month duration (intermittent)
(v) Lymphadenopathy in ≥ 3 sites > one month duration
(vi) Fever > 100 °F for 3 months (intermittent)
(vii) Oral thrush
(viii) Oral hairy leukoplakia (pallisading of mucosa at the side of tongue)

[*] Although the average time for seroconversion is 6 weeks to 12 weeks, it may take up to 6 months (and occasionally more than 6 months). **During this time the patient is highly infectious.**

(ix) Unusual and severe manifestations of certain skin diseases, esp. seborrhoeic dermatitis, eczema, tinea corporis.

(x) Unexpected weight loss (> 10% body weight).

AIDS: Patients with conditions indicative of severe immune deficiency in the absence of any known cause for the immune deficiency. They present with symptoms of opportunistic infections, tumors and/or CNS disease e.g., pneumocystis carinii (jiroveci) pneumonia (shortness of breath), or Kaposi's sarcoma (purplish to bluish to black skin lesions).

Dementia: May be an early or late manifestation and may occur with or without other HIV related conditions.

B. AETIOLOGIC AGENTS

- Human Immunodeficiency Virus (HIV)
- An RNA retrovirus. Two (2) types are known to cause disease in man i.e. HIV-1 (commonly) and HIV-2 (uncommonly).

C. ASSESSMENT OF HIGH RISK BEHAVIOUR

1. Sex with persons with AIDS, HIV infection, known epidemiologic factor (risk behaviour or from HIV endemic region)
2. Paid or transactional sex
3. Anal sex
4. Multiple sex partners (more than two sex partners within the last month or partner has several partners)
5. Sex without condoms
6. History of genital ulcer disease
7. History of transfusion of blood or blood products
8. Cocaine or intravenous drug use (behaviour and immune system change)
9. Infections, tattooing, medical or surgical procedures with non-sterile instruments (including needle-sticks in health personnel).

INDICATOR SYMPTOMS AND SIGNS OF HIV

1. Skin

- Severe pruritus
- Genital or anal ulcer(s), more than 1 month
- Seborrhoeic dermatitis
- Shingles
- Severe or recurring skin infection(s)

2. Gastrointestinal

- Oral thrush (white plaques in mouth)
- Diarrhoea, of more than 1 month
- Oral hairy leucoplakia

3. Neurological

- Increasing headache
- Nuchal rigidity

4. Respiratory

- Cough, more than 1 month

5. General

- Fever, more than 1 month
- Enlarged glands in $\geq$ 3 sites (including groin)
- Chronic malaise and/or fatigue
- Unexplained weight loss

NOTIFICATION OF CASES OF HIV INFECTION

Notification of all persons with evidence of possible HIV infection (that is, one or more positive HIV ELISA tests) should be made to the MO(H), (or designated deputy) at the Parish Health Department or to the SMO(H), (or designated deputy) at the Epidemiology Unit of the Ministry of Health and Wellness.

In all cases, a **second specimen** should be submitted to the laboratory for confirmatory HIV testing.

The report should be submitted on the standard HIV/AIDS reporting form even where the patient is asymptomatic. If this form is not available, a written report should still be submitted. All HIV reports should be considered confidential and should be sent to the MO(H) or to the SMO(H) in a sealed envelope marked **CONFIDENTIAL.**

IMAGES OF COMMON STIS CONTINUED

No. 27 Oral Candidiasis

No. 28 Herpes Zoster

No. 29 Generalised Lymphadenopathy

No. 30 Dermatitis

No. 27: Oral Candidiasis – is commonly seen in patients with HIV infection.

No. 28: Herpes Zoster – in HIV infection, the vesicles and **bullae** of HZV tend to be extensive and involve several dermatomes.

No. 29: Generalised Lymphadenopathy – In STD patients is seen most often in syphilis and HIV infection. Glands are firm, mobile and non-tender.

No. 30: Dermatitis – Atypical and severe dermatitis is seen commonly in HIV patients.

Recommended Treatment

There is no cure or vaccine available for this syndrome at the present time. Antiretroviral therapy significantly reduces mortality and improves the quality of life of persons living with HIV. Current ART treatment regimens are outlined in **Table 19.** Other important aspects of management of HIV include:

- Preventative counselling and education
- Proper nutrition and lifestyle changes
- Prophylaxis against opportunistic infections
- Symptomatic and/or specific management of opportunistic infections (OI)
- Adherence counselling
- Psychosocial support
- Case Management

Table 19. Antiretroviral Therapy for Adults and Adolescents with HIV/AIDS in Jamaica		
First Line antiretroviral regimens for adolescents and adults		
First choice	Tenofovir/Lamivudine (TDF/3TC)	Dolutegravir (DTG)
Second choice	Tenofovir/Lamivudine (TDF/3TC)	Efavirenz (EFV)
Second Line antiretroviral regimens for adolescents and adults		
First choice	Abacavir/Lamivudine (ABC/3TC)	Atazanavir/ritonavir (ATZ/r)
Second choice	Abacavir/Lamivudine (ABC/3TC)	Lopinavir/ritonavir (LPV/r)

For patients failing to attain viral suppression on the above regimens, despite adherence counselling and psychosocial support, consideration for Third Line therapy should be guided by:

- HIV Drug Resistance Testing
- Referral to and/or discussion with HIV Physician or Clinical Mentor

Third Line drug options include:

- Darunavir
- Etravirine

- Raltegravir
- Ritonavir

The basic strategy in antiretroviral (ARV) therapy for HIV infection relies on attacking the virus with drugs targeting several critical stages in the life cycle of the virus. However, the details of ARV therapy and patient follow-up are further outlined in the **Clinical Management of HIV Disease: Guidelines for Medical Practitioners 2017, Ministry of Health and Wellness, Jamaica**. Symptomatic patients requiring secondary care should be referred to a regional centre listed in Appendix VII.

In a patient with a history of high risk behaviour and presenting with features of any part of the spectrum of HIV infection, a HIV rapid rest should be offered. **Pre-test and Post-test counselling are important components of HIV testing.**

Note:

Complete guidelines for the diagnosis and treatment of HIV infection are described in the Clinical Management of HIV Disease: Guidelines for Medical Practitioners 2017, Ministry of Health and Wellness, Jamaica.

SELECTED REFERENCES

1. Clinical Management of HIV Disease: Guidelines for Medical Practitioners 2017, Ministry of Health and Wellness, Jamaica.
2. HIV Epidemiology Report for the Ministry of Health, Jamaica; 2017.

4.7 HUMAN T-LYMPHOTROPHIC VIRUS TYPE I/II (HTLV-I/II)

A. Case Definition

Human T-Lymphotropic Virus Type I/II **(**HTLV-I/II) was the first identified human retrovirus isolated in 1978. HTLV-I is the causative agent of Adult T-cell Leukemia/Lymphoma (ATLL) and HTLV-I Associated Myelopathy/Tropical Spastic Paraparesis (HAM/TSP) initially described in Jamaica as Jamaican neuropathy. HTLV-I has been associated with infective dermatitis in infants, and inflammatory autoimmune syndromes including polymyositis, uveitis, Sjogren's syndrome, chronic arthritis, thyroiditis, and T-cell bronchoalveolitis.

The prevalence of HTLV-I in the Caribbean ranges from 0.3% to 7% and increases with age with adult females having a higher prevalence than males. HTLV-I transmission occurs from mother-to-child primarily through breast milk, by sexual contact, and by blood transfusion. The majority of persons with HTLV-I infection remain asymptomatic throughout their life; however, approximately 5% of infected persons may develop ATLL, HAM/TSP or one of the associated autoimmune syndromes. ATLL is an aggressive form of leukemia/lymphoma usually leading to death and arises in persons infected during infancy or early childhood. HAM/TSP is a debilitating condition causing spastic paralysis. Given the severity of these conditions for which there is no cure, it makes good sense to implement measures to prevent HTLV-I transmission.

PREVENTION OF HTLV-I TRANSMISSION

There are cost-effective policy options in endemic areas such as the Caribbean to prevent the transmission of HTLV-I. These include screening of blood donors as is currently done. HTLV-I infected persons should not donate blood, organs, semen, or milk due to the risk of transmission. **Blood donors who test HTLV-I positive need to be informed and counseled** and their sexual partners tested and counselled.

Antenatal mothers should also be tested for HTLV-I and counselled if found to be positive. Patients are counseled to avoid breastfeeding. The virus is absent in cord blood but present in the cellular components of breast milk and poses a risk of transmission of between 20-25% of babies born to seropositive mothers. Breaking the cycle of transmission through breast-feeding is important because most cases of ATLL are thought to be due to HTLV-I infection during infancy or early childhood.

Sexual transmission of HTLV-I is generally low (estimated to be 0.9% per 100 person years) and is more efficient from males to females than from females to males, hence the higher prevalence among adult women. **Adults who are HTLV-I positive should be encouraged to use condoms**. While the risk of HTLV-I transmission through sexual intercourse is low, the risk increases with multiple sex partners, anal sex and co-infection with HIV, herpes or other sexually transmitted infections.

Conclusion

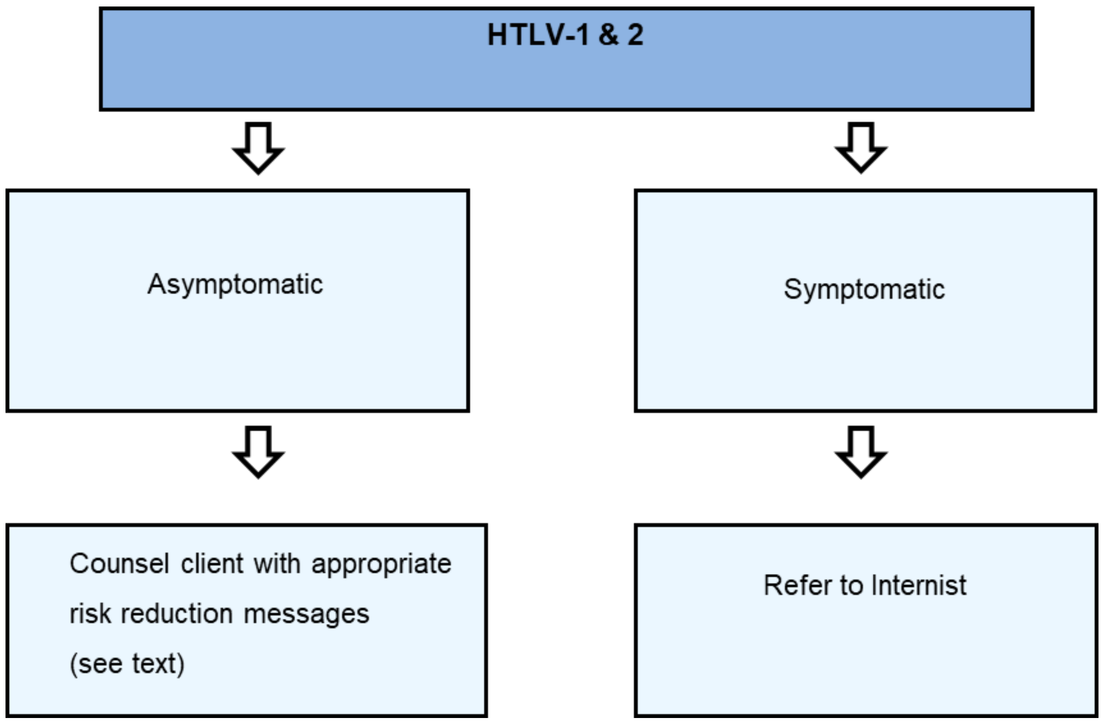
HTLV-I is an important public health concern in the Caribbean, given its potential devastating effects and the fact that its prevention is cost-effective and practical. Although the lifetime incidence of associated disease is low, around 5-10% among infected persons, the effects of disease are chronic and debilitating. It is therefore important to implement measures to prevent HTLV-I transmission including testing blood donors and antenatal mothers, limiting breast-feeding in infected mothers to under 4 months and promotion of safe sexual behaviour.

Counseling messages for the HTLV-I infected adult

Most persons who have HTLV-I remain asymptomatic for life. Approximately 5% of persons with HTLV-I infection may develop a serious disease such as leukemia/lymphoma or spastic paralysis. All blood donors in Jamaica are tested for HTLV-I so the blood supply is safe. Mothers who are HTLV-I positive can still breast feed their baby but should stop at 4 months since most infections due to breast feeding occur after 6 months. HTLV-I transmission through sexual contact is low. Condoms are effective in preventing sexual transmission of HTLV-I.

ALGORITHM: HUMAN T LYMPHOTROPHIC VIRUS TYPE I/II (HTLV-I/II)

Figure 4.5



SECTION 5

Syphilis

SYPHILIS IN PREGNANCY

Jamaica is approaching the status of having eliminated congenital syphilis if transmission rates can be sustained below 0.5 cases per 1,000 live births. Based on antenatal clinic surveillance data for Jamaica, 88% of pregnant women receive testing for syphilis, and 2015 data indicated a prevalence of 1.5% (Pan American Health Organisation, 2015). Jamaica reported 28 cases of congenital syphilis in 2015, an increase from 3 in 2014 and 8 in 2013, giving a rate of 0.56 per 1,000 live births for 2015.

Increased surveillance and improved management of syphilis started during 1990. The control strategies used during the succeeding ten years consisted of the following:

1. Increased passive surveillance of infectious syphilis
2. Improved contact investigation of infectious syphilis cases
3. Active surveillance of congenital syphilis cases at hospitals
4. Increase in number and type of primary health care facilities
5. Training of health care staff at all levels for sustainability
6. Use of a rapid Treponemal test (SD Bioline Syphilis) for syphilis screening at STI and ANC clinics
7. Decentralisation of syphilis screening to the parishes
8. Syndromic case management for STI cases
9. Same day treatment of syphilis cases
10. Use of Laboratory Technician Assistants with short-term task-specific training to staff STI labs at parish level.

Table 20. Stages and Test Results for Syphilis

Stage of syphilis	Treponemal test result e.g. TPPA*, POC kits	Non-Treponemal test result e.g. VDRL**, TRUST***
Primary (sore) (Incubation period: 9-90 days)	Reactive (78 -86 %)	Reactive (76-84%), > R1:8
Secondary (rash) (usually occurs 2-8 weeks after primary stage)	Reactive (100%)	Reactive (100%), > R1:8
Early Latent (infection of less than 1 year duration without clinical signs)	Reactive (97 – 100%)	Reactive (95 – 100%), > R1:8
Late Latent (infection of 1 or more years duration without clinical signs)	Reactive (100%)	Reactive, < R1:8
Tertiary	Reactive (94 – 96%)	Reactive (71- 73%), < R1:8

Notes to Table 20:

*TPPA-Treponema pallidum particle agglutination assay

POC-Point-of-care

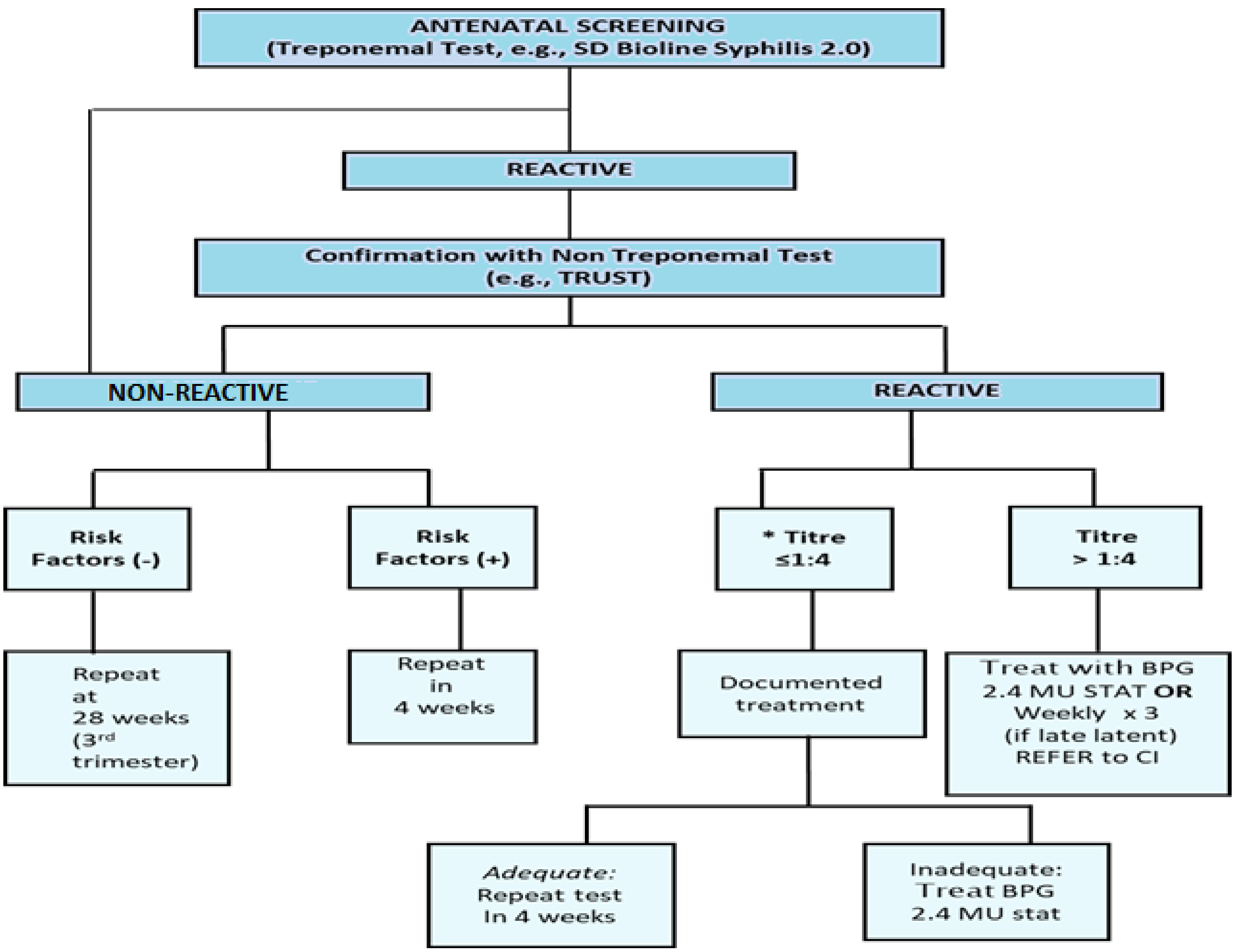
VDRL- venereal disease research laboratory; TRUST, toluidine red unheated serum test

Table 21. Guide to the Interpretation of Serologic Tests for Syphilis

Treponemal E.g., SD Bioline,TPPA	Non-Treponemal e.g., TRUST, VDRL	Interpretation
Reactive	Reactive	Syphilis treated or untreated. If titres R 1:8 initially or more, likely infectious syphilis. If pre-treatment titre was initially high and a four-fold change attained after treatment, may not be infectious.
Reactive	Non-reactive	Treated syphilis or late latent syphilis but may be primary or very early latent syphilis
Non-reactive	Reactive	Biological false positive

ALGORITHM – MANAGEMENT OF MATERNAL SYPHILIS

FIGURE 5.1



It is HIGHLY RECOMMENDED that the stat dose of IM BPG be given in the pregnant mother, even if serofast.

*If history or clinical assessment suggests active infection, give full treatment and refer to Contact Investigator (CI) for partner management.

5.2. CONGENITAL SYPHILIS

A. CASE DEFINITION

Clinical Description

A condition caused by infection in utero with Treponema pallidum. A wide spectrum of severity exists, and only severe cases are clinically apparent at birth. The majority develop signs and symptoms 4-6 weeks post-partum. An infant (<2 years) may have signs such as hepatosplenomegaly, characteristic skin rash, condylomata lata, "bug eye"*, pseudoparalysis, anaemia, or oedema (nephritic syndrome and/or malnutrition). An older child may have stigmata such as interstitial keratitis, 8th nerve deafness, anterior bowing of shins, frontal bossing, mulberry molars, Hutchinson's teeth, saddle nose, rhagades, or Clutton's joints.

Note: Diagnosis in a new born can be made with confidence in the following situations:

a) Positive darkfield microscopy of skin lesions, umbilical or placental serum.
b) Mother has reactive test (NT and TT), and the infant shows classic signs of disease (see above).
c) Infant asymptomatic but titre significantly higher (≥4 fold) than mother's.

Surveillance Reporting

All infants (and stillbirths) born to women with untreated or inadequately treated syphilis, regardless of neonatal symptoms or findings.

N.B. If the mother is seen for the first time during delivery, maternal blood should be taken and not cord blood. The latter tends to give both false positive and false negative results due to contamination with Wharton's jelly.

* Described by Dr. E. Read, Bustamante Hospital for Children, Jamaica

TREATMENT

A. MATERNAL

Table 22. Maternal Syphilis –Treatment

CONDITION	TREATMENT
Primary	BPG 2.4* mega units IM stat for 2 dose**
Secondary and Early Latent (≤ 1 year)	BPG 2.4* mega units IM stat for 2 doses
Late Latent and Tertiary	BPG 2.4 mega units IM for 3 doses at 1 week apart.
Penicillin-Allergic Patients with Early Syphilis	*Refer to infectious diseases specialist for penicillin desensitisation.

Erythromycin 500 mg PO qid x 4 weeks is no longer used. However, erythromycin passes the placental barrier very poorly and treatment completed at any stage of pregnancy requires careful follow-up of both mother and infant.

* For adequacy in mother, BPG treatment must be completed at least 30 days before delivery.
**Although 1 dose is effective, 2 is recommended locally to provide cover for reinfection, delay in getting contact(s) to treatment and other cultural issues.

Follow-up (FU):

Monthly: (especially for non-penicillin treatment). In early disease, titres should show at least a 4-fold decrease within 3 months.

B. CONGENITAL

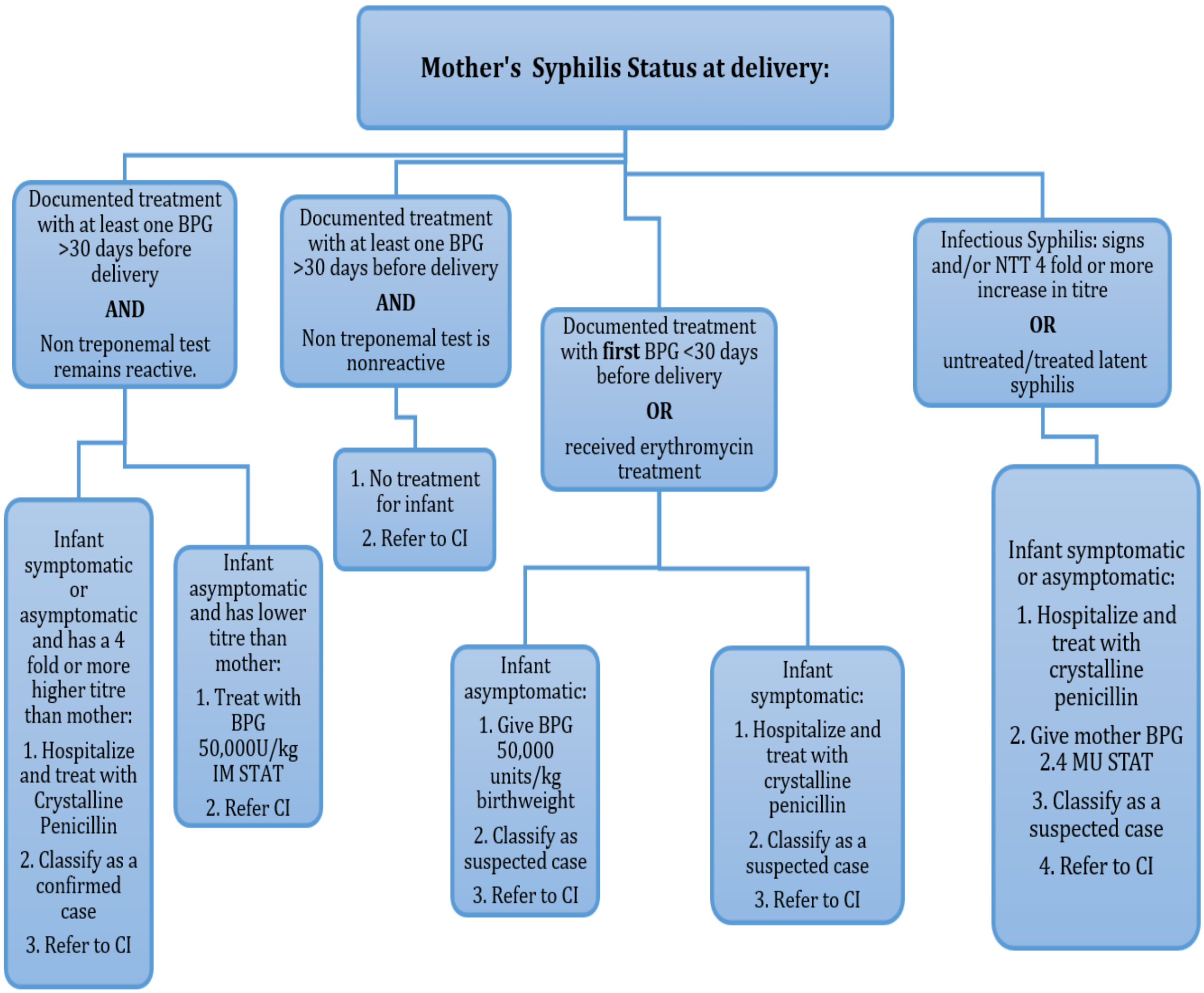
Table 23. Congenital Syphilis – Treatment

Condition	**Treatment**
Early Congenital Syphilis < 2 years	Aqueous Crystalline Penicillin G (ACPG): 50, 000 units/kg/dose IV every 8 – 12 hours (100,000-150,000 units daily) for 10-14 days. **OR** Aqueous Procaine Penicillin G (APPG): 50,000 units/kg IM stat daily for 10-14 days. When mother has been treated with erythromycin during pregnancy or BPG< 30 days before delivery, this is a probable case of congenital syphilis which requires admission for confirmation. (**N.B.** BPG passes the blood/brain barrier poorly).

ALGORITHM FOR THE MANAGEMENT OF CONGENITAL SYPHILIS

FIGURE 5.2 a

N.B: Educate and Counsel all Mothers (4C's). Advise for HIV screen.



5.3 OPHTHALMIA NEONATORUM (ON)

A. Case Definition

A mucoid, mucopurulent or purulent discharge from the eyes of the newborn (≤1 month). There may be redness and swelling of the conjunctivae and palpebrae. The latter may stick together ("sticky eyes"). In severe cases, there is chemosis of the conjunctivae, tense oedema and swelling of both eyelids.

ON is a Class 1 Notifiable Disease and as such, a Class 1 Notification form needs to be filled out.

B. Aetiologic Agents

- **C. Trachomatis (CT)** – Usually late onset (≥ 1 week) and subacute infection. Recurrent infection may show follicular hyperplasia of lower palpebral conjunctiva.

- **N**. **Gonorrhoea (NG)** – Usually acute and early onset (2-5 days). May be late onset (up to 3 weeks) and indolent, depending on virulence of strain and suppression by prophylaxis (esp. use of chloramphenicol eyedrops). In acute cases, when the palpebrae are adherent and the pus under pressure, corneal ulceration may occur which in rare cases may progress to panophthalmitis and blindness.

- **Staphylococcus aureus,** Streptococcus pneumonia, Haemophilus sp. and Pseudomonas sp. are other common causes of ON.

- **Chemical conjunctivitis –** (from 1 per cent silver nitrate drops – Credé's Method) is easily distinguished from infectious causes. It develops 6 – 8 hours after birth, is self-limiting and clears in 24 - 48 hours without treatment.

N.B: Acute gonococcal ophthalmia neonatorum should be treated as a medical emergency.

Prophylaxis for infants

Since prevalence is high, instillation of one of the following prophylactic agents into the eyes of all newborn infants is recommended:

- 1% tetracycline ophthalmic ointment **OR**
- 0.5% erythromycin ophthalmic ointment

Credé's Method: after infant's head is crowned (or within 1 hour of delivery), wipe eyes outwardly using sterile moist swabs, open eyelids carefully and instil drops or ointment.

C. Recommended Treatments

Table 24. Ophthalmia Neonatorum – Treatment	
Gonococcal ON	Ceftriaxone 50 mg/kg, IM stat (maximum 150 mg) **PLUS** Tetracyline eye ointment 1% in each eye hourly for 1 day, then tid to 10 days. **N.B** Asymptomatic infants born to mothers infected with NG, treat as above.
Chlamydial ON	Erythromycin syrup, 50 mg/kg per day P.O., in 4 divided doses x 3 weeks but omit topical therapy.

Note:

1. Some cases of ON may be asymptomatic

2. Use suitable infection precautions e.g. careful hand washing, efficient sterilisation of thermometers etc. in order to prevent nosocomial spread of infection among other infants attending the clinic. Avoid cross-contamination with the use of the tetracycline tube.

Treatment of NG and CT in pregnant women is the best method of preventing ON caused by these organisms.

No. 31 Congenital Syphilis

No. 32 Ophthalmia Neonatorum

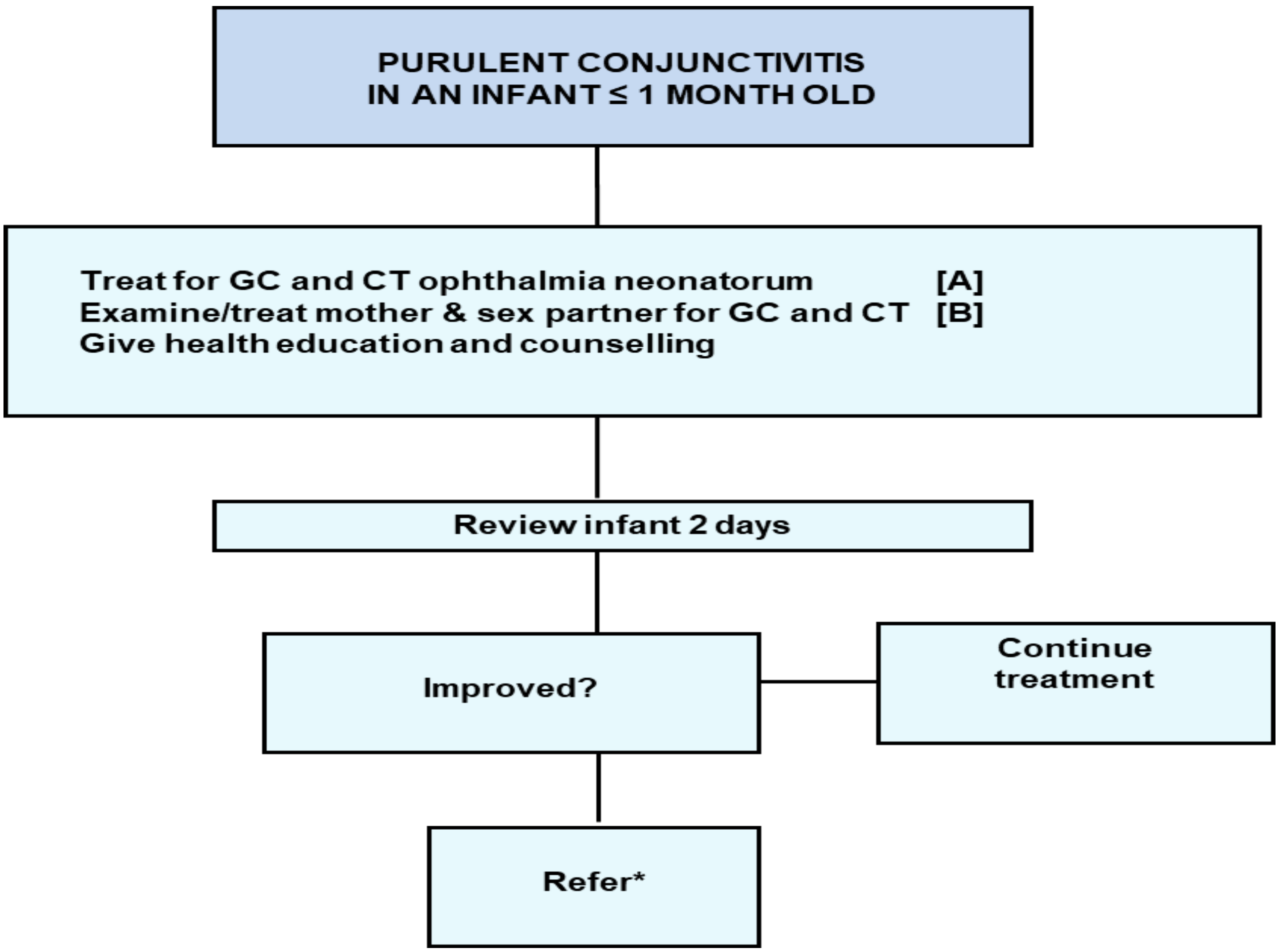
No. 31: Congenital Syphilis – Is a preventable cause of failure to thrive, skin rashes and eighth nerve deafness, among others. All pregnant women must be screened for syphilis.

No. 32: Ophthalmia Neonatorum – Due to gonococcus in a 3-day old baby. Emergency treatment is required to prevent corneal scarring and blindness.

ALGORITHM: MANAGEMENT OF OPHTHALMIA NEONATORUM

FIGURE 5.2b

(Without Laboratory support)



*ON is a medical emergency and best managed in hospital

[A] Because on occasion erythromycin syrup will not be available, and has inferior activity against **N. gonorrhoeae,** recommended treatment consists of tetracycline eye ointment combined with a systemic treatment effective against **N. Gonorrhoeae (see table 24)**

[B] In the absence of a confirmed diagnosis, the decision to notify partner(s) should take into account local cultural and epidemiological factors.

SECTION 6

STI in Adolescents

STIS IN ADOLESCENTS

In Jamaica, adolescents 10 to 19 years old account for approximately six per cent of HIV infections (MOH, 2017). Based on Ministry of Health and Wellness STI sentinel surveillance data, 16 – 19 year-olds comprised 14% of persons with STIs (trichomonas, chlamydia, gonorrhoea, and syphilis). HIV/AIDS-related deaths among females in the 10 – 19 years age group is over three times deaths among males between 1982 to 2017.

Major causes for this high incidence include:

- *Early sexual initiation among adolescents*: median age of sexual debut was 15 years for males and 17 years for females (Knowledge, Attitudes, and Behaviour Survey, 2017), similar to what was found in 2012.

- *Incorrect or inconsistent condom use*: the mean number of sexual partners reported by persons ages 15 – 24 years was 5.87. Condom use the last time they had sex was reported by 44% of persons with multiple partners (overall for ages 15 – 49 years); and only 17% of this high risk group reported always using a condom the last 10 times they had sex (KAPB, 2017).

- *Biological susceptibility in females*: the relatively immature vaginal and cervical mucosa pre-menarche lack the protective effects of oestrogens and acidity, thereby being more susceptible to STIs.

- *Inadequate treatment-seeking behaviour*: this results from a variety of factors such as:

  - ► Lack of knowledge about STIs, including HIV

  - ► Gender-power relation and economic dependency in older adolescents

- ► Stigma of STI/HIV with the associated feelings of fear, guilt and shame
- ► Asymptomatic nature of many STI/HIV, especially in females
- ► The fact that many persons, especially females, do not recognise their symptoms as due to STIs, or think that they are normal (e.g., females with vaginal discharges)
- ► Myths about "virgin cure", "obeah cure", "illegal treatment", etc.

- *The attitude of health providers* –real and perceived: main complaints by adolescents are that some health staff are unfriendly, judgmental, moralistic, discourteous, and inattentive. They exhibit a "holier than thou" attitude, lack confidentiality and empathy, and show a negative body language.

- *Low assessment of personal risk*: among persons ages 15-49 years with multiple partners, 73% perceived themselves as being at low or no risk for HIV.

- *Focus on symptomatic case management*: a focus on symptomatic case management to the exclusion of counselling and other vital behaviour change components of care specially oriented to adolescents.

- *Invulnerability*: many adolescents knowingly indulge in behaviours at high risk to their health because they feel that "it cannot happen to me".

Priority strategies needed to improve adolescent health services:

- *Training and retraining of health providers:* at all levels to recognise, and respond to the special health needs of adolescents.

- *Policy and legal climate:* intensify the rate of change for policy formulation. Enforce existing laws for the rights of adolescents.

- *Integration of adolescent health services*: such as FP, STI/HIV, and MCH.

- *Youth friendly centres:* making clinics accessible and youth-friendly, and establishing centres which combine health care and promotion with other aspects of adolescent social, educational and recreational life.

- *Supportive environment:* empowering communities through Community Based Organisations, youth/parent/teacher organisations etc.

- *Research activities:* aimed at guiding and informing the above interventions.

REFERENCES

1. UNICEF. (2018). Situation analysis of Jamaican Children.
2. Ministry of Health. (2017). 2017 HIV/AIDS Knowledge, Attitudes and Behaviors Survey, Jamaica.
3. Serbanescu, F., Ruiz, A., Suchdev, D. B. and National Family Planning Board, Jamaica. (2010). Reproductive Health Survey Jamaica 2008: Final Report. Atlanta, GA.

PART 2

STI Case Management Checklists

STI TREATMENT (Syndromic/Aetiologic) CHECKLIST **(First line treatment bolded)**
GENITAL DISCHARGE
Gonorrhoea (GC): **Ceftriaxone 250 mg IM stat** **OR** Ciprofloxacin 500 mg PO stat **OR** Ofloxacin 400 mg PO stat **OR** Norfloxacin 800 mg PO stat *Chlamydia (CT):* **Doxycycline 100 mg PO bid x 7 days** **OR** Tetracycline 500 mg PO qid x 7 days **OR** Erythromycin 500 mg PO qid x 7 days **OR** Ofloxacin 300 mg PO bid x 7 days **OR** Azithromycin 1 g PO stat. [1] *Trichomoniais (TV) and/or Bacterial vaginosis (BV)*[2] **Metronidazole 500 mg PO bid x 7 days** **OR** Metronidazole 2 g PO stat **OR** Clindamycin 300 mg PO bid x 7 days

GENITAL DISCHARGE
Candidiasis (CA): **Clotrimazole 200mg vaginal tablet od x 6/7** **OR**
Miconazole 200 mg vaginal suppository [Insert dose once nocte x 3 nights] **OR** Fluconazole 150 mg PO stat (acute and recurrent infection) **OR** Ketoconazole 200 mg PO daily x 14 days (in chronic/resistant or systemic cases)
PID (ambulatory)
Ceftriaxone[3] 250 mg IM stat **PLUS** **Doxycycline 100 mg PO bid x 14 days** **PLUS** **Metronidazole[4] 500 mg bid PO X 14 days**
GENITAL ULCERS
Syphilis: **Benzathine penicillin G 2.4 MU IM stat** **(repeat dose in 1 week)** **OR** Doxycycline 100 mg PO bid x 15 days **OR** Erythromycin 500 mg PO qid x 15 days *Herpes simplex (HSV):* Symptomatic management for most cases. **Valacyclovir 1 g PO bid x 7 – 10 days;** (500 mg bid x 5 days for recurrences)

GENITAL ULCERS
Chancroid:
Cotrimoxazole 160/800 MG PO bid x 7 days
OR
Erythromycin 500 mg PO qid x 7 days
OR
Ceftriaxone [5] 250 mg IM stat
(Saline sitz baths daily to promote healing)
Granuloma Inguinale (GI):
Doxycycline 100 mg PO bid PLUS
Streptomycin 1 g IM daily for 2-3 weeks
PLUS
Cotrimoxazole 160/800 mg PO bid x 2-3 weeks
Lymphogranuloma venereum (LGV)
Doxycycline 100 mg PO bid x 2-3 weeks
OR
Cotrimoxazole 160/800 mg PO bid x 2-3 weeks

Notes to Table.

1. **If used for both CT and GC, use 2 g**
2. **First line for TV but not BV. Metronidazole 2 g PO stat for treatment naïve with no history of STI and uncomplicated males.**
3. **Or any equivalent drug for gonorrhoea**
4. **Or Clindamycin 450 mg qid (anaerobic agents); this part is not required for partner management**
5. **May not be effective in patients with HIV**

Vaginal Discharge Differentiation Checklist

	Normal	Trichomoniasis	Candidiasis	Bacterial Vaginosis
Symptom presentation	None	Itch, discharge, 50% asymptomatic	Itch, discomfort, dysuria, Thick discharge, no odour	Odour, discharge, itch
Vaginal discharge	Clear to white; May vary with cycle	Frothy, gray or yellow-green; malodorous	Thick, clumpy, white "cottage cheese"	Homogenous, adherent Thin, milky white; Malodorous "foul fishy"
Clinical findings		Cervical petechiae "strawberry cervix"	Inflammation and Erythema	
Vaginal pH	3.8 - 4.2	> 4.5	Usually ≤ 4.5	> 4.5
KOH "whiff" Test	Negative	Often positive	Negative	Positive

	Normal	Trichomoniasis	Candidiasis	Bacterial Vaginosis
Saline wet mount	Lactobacilli predominate	Motile flagellated protozoa, WBCs may be present	Few WBCs	Clue cells ($\geq$ 20%), no/few WBCs
KOH wet mount	Yeast may be seen		Pseudohyphae or spores if non-albicans species	

CHECKLIST- Major Clinical Features of Genital Ulcers

	AETIOLOGY				
FEATURE	**Syphilis**	**Chancroid**	**Herpes**	**LGV**	**GI**
Organism	Treponema pallidum	Haemophilus ducreyi	Herpes simplex (primary)	Chlamydia trachomatis	Klebsiella granulomatis
Incubation	9 – 90 days	1 – 14 days	2 – 7 days	3 – 21 days	8 – 80 days
Primary lesion	Papule	Papule or pustule	Vesicle	Papule, pustule or vesicle	Papule
Number of lesions	Usually 1. May be multiple	Usually 1 – 3. May be multiple	Multiple. Lesions May coalesce[1]	Usually 1. Evanescent	Multiple. Coalescing to form large beefy ulcers. Variable
Pain	Rare	Common	Common	Variable	Rare (except pseudobubo)
Induration	Firm	Usually soft	None	None	Firm (esp. chronic lesions)
Diameter (mm)	2 – 40	1 – 50	1 – 35	2 – 15	6 – 35

[1] Recurrent cases may be –unilateral, grouped or single, non-indurated, and itchy

FEATURE	Syphilis	Chancroid	Herpes	LGV	GI
Edge	Oval, sharply demarcated, Elevated, round	Undermined, ragged, Irregular	Erythematous	Elevated, oval, round	Elevated, rolled, irregular
Base	Smooth, non-purulent	Greyish, adherent	Serous, smooth	Variable	Red and granular, bleeds on touch
Depth	Superficial/deep	Excavated	Superficial	Superficial	Usually elevated
Lymph nodes	Firm, mobile, non-tender, Bilateral	Tender, may suppurate, unilocular, unilateral	Firm, tender, often bilateral	Tender, may suppurate, loculated, unilateral	Pseudobubo (dermal infiltration of Donovan's bodies)

HIV COUNSELLING AND TESTING

CHECKLIST FOR HIV SEROPOSITIVE COUNSELING
1. **Introduce** yourself, establish **Rapport** 2. Assure **Confidentiality** 3. Explain the **basis of the test,** the meaning of a Negative or a Positive result 4. Help client to assess own **HIV risk status** 5. Discuss **safer sex** and condom use 6. Assess and reinforce client's **coping skills** for a possible positive outcome 7. Reinforce need for **Revisit** 8. Protect **Unborn Child** 9. Address specific **Questions of Concern** to client 10. HIV testing may also be initiated by clinician while patient accesses health services for other reasons (provider initiated testing and counselling, PITC).

CHECKLIST FOR SERONEGATIVE HIV COUNSELLING
1. Introduce yourself, establish **Rapport** 2. Assure **Confidentiality** 3. Give test result 4. Discuss meaning of **Seronegative Status** 5. Reinforce **Safe Behaviours** 6. Discuss **Follow-up –** need for **Retest** 3 months from last **Unsafe Exposure** 7. Protect your **Unborn Child** 8. Answer any questions or **Concerns** 9. **Facilitate Referral** to support and Preventive Services if necessary.

GUIDELINES FOR COMMENCING WITH ANTIRETROVIRAL THERAPY (ART) DURING PREGNANCY

Antiretroviral therapy reduces the incidence of opportunistic infections and significantly increases quality of life and life expectancy among people living with HIV (PLHIV). All newly diagnosed PLHIV, including pregnant HIV positive women, should have a CD4 count performed upon presentation for disease staging. With the advent of Test and Start in Jamaica since January 2017, all patients, including pregnant mothers, should commence ARVs regardless of CD4 count.

Scenario A: Known HIV Positive Women Who Become Pregnant

Although there are concerns relating to potential effects of ARV drugs on the developing foetus, the benefits are considered to outweigh the risks and **suspending** treatment during the first trimester is **not recommended. Women should continue on their existing regime throughout the course of the pregnancy and beyond.**

Women on ART fixed dose combination of **TDF + 3TC (or FTC) + DTG, the preferred choice for first-line or alternatively, TDF + 3TC + EFV as the other option for first-line,** should continue on this regime unless otherwise contraindicated. Women not previously on ART because of appropriate CD4 counts should also be started on this regime, in keeping with the new recommendations of Test and Start, regardless of CD4 levels.

Review of available data from WHO's technical update on treatment optimisation reassures that there is no increase in birth defects or other significant toxicities with the use of efavirenz (EFV) during pregnancy. Chemoprophylaxis to the infant consists of a

stat dose of Nevirapine (NVP) at birth + 6 weeks of Zidovudine (AZT) twice daily for 6 weeks.

Initial studies had highlighted a possible risk between dolutegravir and neural tube defects in infants born to PLHIV women using the drug at the time of conception based on an initial study in Botswana. New data from two large clinical trials comparing the efficacy of DTG and EFV show that the risks of neural tube defects are significantly lower than what the initial studies had suggested. The WHO recommends the use of DTG as the preferred first-line and second-line treatment for all populations, including pregnant women and those of childbearing potential. Treatment decision needs to be based on an informed discussion with the health provider weighing the benefits and potential risks.

Scenario B: Known HIV Positive Women Presenting Late in Pregnancy

For known HIV positive women who present late in pregnancy and are not on ART, the recommendation is to commence ART **(TDF+3TC+DTG)** and continue, with chemoprophylaxis to the infant (**NVP once daily for 6 weeks and AZT twice daily for 6 weeks**). This period may be extended to twelve weeks if it is known that the mother has started ART less than 4 weeks before delivery.

Scenario C: Newly Diagnosed HIV Women who present Early in Pregnancy

Treatment with ART **(TDF+3TC+DTG)** ideally should begin in the first trimester; or as soon as possible; to increase the chances of prevention of vertical transmission of HIV to the foetus. CD4 counts are not a determining factor to commence treatment.

Scenario D: Newly Diagnosed HIV Women who present Late in Pregnancy

For pregnant women with unknown HIV status, who present at the time of labour without previously accessing HIV testing and counselling, diagnosis at the time of labour and delivery is still an important point of entry to both preventive and treatment services.

- Offer pre-test information and rapid testing during labour
- Administer **TDF + 3TC + DTG or EFV** to women testing positive. Provide their infants with **6 weeks of NVP once daily + 6 weeks of AZT twice daily regardless of birth weight**.
- These women may be offered an emergency Lower Segment Caesarean Section if viral load is >1000 copies/ml
- These women should continue ART post-partum and be referred to the Treatment Centre for a CD4 count, viral load testing and other follow-up post-delivery. ART is now recommended for life regardless of CD4 counts. HIV viral load may be used with the CD4 counts in monitoring the woman's response to treatment. Viral load should be done every three (3) months after treatment initiation.
- Bactrim (Trimethoprim/Sulphamethoxazole) prophylaxis against *Pneumocystis Jiroveci* pneumonia for mothers with CD4 counts below 200 cells/mm^3.

ADVERSE EFFECTS

Adverse effects of ARV drugs vary according to class. Clients should be counselled on the most common side effects when commencing medication. Occurrences of adverse events should be immediately reported, documented and appropriate interventions taken. Pregnancy-associated nausea and vomiting may affect a woman's ability to adhere to ARV and appropriate counselling and symptomatic treatment should be provided. Alternative regimens can be explored in keeping with the national treatment guidelines. **(See MOHW PMTCT Manual and HIV Clinical Manual for further details).**

CHECKLIST FOR MEDICAL HISTORY TAKING

Medical History Checklist	
Demographic 1. Name 2. Age 3. Gender identity (M/F/Transgender M-F/F-M/other) 4. Sexual orientation 5. Address (resident/in transit/ include landmarks, street and street number) 6. Place and date of birth 7. Marital status (single/married/separated/ divorced/widowed/how long) 8. Identification/registration number 9. Living at home (if not, with whom) 10. Occupation (local/peripatetic/unemployed, how long/reason) 11. Telephone no. (home/work/if any) 12. Date first seen **Present Medical History** 1. Reason for present visit 2. Symptoms (duration/severity/course)	**Past Medical History** 1. Previous STI, GU, OBGYN, (diagnosis/treatment/result) 2. History of allergy (esp. penicillin) 3. Recent antibiotic treatment (past 2weeks: type) 3. Contraception (type, usage) 4. Other illnesses 5. Mental status **Family History** 1. Health of parents (cause of death where relevant) 2. Sibling birth order (if relevant) 3. Position of patient in family (where relevant) 4. Cause of death of siblings (where relevant) **Female Patients** 1. Menstrual history (recent changes in cycle)

3. Treatment taken (result/sample drug)	2. Pregnancy history (number and health of live children, causes of death, dates of miscarriages, abortions, stillbirths) 3. Gynaecologic history (pelvic history etc.)

CHECKLIST FOR SEXUAL HISTORY TAKING

<table>
<tr><td colspan="2">SEXUAL HISTORY CHECKLIST</td></tr>
<tr><td>

Exposure to Infection

1. Dates of sexual exposure (past 3 months or longer)
2. Marital, regular consort, other
3. Number and type of sexual partners (heterosexual/homosexual/bisexual)
4. Sites of sexual exposure (genital/oral/anal/other)
5. Place of sexual exposure (same country/abroad/where)

Personal Sexual Behaviour

1. Age at sexual initiation
2. No. Lifetime sex partners
3. No. Sex partners past year
4. Sex with a new or different partner past 3 months
5. Previous STI in past year
6. Sexual activities (insertive or receptive) tried:
    - Vaginal intercourse
    - Oral intercourse
    - Anal intercourse

</td><td>

Partner(s) Sexual Behaviour

Had sex with a partner who has:

- Had sex with other partners
- Had an STI
- HIV infection
- Sex with other men
- Injected/used drugs.

Other Personal Risk Factor

1. HIV infected
2. Blood transfusion (when?)
3. Exposure to infected body fluids
4. Perinatal infection (children).
5. Sexual assault

Personal Drug Use

1. Frequency and type:
    - alcohol
    - ganja
    - crack/cocaine
    - injectable
    - other

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</table>

• Non-penetrative activites e.g., masturbation (alone or shared) 7. Vaginal intercourse during menses 8. Sexual intercourse with: • Opposite sex • Same sex • Both 9. Given or received sex for money, goods or drugs	2. Mixed drugs with sex **Personal Protective Behaviour** 1. Protective measures used against STI/HIV 2. Condom use – when, how often, with whom, method of use? 3. Low-risk/safe sexual activities practised – how often, with whom, why?

APPENDICES

APPENDIX I

STI MEDICAL RECORD	DATE: ____/____/____ dd mm yyyy	D.O.B.: ___/___/___ dd mm yyyy	AGE: ______ yrs.
MEDICAL RECORD ____________	( ) New patient/First visit ( ) Follow-up visit ( ) Old patient/new episode __ ≤ 6 months __ >6months < 12 months __≥12 months		

1. NAME: ____________________ ____________________ ______________ SEX : M ( ) F ( )
Last First Middle

2. ADDRESS: ______________________________ PARISH: ____________Tel: ______________

	Reason for visit	Duration
□ Yes □ No	Dysuria	
□ Yes □ No	Genital/Urethral Discharge	
□ Yes □ No	Genital ulcer/sore	
□ Yes □ No	Rash	
□ Yes □ No	Itching	
□ Yes □ No	Abdominal pain (LAP)	
□ Yes □ No	Warts	
□ Yes □ No	Contact to	
□ Yes □ No	Other	

LAB TESTS ORDERED
□ Blood □ TRUST/VDRL □ TPPA □HIV □Hepatitis____
[] Wet Prep: ____________________
[] Cultures: ____________________
[] PCR: ____________________
[] Other ____________________

CLINICAL HISTORY
Partner symptoms □Yes □ No ____________
Antibiotic (last two weeks) □Yes □ No ____________
Allergy (esp penicillin) □Yes □ No ____________
STI last year □Yes □ No ____________
Douche last month ____________
Last Pap Smear dd / mm / yyyy

SEXUAL HISTORY	Gender identity: □ Male □ Female □ Transgender (M to F) □ Transgender (F to M) □ Other
Last sexual intercourse: dd / mm / yyyy	LMP : dd / mm / yyyy

No of persons: Last 6 months____ How many were in the last 1 month:____
How many were:[ ]Male [ ]Female [ ]Trans (MtoF) [ ]Trans (FtoM)
Of all these persons, how many were new: _______

Anal sex in last 6 months: □ **No;** □ **Yes**, (If yes, answer anal receptive ques)
Anal Receptive and/or □ **Yes**, Anal Insertive

Condom use in last 6 months with vaginal sex: □Always □Sometimes □Never
Condom use in last 6 months with anal sex: □ Always □ Sometimes □ Never

Did you exchange sex for money as a source of income or for goods in last 6 months? □Yes, money; □Yes, goods/things; □No exchange

In the last 6 months, have you paid for sex with money/goods? □Yes □ No

When did you last test for HIV? □ Never □ ≤ 6 months □ 6 - 12 months □≥12 months

What was the result of your last test: □ Positive □ Negative □ Did not get result
If positive: On Treatment □Yes □ No

Contraception: □Pill □Injection □Condom □ TL □Other □None

PHYSICAL EXAMINATION Risk Score Females: ____________

□Yes □ No	Oral/Pharyngeal	□ Normal □Abnormal, specify ____________
□Yes □ No	Rash	____________
□Yes □ No	Lower abdominal tenderness	____________
□Yes □ No	Genital Ulcer	□ None □single, multiple, vesicles, clean based, necrotic____________
□Yes □ No	Warts	____________
□Yes □ No	Vaginal/urethral discharge	□Normal □Abnormal-white, yellow/green, mod./profuse/scant, curdy, frothy, homogenous, odor, other_____
□Yes □ No	Cervix	□ Normal □ Abnormal- erythema, erosion, ectropy, contact bleeding, polyps,
□Yes □ No	Cervical discharge	□ None □ Abnormal – mucopus, bloody
□Yes □ No	Cervical motion tenderness	____________ □ Amine □ pH ____
□Yes □ No	Anal DRE/ Anoscopy	□ Abnormal—ulcer, warts, discharge, blood, other ____________
	OTHER: ____________	

DIAGNOSIS AND TREATMENT
STI: □ No □ Yes Syndromic Dx: □ Genital discharge □ Genital ulcer □ PID □ Other, specify __________
Clinical dx: ____________ Therapy: □ Given on site _______ □ Prescription __________

□Azithromycin ___g PO STAT	□Metronidazole ___ mg PO BID x ___days	□Benzathine Penicillin G 2.4 MU: IM weekly x ___
□Ceftriaxone 250/500 mg IM STAT	□Doxcycline 100 mg PO BID x ___ days	□Erythromycin 500 mg PO QID x ___ days
□Ciprofloxacin 500 mg STAT	□Acyclovir 400mg 3 x daily	□Cefixime 400/800 mg
□Other ____________		

Recommendation: ______________________________
Clinician (signature): ______________ □FNP □MO: (print): ______________ Return date: dd / mm / yyyy
Health education: □Drug compliance □Disease awareness **Partner:** □Notification □Treatment □Reduction □Condoms, # issued___
Contact investigator (signature): ______________ (print): ______________
Referral to: □social worker □hospital clinic □other ______ Date: dd / mm / yyyy

APPENDIX II

Notifiable Sexually Transmitted Infections

The following conditions should be reported to the Parish Medical Officer (Health)* and/or the Epidemiology Unit, Ministry of Health and Wellness, ARC Building, Kingston 5

CLASS 1

An individual case report should be made immediately on suspicion. HIV/AIDS must be reported in a strictly confidential manner using the Class 1 Notification Form.

HIV/AIDS	**-**Notify. Complete HIV/AIDS report form.
Congenital Syphilis	**-**Notify. Treat mother and partners.
Ophthalmia Neonatorum	**-**Inform staff of ANC that mother attended. Check ANC procedures (STS, Credé, etc.) Take corrective action as indicated.

* See Appendix VII for addresses and telephone numbers.

CLASS I REPORTING FORM - INDIVIDUAL NOTIFICATION (ON SUSPICION)

Date of Report: _____ / ____ / _____ (DD/MM/YY) NEW CASE / PREVIOUSLY REPORTED CASE (Circle One)

Diagnosis: ______________________________ ______________________________

Case Demographic Information

Name (including pet name): ______________________________ Sex: ______ Age: ______ D.O.B ____ / ____ / ____ (dd/mm/yy)

Address: (Include Landmark) Lot #: ________ Street: ______________________________ (Name) Street Type: ______________ (Drive, Road, Close etc)

Community: ______________________ Neighbouring Community/District: ______________________ Parish: ______________

Workplace/School: ______________________________ Occupation: ______________________________

(H) Phone #: ______________ (Wk) Phone #: ______________ History of overseas travel in past 4-6 weeks? Y / N Specify area/country: ______________

Name of NOK/Parent: ______________________________ Relationship to case: ______________

Address of NOK/Parent: ______________________________ Phone No.: ______________

Clinical information:

Symptoms: ______________________________ Hosp./Facility Name: ______________

Medical Record # ______________

Date of onset: ____ / ____ / ______ (dd/mm/yy) Date seen: ____ / ____ / ______ (dd/mm/yy) Case admitted to Hosp?: Y / N (Circle one)

Specimen Taken Y / N Type: ______________ Date of Admission: ____ / ____ / ______ (dd/mm/yy)

Specimen Date: ____ / ____ / ______ (dd/mm/yy) Laboratory: ______________ Ward: ______________

Result (s): ______________________________ If dead, Date of Death: ____ / ____ / ______ (dd/mm/yy)

Notifier Information

Name of notifier: ______________ Phone #: ______________ Received by MO(H) ____ / ____ / ______ (dd/mm/yy)

Address: ______________ Email: ______________ Parish MO(H) Signature ______________

Comments: ______________ Forwarded to R.S.O ____ / ____ / ______ (dd/mm/yy)

Forwarded to Surveillance Unit ____ / ____ / ______ (dd/mm/yy)

Ministry of Health, Surveillance Unit, September 2006

CLASS 3

Report by syndrome each month, line-listing by age and sex:

- Urethral discharge
- Vaginal discharge
- Pelvic discharge
- Genital Ulcers (total and specify by clinical appearance):
    - ► Herpes genitalis
    - ► Chancroid
    - ► Syphilis
    - ► Lymphogranuloma venereum
    - ► Granuloma inguinale

GUIDE FOR HOSPITAL SURVEILLANCE

A PHN should visit the hospital routinely each week, especially the paediatric and medical wards. Additional visits should be made as necessary. The MO(H) should also visit the hospital twice monthly and keep in touch with his/her colleagues.

APPENDIX III

STI Case Reporting and Confidentiality

(Clinician's Responsibility)

SURVEILLANCE

Disease surveillance activities, which include the accurate identification and timely reporting of STIs, form an integral part of successful disease control. Reporting assists local health authorities in identifying sexual contacts who may be infected.

REPORTING

Reporting may be provider and/or laboratory-based. Cases should be reported in accordance with local statutory requirements and in as timely a manner as possible. Clinicians who are still unsure about these requirements are encouraged to seek the advice of their parish health department or STI clinic.

CONFIDENTIALITY

STI reports are held in strictest confidence. Before any follow-up of a positive STI test is conducted by the programme representatives, these personnel consult with the provider to verify the diagnosis and treatment. All parish health authorities offer follow-up services for selected STIs.

APPENDIX IV

Priority STI Conditions to be Referred to the Contact Investigator

1. Pregnant women with reactive serologies
2. HIV in pregnancy
3. Infectious syphilis in pregnancy
4. Congenital syphilis
5. Ophthalmia neonatorum
6. Pelvic Inflammatory Disease in adolescents (10-19 years)
7. Resistant gonorrhoea or any unusual occurrence/re-emerging strains
8. STI in Childhood

These STI conditions should receive priority for investigation. They should be referred to a trained STI Contact Investigator for partner notification and management.

APPENDIX V

Importance of Referral to a Contact Investigator /Partner Notification and Management of Sexual Contacts

Breaking the chain of transmission is crucial to STI control. Further transmission and reinfection are prevented by referral of sex partners for diagnosis and treatment (**partner notification or referral**). This activity may be undertaken by:

1. **The Client** as a direct result of patient counselling, which may include motivating the patient to assume an active role in bringing contact(s) for evaluation and treatment (**contact interviewing**); OR

2. **The Health Provider** may implement an active search for STI contacts (**contact tracing**).

Patients should ensure that their sexual partners, including those without symptoms, are referred for evaluation. Partners of patients with STI should be examined. Treatment should not be provided for partners who are not examined, except in rare instances such as when the partner is at a site remote from medical care. The appropriate management of STI patients must include the application of full treatment regimen to all known contacts; particularly the regular sex partner(s), e.g., husband/wife/significant other. In addition, **the source of infection** and those **contacts at high risk** of having been infected must be treated.

Contacts at high risk are those exposed during the "critical period" of a specified disease. This period is the maximum of the incubation period of the disease plus the length of time that the patient has had symptoms.

Appropriate referral for sex partners should be provided if care will not or cannot be provided by the initial health care provider.

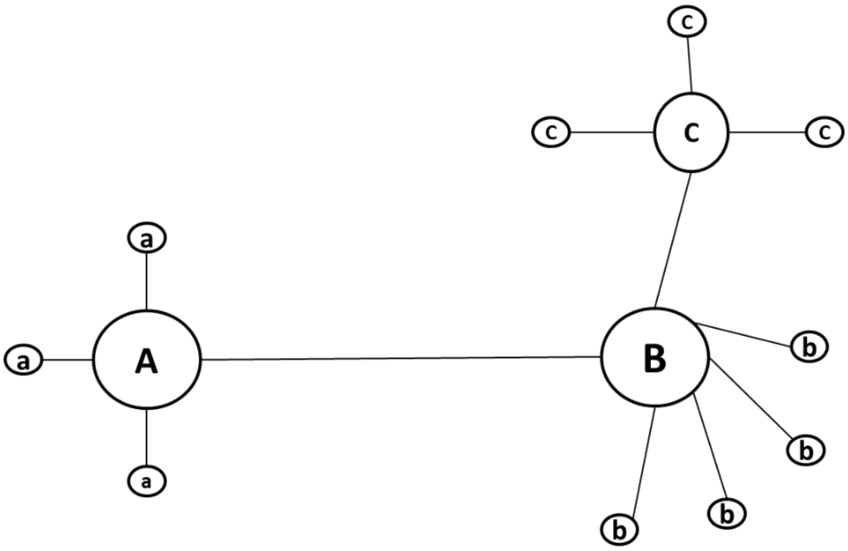
A patient's (A) risk may be due to having:

1. Multiple partners (a).
2. A single partner (B): with multiple partners (b), or with a single partner (C) with multiple partners (c).
3. Any combination of the above.

Partner (C) is usually the most difficult to elicit. (See Figure V.1)

Figure V.I

STI Chain of Transmission



APPENDIX VI

Parish STI Centres and Health Offices

REGION	PARISH	CLINIC	TYPE	CLINIC ADDRESS	TELEPHONE (FAX)	HEALTH OFFICE ADDRESS TELEPHONE (FAX)
SERHA	**Kingston & St. Andrew**	Comprehensive H/C	V	55 Slipe Pen Rd. Kgn 5	876-924-9473, 876-922-2095 876-3053, 876-924-9380	Marescaux Rd, Kgn 5 876-926-1550-2 876-929-4160, 876-920-3610, 876-754-4482, 876-920-8103, 876-754-4483
		Windward Road H/C	V	18 Paradise Str, Kgn 16	876-938-3901	
		Maxfield Park H/C	V	87 Maxfield Ave., Kgn 13	876-968-7585, 876-968-7615	
		Duhaney Park H/C	III	122a Baldwin Cres, Kgn 20	876-933-3484, 876-933-7897	
		Edna Manley H/C	III	35a Grants Pen Rd, Kgn 8	876-931-6377	
	St. Thomas	Morant Bay H/C	IV	Lyssons Rd, Morant Bay	876-7036062	Morant Bay 876-703-6181-2 876-982-1619, 876-703-6183
	St. Catherine	St. Jago H/C	IV	St. Jago Park, Spanish Town	876-907-5280	St. Jago Park, Spanish Town 876-984-2282/3318 876-907-5281-2: 5284-5 876-984-5916
		Linstead H/C	III	Linstead, St. Catherine	876-985-2203	
		Old Harbour H/C	III	Old Harbour, St. Catherine	876-983-2313	
		Greater Portmore H/C	V	5W Greater Portmore St. Catherine	876-989-4638	

Parish STI Centres and Health Offices (Cont'd)

REGION	PARISH	CLINIC	TYPE	CLINIC ADDRESS	TELEPHONE (FAX)	HEALTH OFFICE ADDRESS TELEPHONE (FAX)
NERHA	**Portland**	Port Antonio H/C	IV	Smatt Rd, Port Antonio	876-993-2557, 876-993-2873	Smatt Rd, Port Antonio 876-993-2557/2873, 993-9426
		Buff Bay Comm. Hospital	CH	Buff Bay, Portland	876-996-1478	
	St. Mary	Annotto Bay H/C	III	Anotto Bay, St Mary	876-996-2034	Port Maria, St. Mary 876-994-9711/9979/2643/2 876-994-2689
		Gayle H/C	III	Gayle, St. Mary	876-975-8054	
	St. Ann	Port Maria Hospital	H	Port Maria, St. Mary	876-994-2330	
		St. Ann's Bay H/C	IV	St. Ann's Bay, St. Ann	876-972-9701/972-2190	4 Windsor Rd, St. Ann's Bay 876-972-2215, 876-972-2227 876-972-1337
		Ocho Rios H/C	III	Ocho Rios, St. Ann	876-974-2691	
		Browns Town H/C	III	Brown's Town, St. Ann	876-975-2338	
		Alexandria Polyclinic	P	Alexandria, St. Ann	876-975-1007	

Parish STI Centres and Health Offices (Cont'd)

REGION	PARISH	CLINIC	TYPE	CLINIC ADDRESS	TELEPHONE (FAX)	HEALTH OFFICE ADDRESS TELEPHONE (FAX)
WRHA	Trelawny	Falmouth H/C	IV	Falmouth, Trelawny	876-954-3689	Falmouth, Trelawny 876-954-3689/4904 876-954-3563
		Dewar H/C	III	Duncan's Trelawny	876-954-9304	
		Stettin H/C	III	Albert Town, Trelawny	876-610-0785	
	St. James	Montego Bay H/C	V	Creek St, Montego Bay	876-7820-4	Creek St 876-979-7820-4/7813 876-979-7802
		Catherine Hall H/C	III	Catherine Hall, Montego Bay	876-952-0396	
		Cambridge H/C	II	Cambridge, St. James	876-912-2384	
	Hanover	Lucea H/C	IV	Lucea, Hanover	876-956-2604	Lucea, Hanover 876-956-2604 876-956-9637/9688
SRHA	Westmoreland	Sav-la-Mar	IV	Sav-la-Mar, Westmoreland	876-955-2218	Sav-la-Mar 876-955-2308/2929
	St. Elizabeth	Black River H/C	IV	Black River, St. Elizabeth	876-965-2712/2701	High St, Black River 876-965-2266/2701
		Santa Cruz H/C	III	Santa Cruz, St. Elizabeth	876-966-2101	
	Manchester	Mandeville H/C	IV	Mandeville, Manchester	876-962-2288	Mandeville 876-962-2288/7033 876-625-3433, 876-962-2171
		Porus H/C	III	Porus, Manchester	876-904-0943	
	Clarendon	May Pen H/C	III	May Pen, Clarendon	876-902-3642/3640	Denbigh, May Pen 876-986-9712/4548/7869 876-986-9713
		Race Course H/C	III	Race Course, Clarendon	876-978-5307	
		Chapelton Comm Hospital	CH	Chapelton, Clarendon	876-785-0125, 876-987-2215	

APPENDIX VII

Advice to Health Provider re Referral of Patients

Evaluation of reactive test for syphilis

Minimum data required from referring agency

1. Presumptive diagnosis and stage of disease
2. Dates and titres of screening tests
3. Date and result of confirmatory test
4. Type, dosage and schedule of drug used for treatment
5. Dates when treatments given
6. Was contact(s) seen, if so disposal as under #s 1-5 above
7. Any other relevant data

N.B. Without this minimum information, it is impossible to evaluate your patient properly.

APPENDIX VIII

STI Patient Education - Behavioural

Every patient with an STI and especially those with genital ulcers should have a serologic test for syphilis and should be offered confidential counselling and testing for HIV infection. Some of the important messages each health provider needs to emphasise to patients are listed below:

- Understand how to take any prescribed oral medication. If doxycycline is prescribed, take it 1 hour before or 2 hours after meals. Avoid dairy products, antacids, iron and other mineral- containing preparations and sunlight.
- Avoid alcohol 24-48 hours before and after taking metronidazole.
- Do not share own medications with sex partner(s).
- Dangers of self-medication (asymptomatic diseases, resistance, drug reaction, etc.).
- Know what medication has been prescribed by health provider and/or bring container with them if they go to another medical centre.
- Return for clinical assessment (or laboratory test-of-cure, TOC) on appointment date after completing therapy.
- Refer/bring in all sexual partners for examination and treatment
- Avoid sexual intercourse until patient and partner(s) cured.
- Know proper use of condoms to prevent future infections.

- If IUCD is to be removed, consult with patients' physician or F.P. agency.
- Know basic facts about disease and how to control spread.
- Stress asymptomatic infections.
- Disease recognition/suspicion.
- Risk reduction (see Section 3).
- Stress confidential nature of interview.
- For HSV and HPV, annual Pap Smear for women.
- STI the dangers of drug abuse (especially injectibles and needle sharing) in the transmission of viral STIs, especially HIV.
- Disinfestation of clothing and linen for "crabs" and "mites"- washing, dry cleaning, ironing, exposure to sunshine or avoid using for 1-2 weeks.

APPENDIX IX

Anaphylactic Reaction to Penicillin

IMPORTANT: Question patient about allergic reactions. If possible, this should be done by more than one provider.

SIGNS AND SYMPTOMS

1. Generalised flush
2. Paroxysmal cough
3. Urticaria
4. Wheezing
5. Orthopnea
6. Shortness of Breath
7. Vomiting
8. Anxiety
9. Cyanosis
10. Shock

IMMEDIATE TREATMENT

ADULT	CHILD
Adrenaline 1/1000 0.5 – 1 ml IM stat	Adrenaline 1/1000 0.1 – 0.3 mls SC/IM stat
Repeat in 5 mins, as necessary. If in shock, administer IV slowly diluted in 5 – 10 ml saline. Leave needle in place and set up drip Immediately on appearance of signs of a reaction, lie patient flat, loosen, constricting clothing and administer adrenaline (maintains blood pressure and dilates bronchi, alleviating wheezing).	

Additional procedures may be necessary and include:

- For urticaria, angioneurotic oedema, conjunctival congestion: DIPHENHYDRAMINE (Benadryl): 25 – 50 mg IM or IV.

- To reduce vascular permeability, maintain blood pressure and reduce shock: HYDROCORTISONE: 100 mg IM or IV.

- For pulmonary oedema (cough, dyspnoea, respiratory distress): AMINOPHYLLINE: 500 mg IV.

- For cyanosis:
  OXYGEN: 6 L/min (maintain a clear airway)
- For shock:
  Raise foot off bed. NORMAL SALINE infusion. (Adult 1000 ml over 15-30 mins. Child 20 ml/kg over 15-20 mins.)
- Moderate to severe shock:
  Cortisone up to 200 mg (IV/drip). If no response, vasopressors. Adjust rate to keep B.P. at 100 systolic.

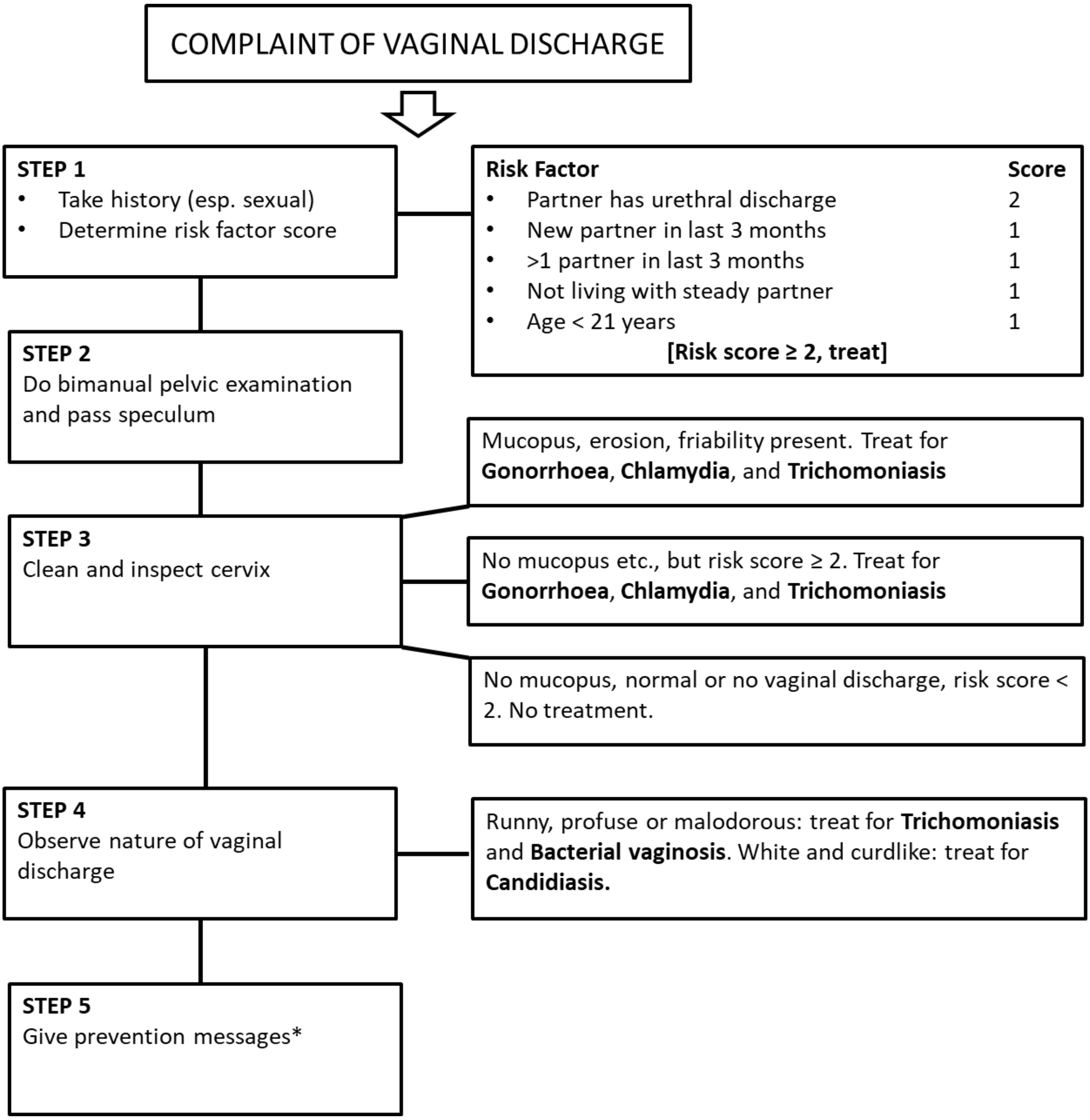
NOTE: IV drip, tracheotomy, respirator, referral to hospital (ICU) may be necessary. Sensitivity testing with penicillin and penicilloyl give little more information than can be had from careful interrogation and may be dangerous. Patients may react to penicillin injection while giving a negative skin test and vice versa.

Very Important: Resuscitants must be within reach and ready for immediate administration. Being locked away in a cabinet to which the key cannot be found when needed is to be avoided.

APPENDIX X

Algorithms for diagnosis

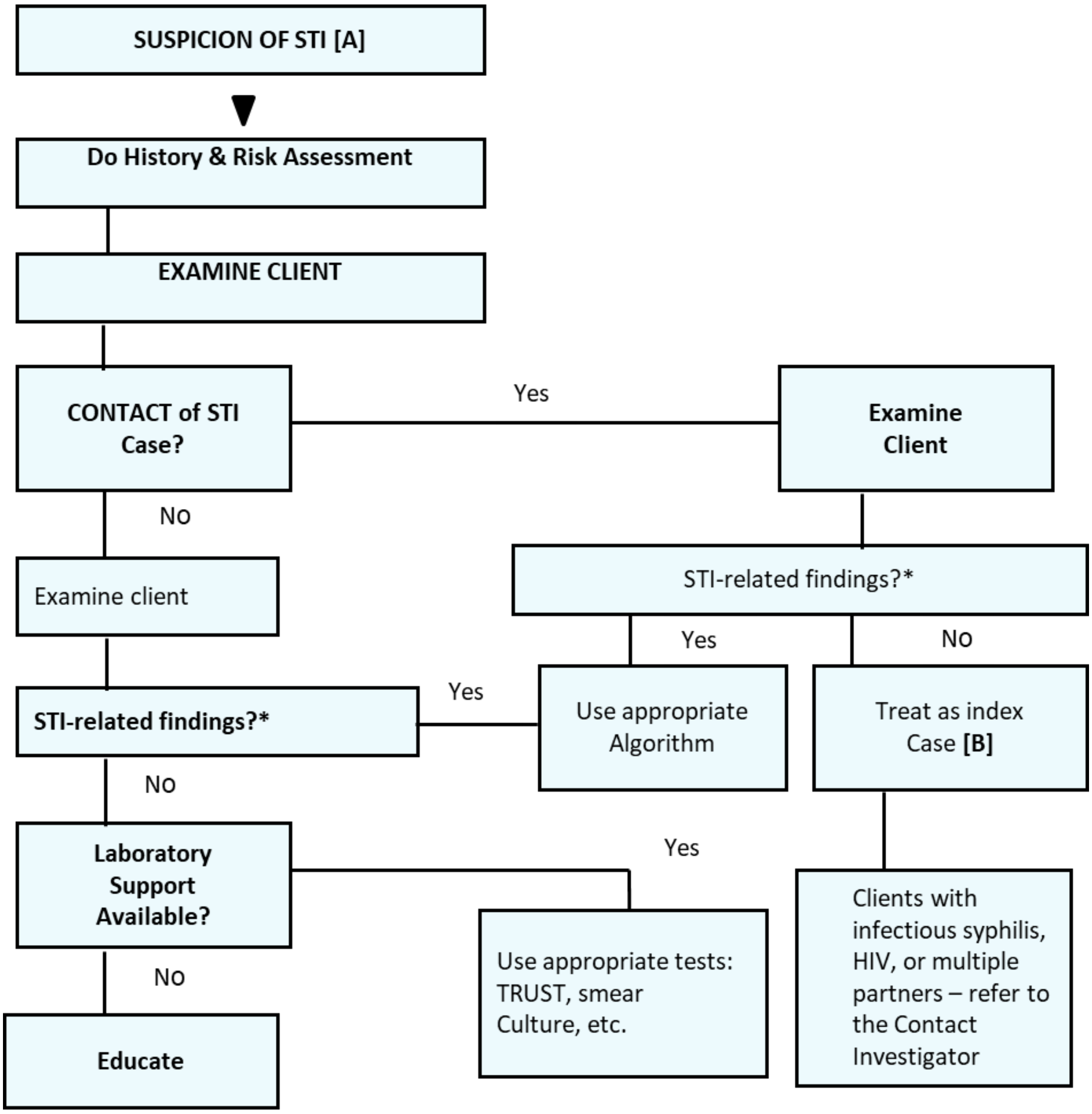
1. Algorithm for Management of Cervico-Vaginal Discharge



*Comply with medication; Condoms; Contacts; Counsel re risk reduction.

[Screen all patients for syphilis and HIV. Pap smear and refer persistent cases of discharge to gynaecologist.]

2. Algorithm for Management of STI Screening



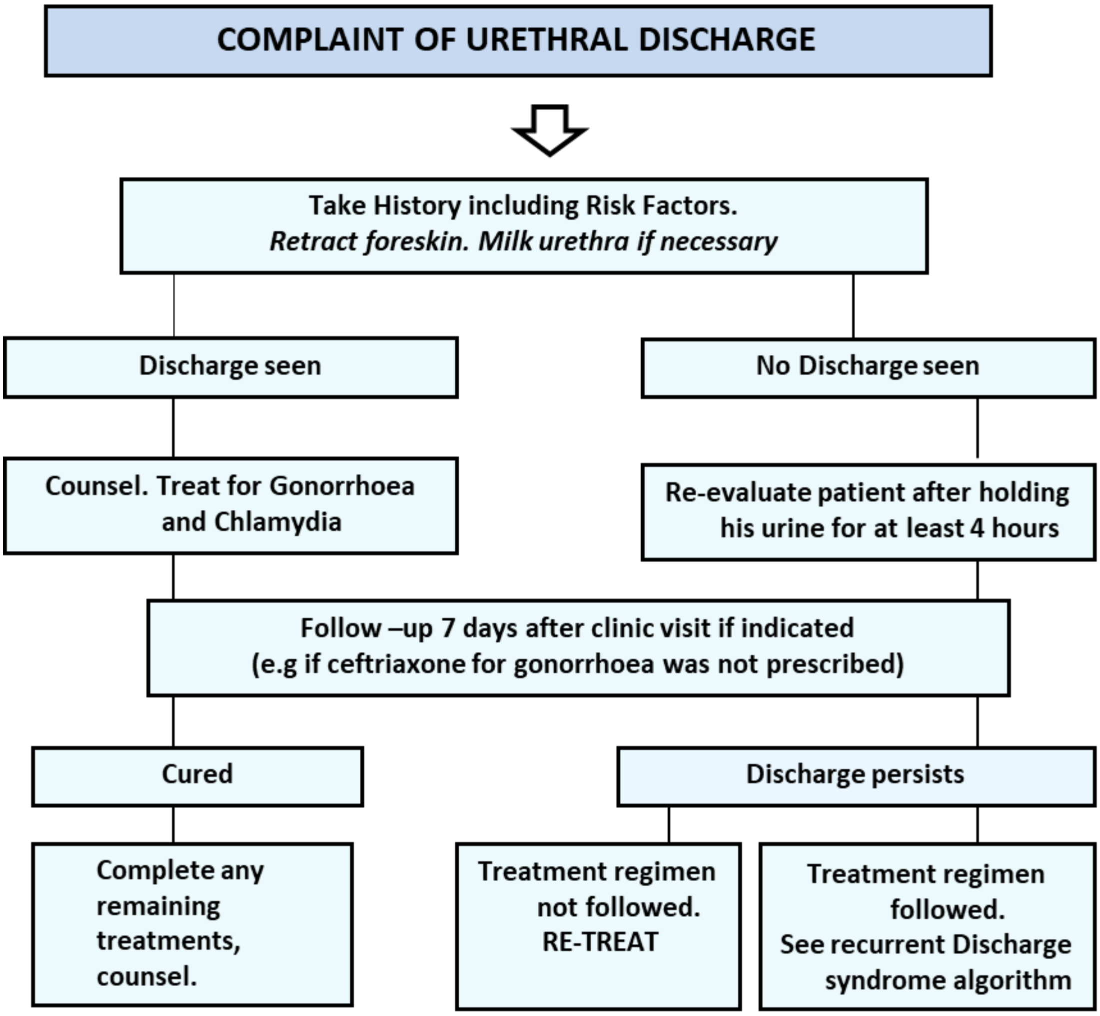
Notes:

*See **Table 8** (Summary of Common STI-related Findings)

[A] If the patient has signs, use the appropriate algorithm

[B] Appropriate explanation should be given as to why (s)he needs treatment

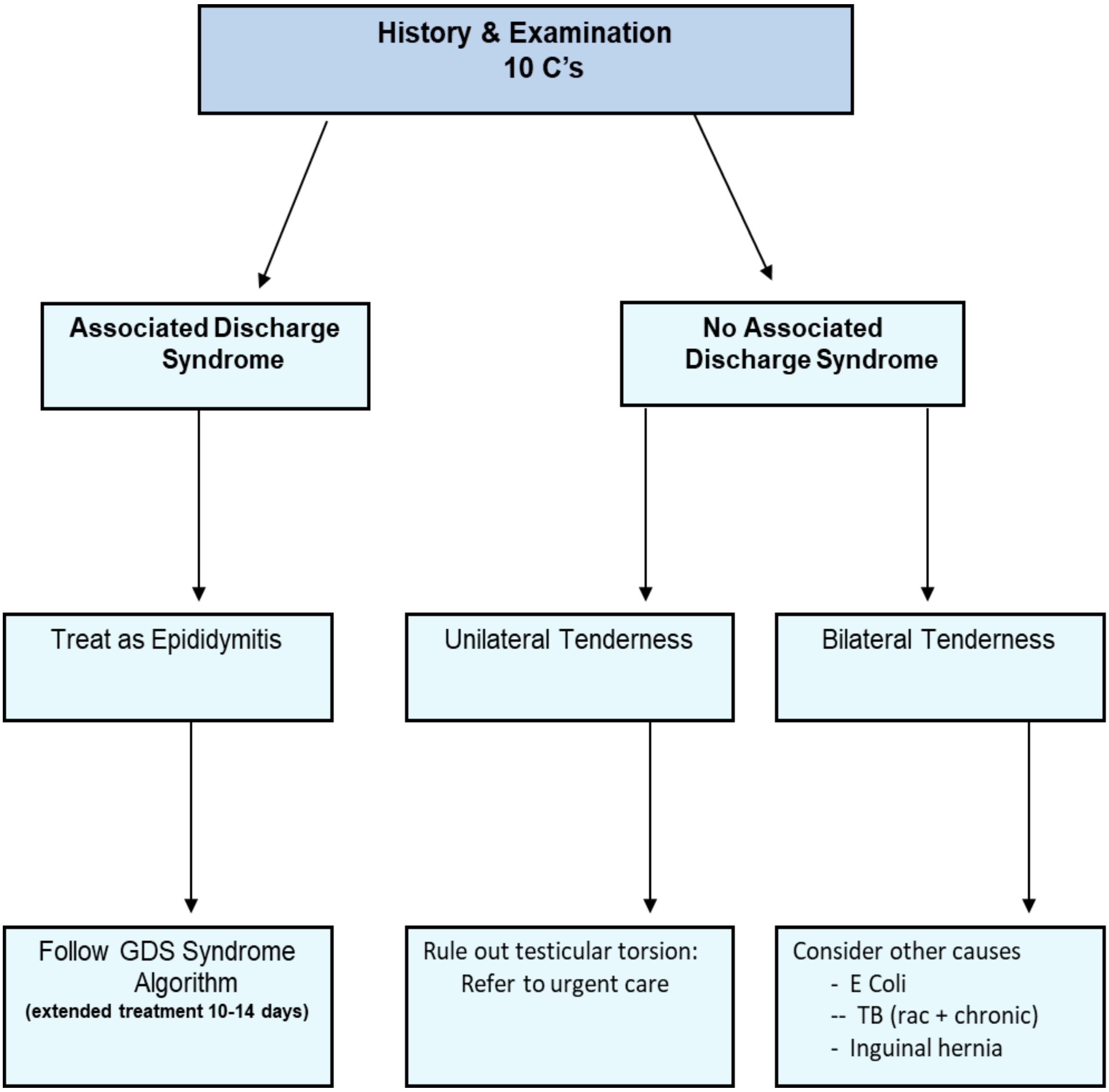
3. Algorithm for Management of Urethral Discharge



Notes:

- Early morning bead of pus; urinary “threads” in first urine passed; WBCs>4/HPF on Gram stained smear - suggests chlamydial/non-gonococcal urethritis.
- Refer to urologist or hospital, especially patients >35 years: non-STI causes of urethritis may be present.

4. Algorithm for Management of Testicular Pain



5. Algorithm for Management of Persistent Discharge

Re-Evaluate patient: 10 C's History Taking Examination

↓

Appropriate regime not followed OR unprotected sex with untreated partner

↓

Re-treat

Ceftriaxone 250mg stat PLUS

Azithromycin 1g stat

Or

Doxycycline 100 mg bd x 1/52

↓

Appropriate regime followed & no unprotected sex

↓

Gonorrhea swab & culture and

Then treatment

Ceftriaxone 500 mg IM AND

Azithromycin 2 g PO stat

OR

Cefixime 800mg po stat AND

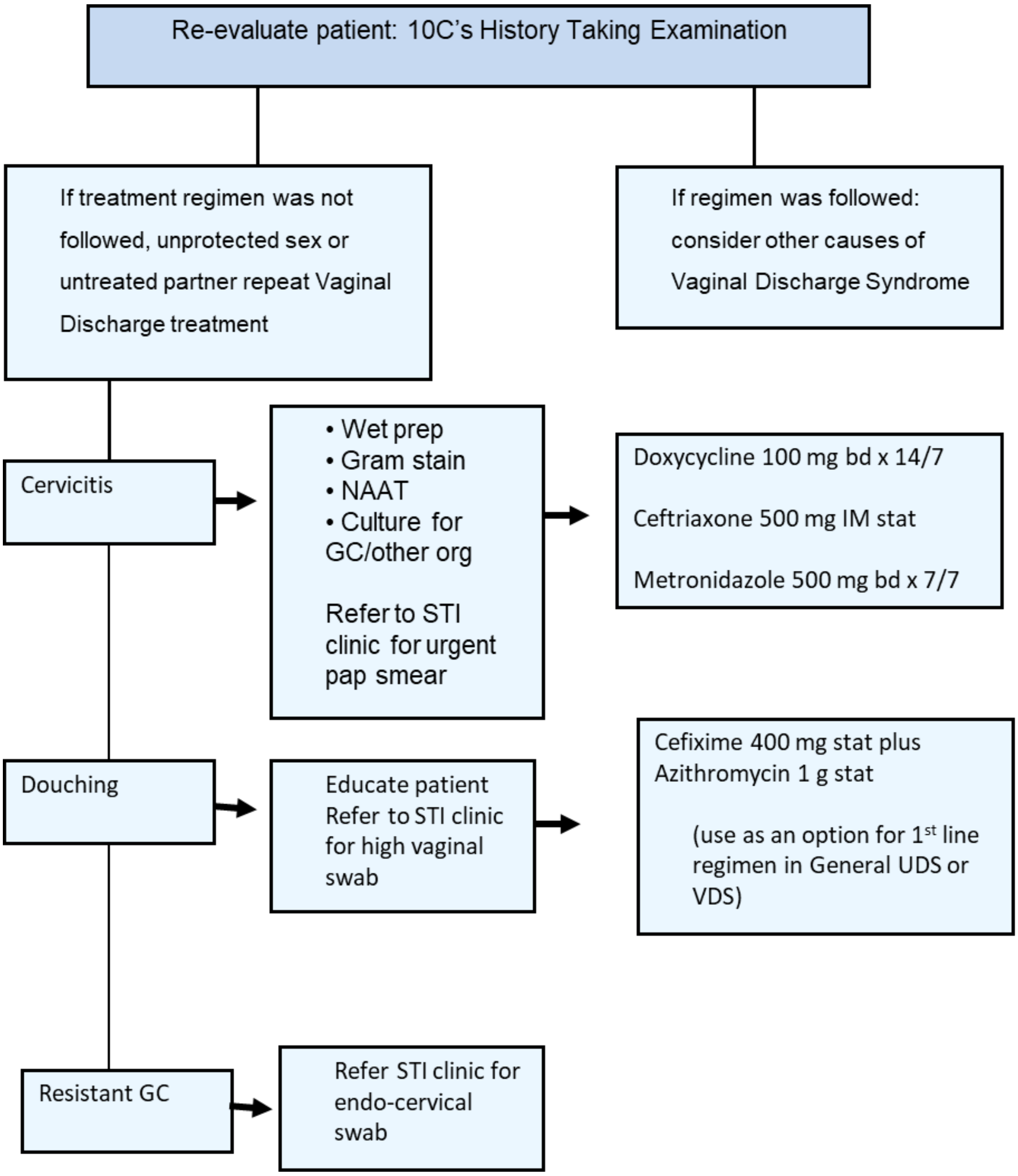
Azithromycin 2g po stat

Refer to STI Clinic

Notes:

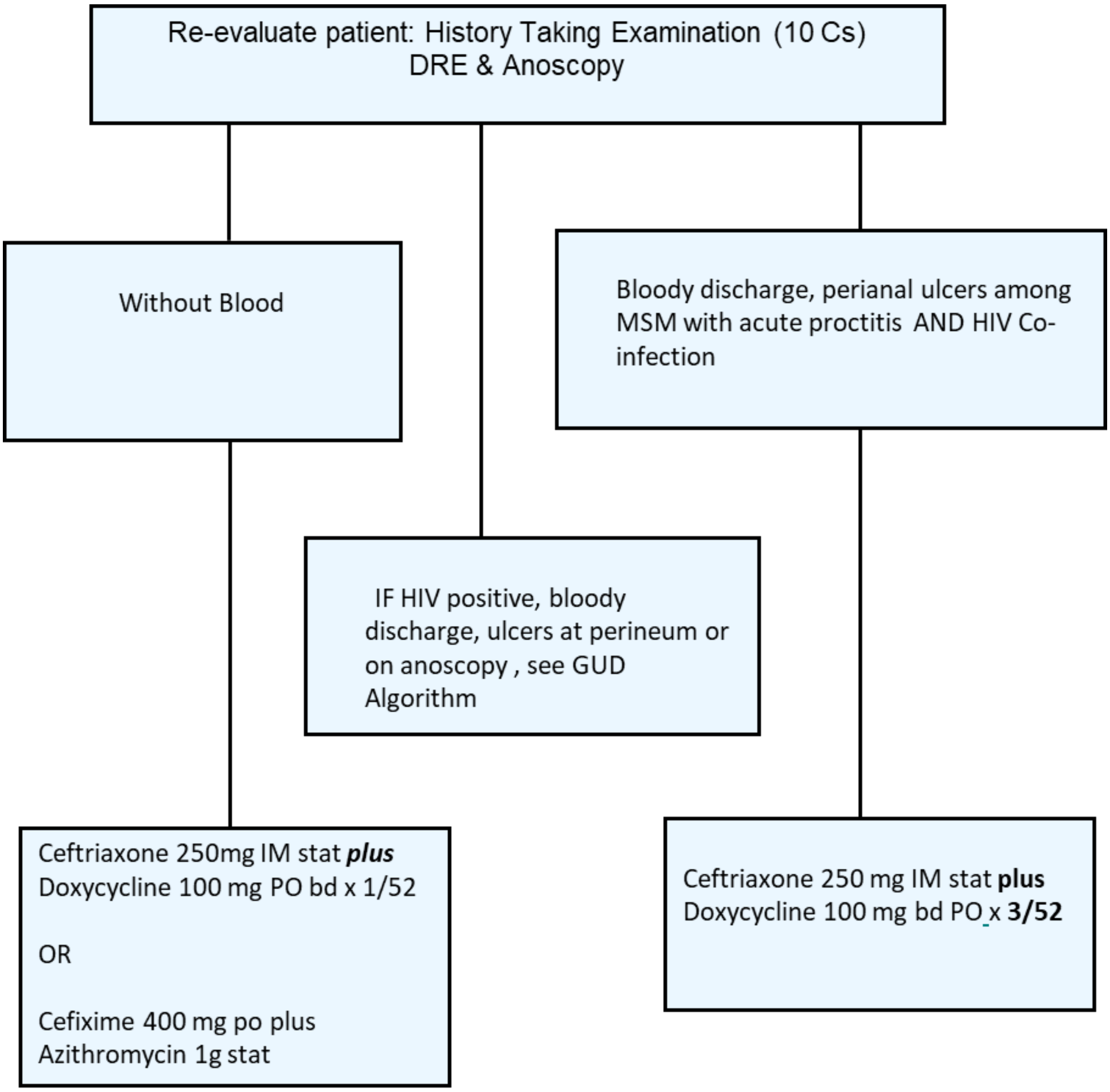
Abstain from all sexual activity OR use a condom with ALL types of sexual activity for 7 days OR until completion of course of medication to prevent spread to partner(s).

6. Algorithm for Management of Persistent Vaginal Discharge

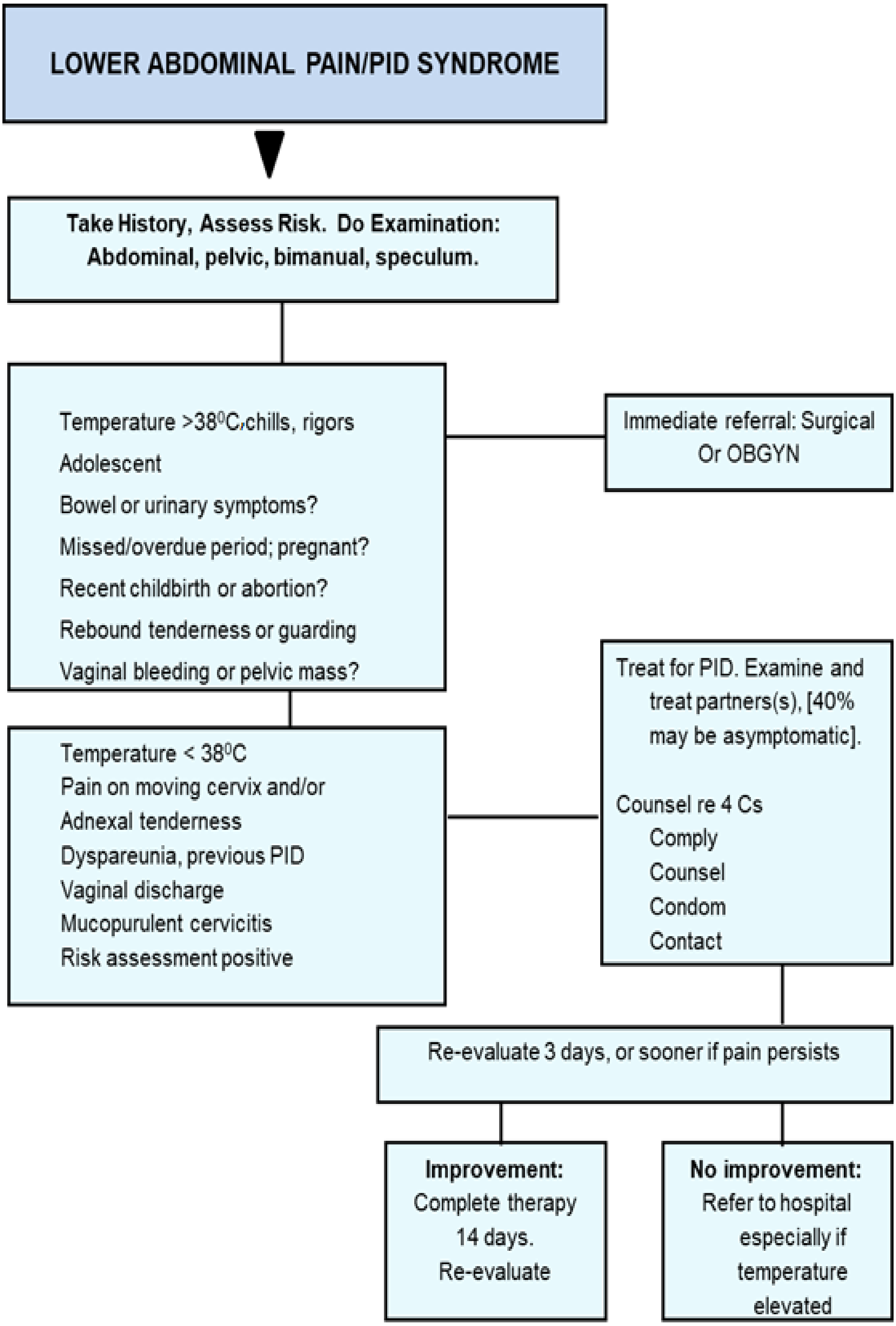


Notes: NAAT: nuclear acid amplification test

7. Algorithm for Management of Anal Discharge

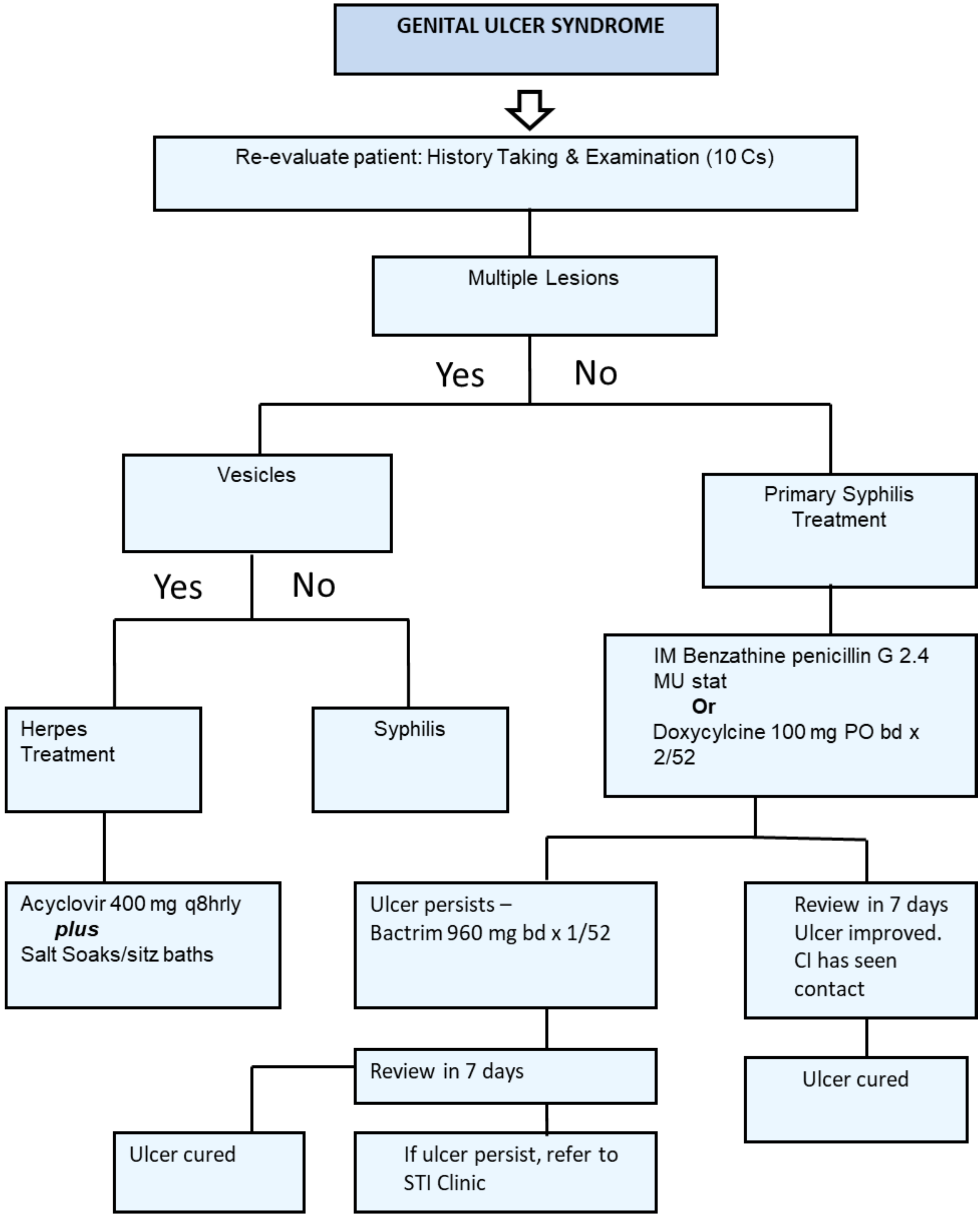


8. Algorithm for Management of Lower Abdominal Pain/PID



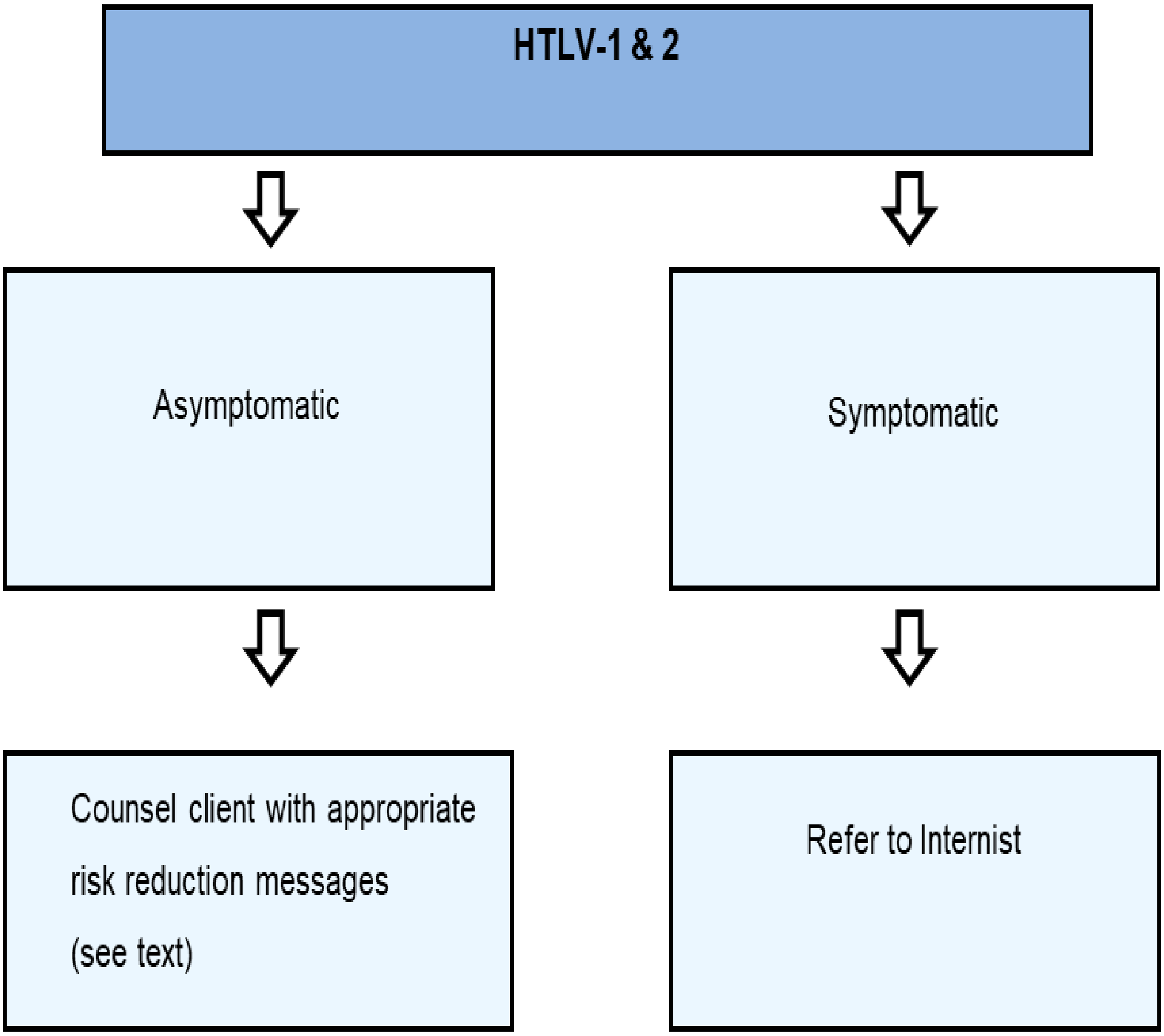
Test all patients for syphilis and HIV; offer pap smear if none within past year.

9. Algorithm for Management of Genital Ulcer Disease (GUD)

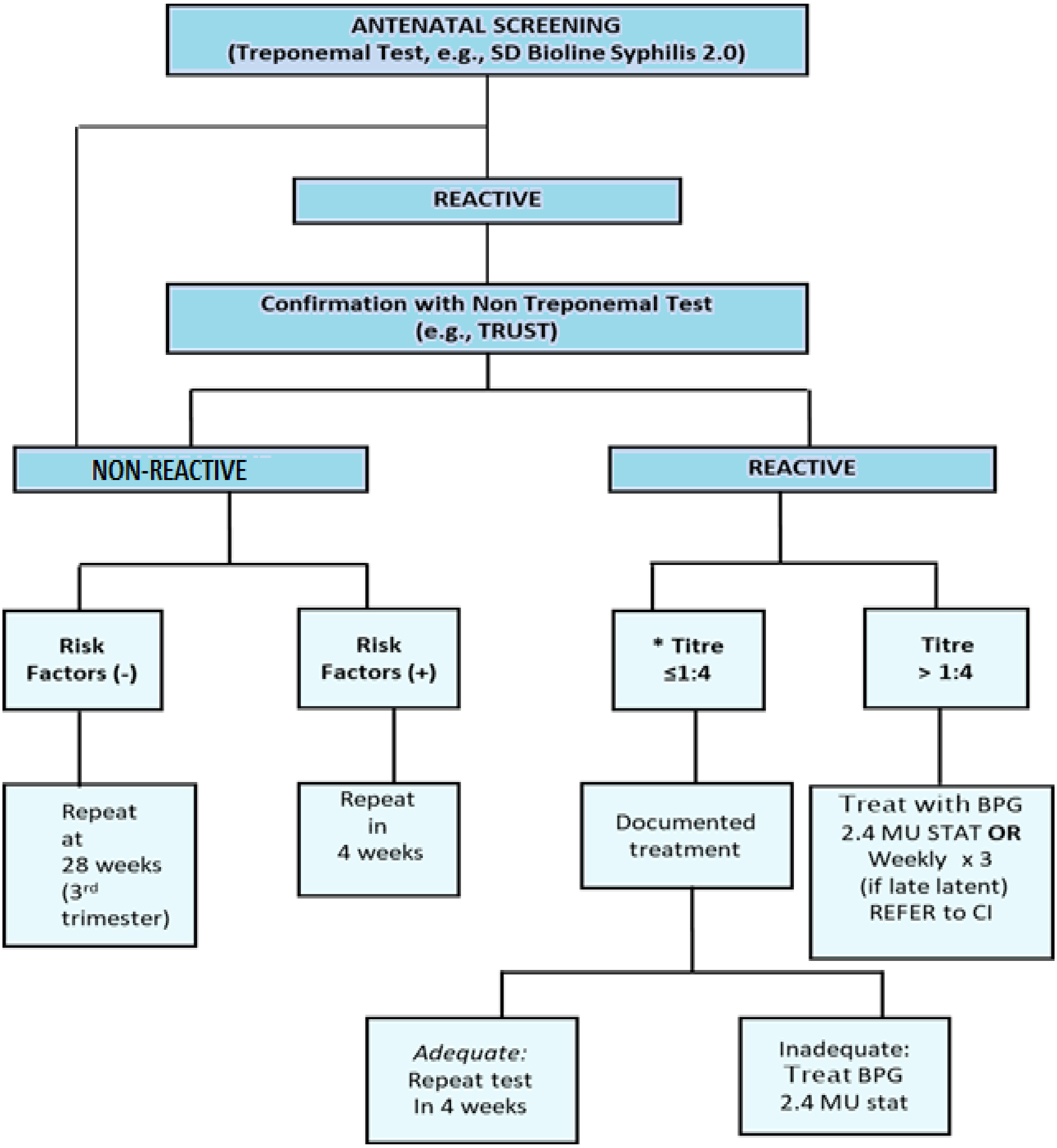


NB: Footnotes for Care of Genitalia: Do not use bleach, disinfectants or other chemicals to area.

10. Algorithm for Management of Human T Lymphotrophic Virus Type I/II (HTLV-I/II)



11. Algorithm for Management of Management of Maternal Syphilis

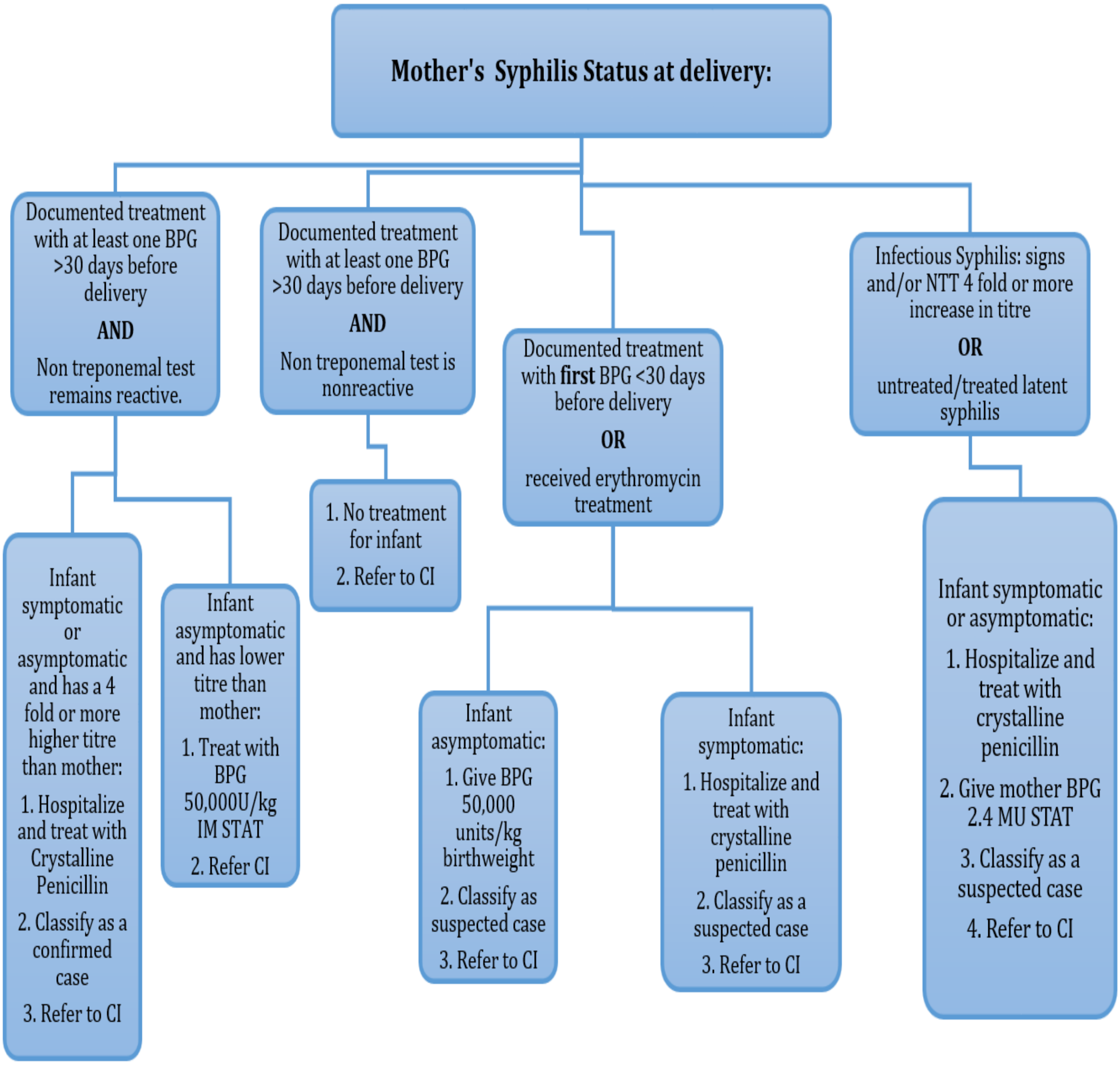


It is HIGHLY RECOMMENDED that the stat dose of IM BPG be given in the pregnant mother, even if serofast.

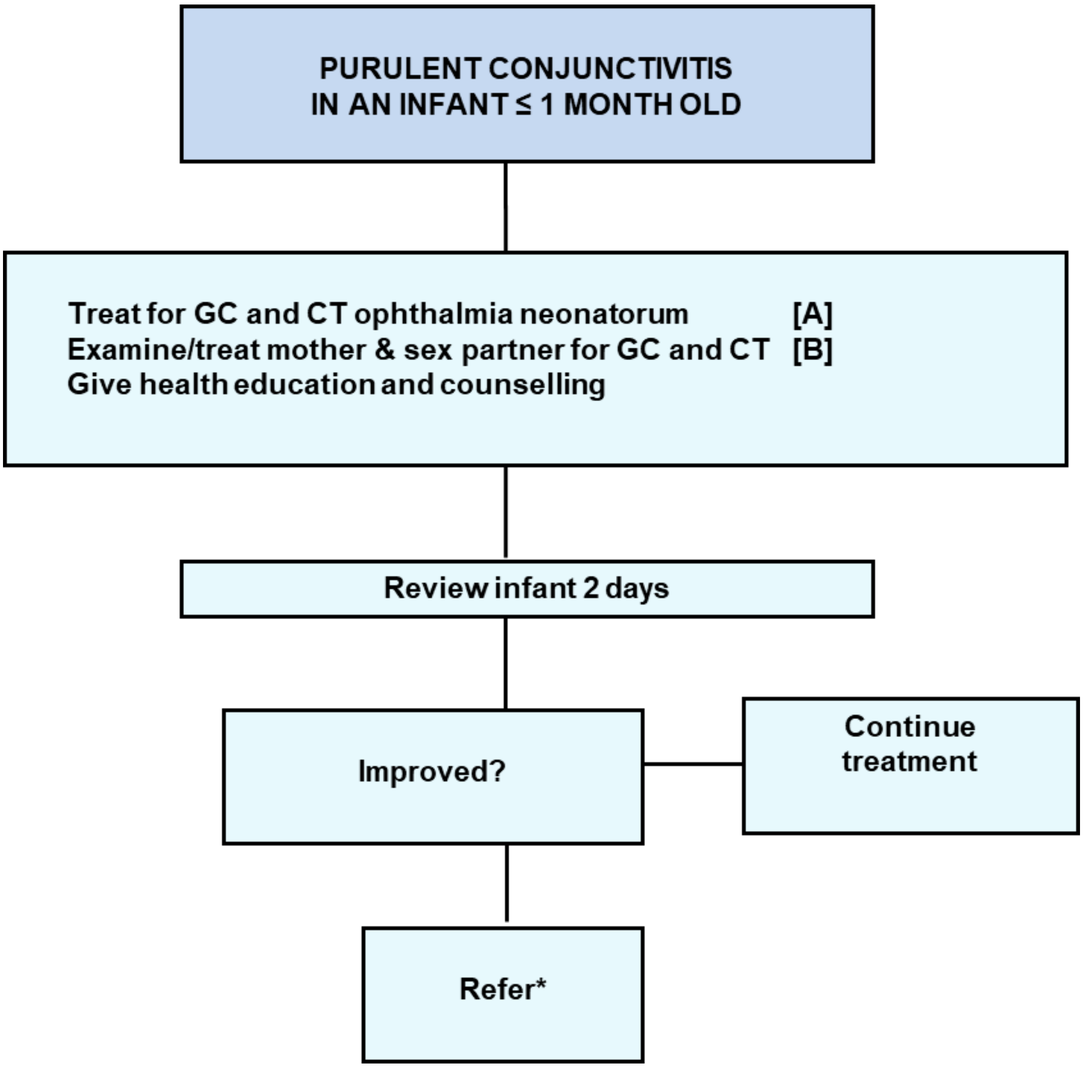
*If history or clinical assessment suggests active infection, give full treatment and refer to Contact Investigator (CI) for partner management.

12. Algorithm for Management of Management of Congenital Syphilis

N.B: Educate and Counsel all Mothers (4C's). Advise for HIV screen.



13. Algorithm for Management of Management of Ophthalmia Neonatorum (Without Laboratory support)



*ON is a medical emergency and best managed in hospital

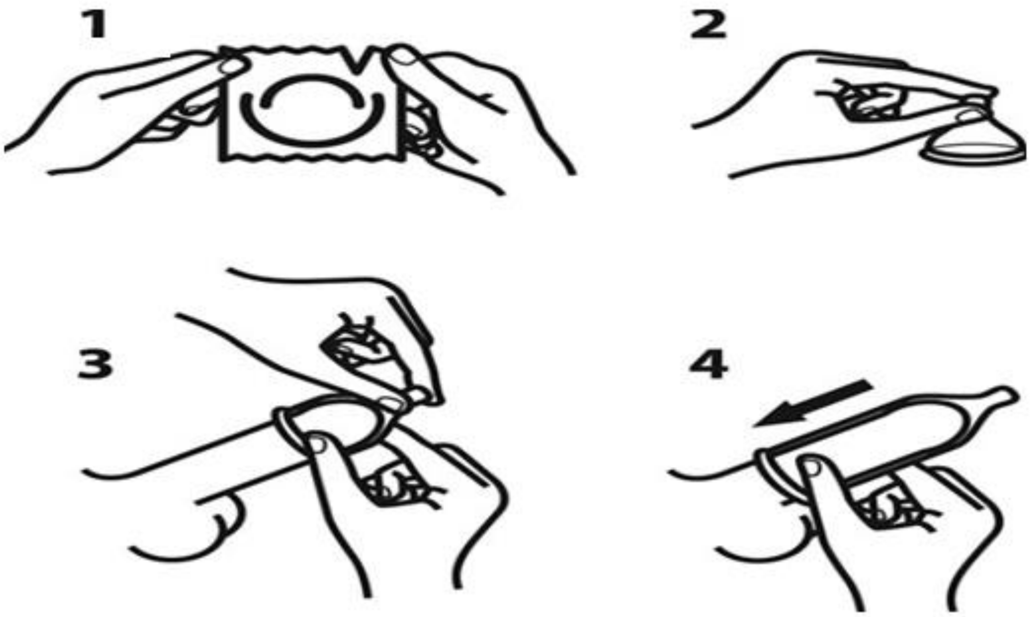
APPENDIX XI

REGIONAL AND CENTRAL REFERRAL HOSPITALS FOR HIV/AIDS

HOSPITAL	**TELEPHONE NUMBER**
Kingston Public Hospital North Street	876-922-0210; 876-922-0227-9 876-922-0530-1; 876-924-9111
Mandeville Public Hospital Hargreaves Avenue	876-962-2067; 876-962-2744 876-962-3370
Cornwall Regional Hospital Mt. Salem, Montego Bay	876-952-5100; 876-952-5105 876-952-5227; 876-952-7571
St. Ann's Bay Public Hospital St. Ann's Bay	876-972-2272; 876-972-0150-2
University Hospital of the West Indies Mona, Kingston 7	876-927-6520; 876-927-6555 876-927-1620
Bustamante Hospital For Children Arthur Wint Drive, Kingston 5	876-926-5721-5
Department of Child Health University Hospital of the West Indies Mona, Kingston 7	876-927-6520; 876-927-6555 876-927-1620

APPENDIX XII

How to Put On and Remove a Condom



For maximum protection, condoms must be used correctly. Health workers should not assume that people know how to use condoms. ***Demonstrate*** the use of the condom using a dildo or similar device, and with clear explicit instructions. Have client repeat the demonstration. ***Dual protection*** (e.g., oral contraception plus condom) in persons at high risk for unwanted pregnancy and STI/HIV, is the national policy, and should be encouraged.

1. Check that it is a latex condom
2. Check date of expiration and/or manufacture
3. Make sure the package is not damaged. Condoms that are sticky, brittle, or otherwise damaged should be discarded.

4. Hold pack by edge and open by tearing from a (ribbed) edge (do not use sharps, teeth or nails)
5. Hold condom so that the rolled rim is facing up
6. Place condom on erect penis
7. Pull foreskin back if penis is uncircumcised
8. Squeeze one inch of the tip of condom between the fingers to push the air out
9. Still holding tip, place it on the head (end) of the penis
10. Gently unroll condom with the other hand, all the way down to the base of the penis, making sure there is the extra space at the tip of the condom
11. Use condom throughout sex – from start to finish
12. Adequate **water-based** lubrication ("K-Y", "For Play", glycerine, or contraceptive gel; not Vaseline or hair oil greases), will protect against breakage. Saliva is best avoided since it dries quickly
13. Immediately after ejaculation, hold rim of condom to base of penis so that it does not slip off
14. Slowly withdraw penis before it becomes soft (flaccid)
15. Remove condom carefully to avoid spilling semen (ejaculate, cum)
16. Tie or wrap condom and discard appropriately (burn, bury, throw in latrine pit, may be flushed down toilet if served by an individual pit, do not put in garbage if likely to be rummaged by street people)
17. Use a condom for each act of sexual intercourse
18. If condom breaks during sex, withdraw penis immediately and put on a new condom. The female should use emergency contraceptive pills
19. Don't store condoms for a long time in your wallet, back pocket, or near heat, because it might break. Store in a cool dry place if possible
20. Condoms are designed so as not to break if used correctly. They protect you against unwanted pregnancy, STIs and HIV/AIDS. To help prevent disease, use them for vaginal, anal, or oral sex.

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